Model Analysis

Table of contents

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```python
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]


    cols = list(dict.fromkeys(target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
```

```python
    #df['campaign'] = "ALL"

    return df
```

```python
def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)
```

```python
    return df,dict_features
```

```python
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test $<$ 0.95 (under performance of the test)

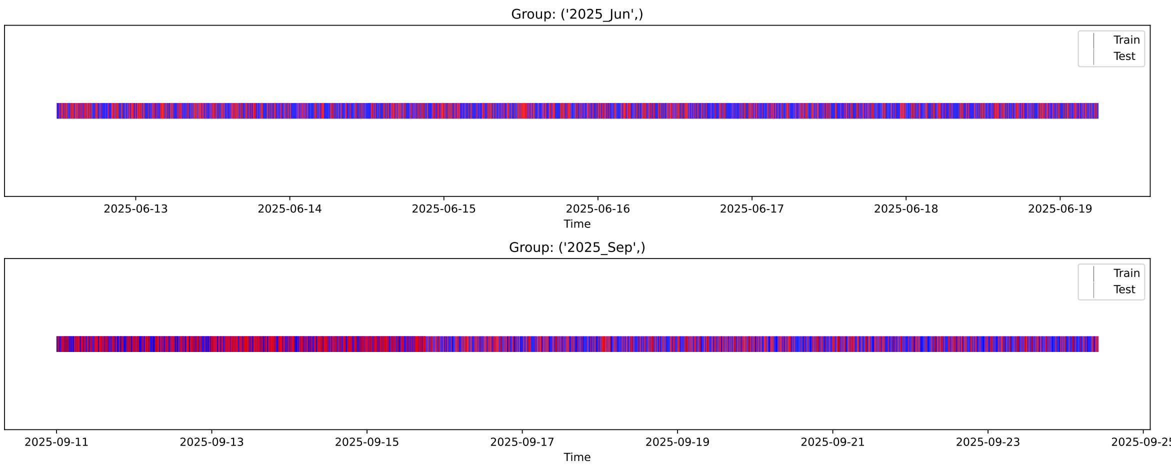
[Green] r_test $>$ 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_train vs X_test

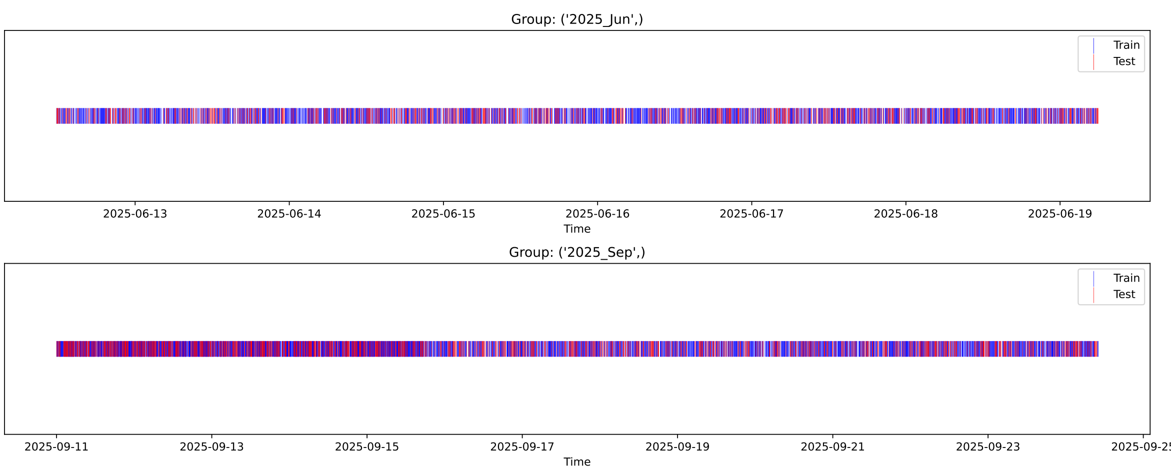
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

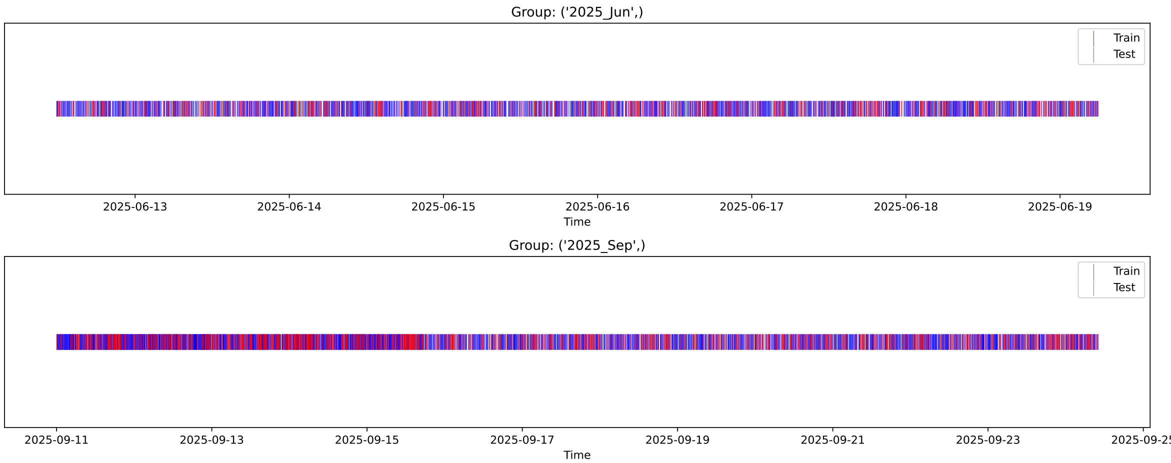
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



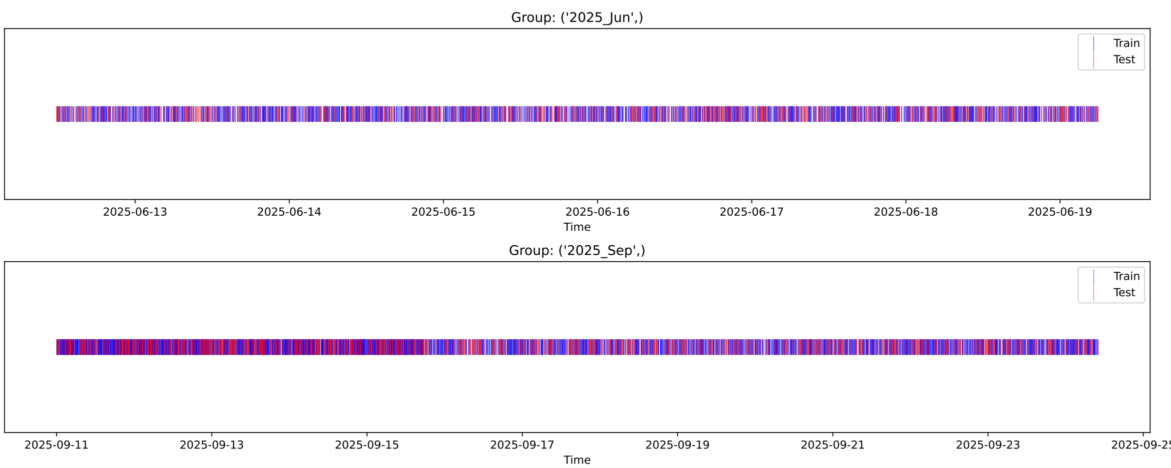
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



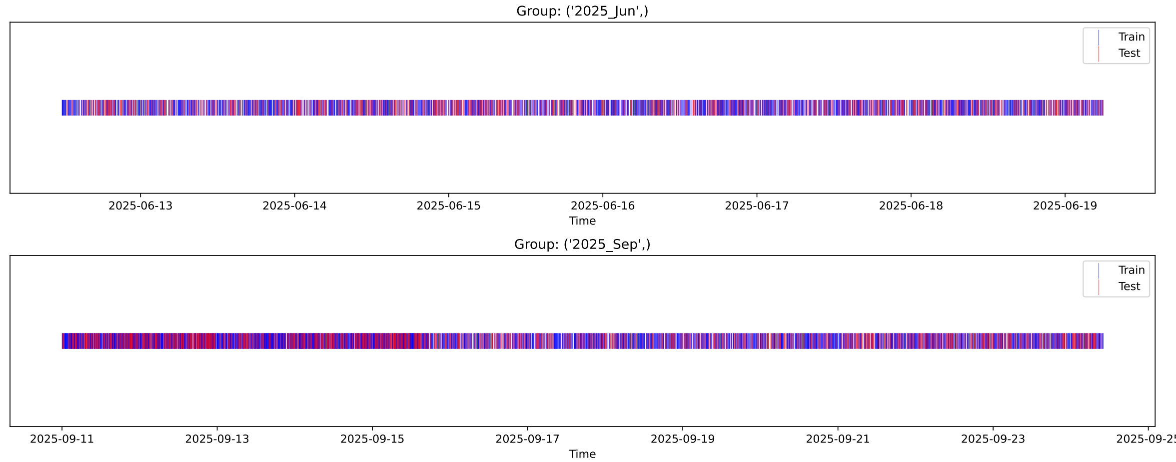
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



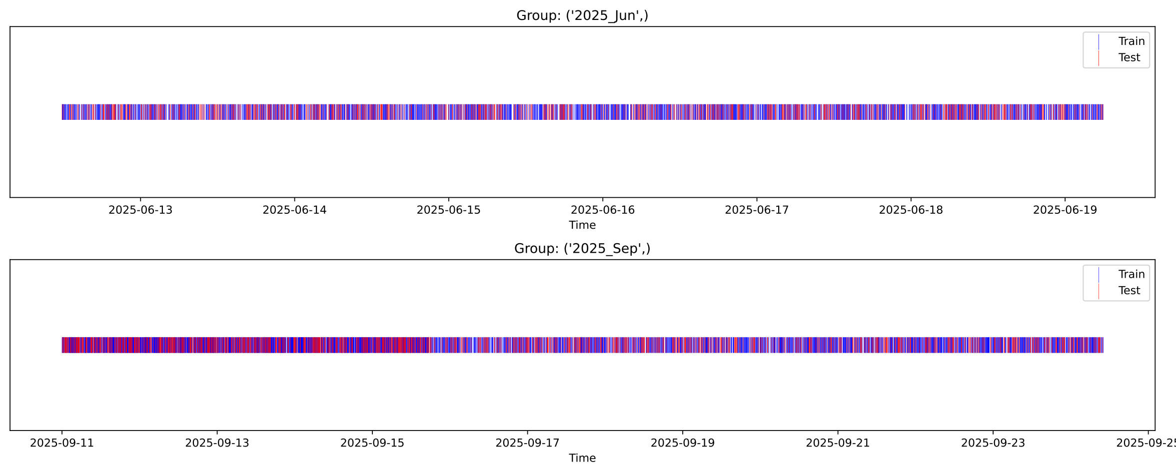
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.339, std=0.798, min=0.959, max=13.845 y_test: mean=2.374, std=0.786, min=1.071, max=7.277

Fold 0 — y_test: mean=2.34, std=0.82, y_train: mean=2.34, std=0.79

Fold 1 — y_test: mean=2.34, std=0.77, y_train: mean=2.34, std=0.81

Fold 2 — y_test: mean=2.36, std=0.85, y_train: mean=2.33, std=0.78

Fold 3 — y_test: mean=2.36, std=0.84, y_train: mean=2.33, std=0.78

Fold 4 — y_test: mean=2.35, std=0.76, y_train: mean=2.34, std=0.81

train scores CV: [0.35126041 0.34129782 0.38941971 0.36572649 0.35147845]

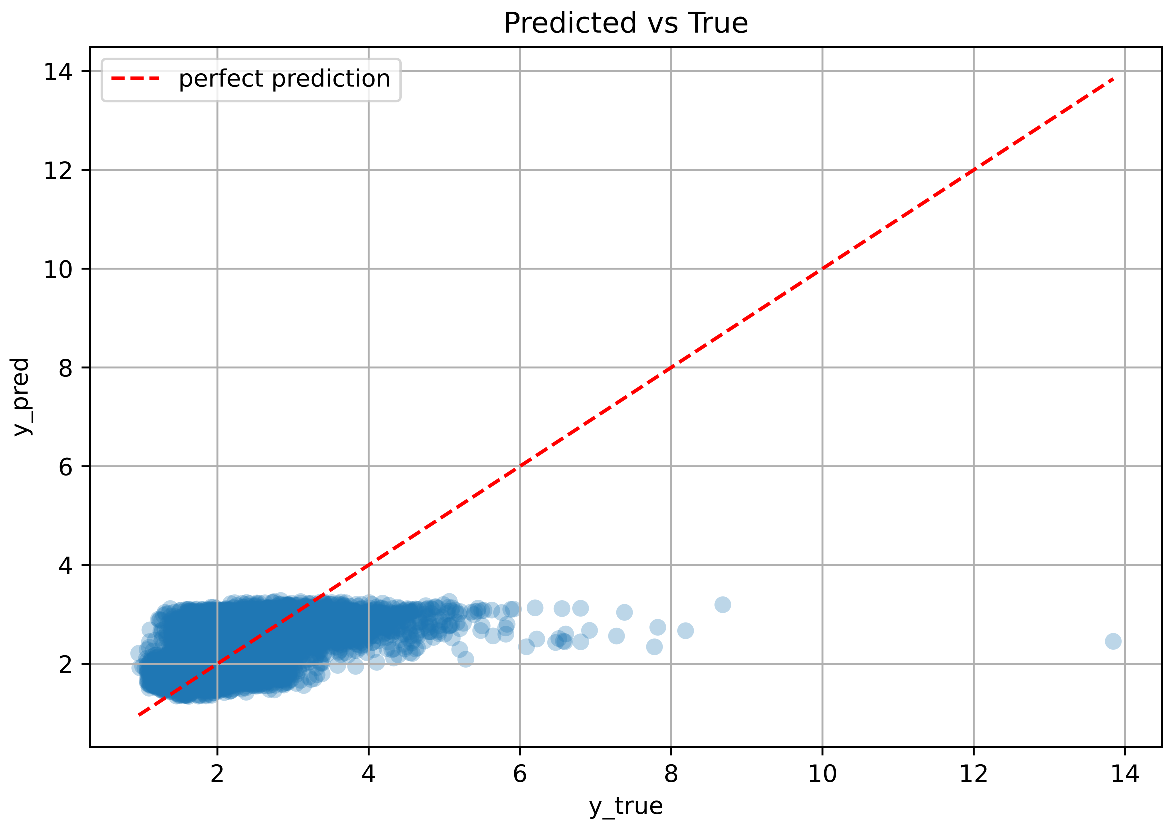
test scores CV: [0.37881413 0.40998904 0.30281638 0.34887505 0.38406497]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3598	0.3649	0.0366	0.3451	0.3453	-0.0197	0.9461	0.9861

Coefficients:

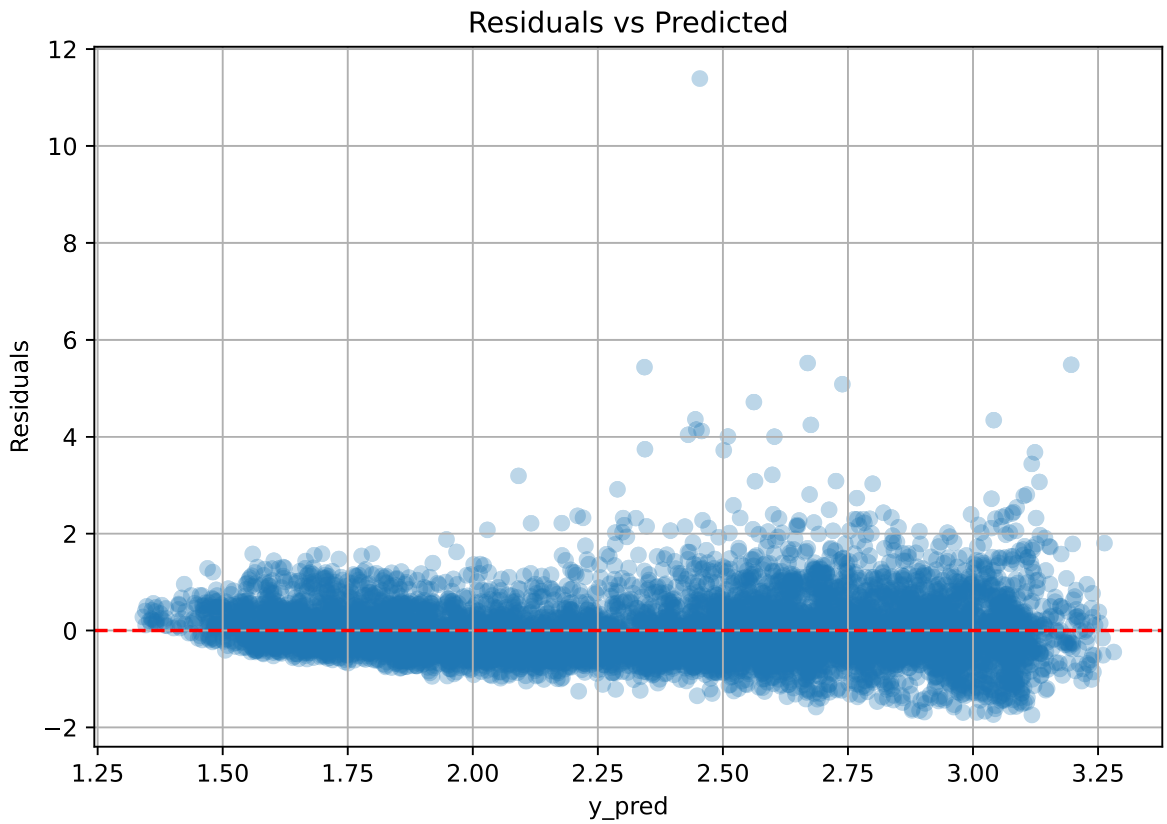
coef_r_CO2 = -0.0491, coef_r_NH3 = -0.0447, coef_r_THI = 0.5970, coef_r_temperature = -0.7372, coef_r_umidity = -0.6671, intercept = 2.3392,

2.3.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

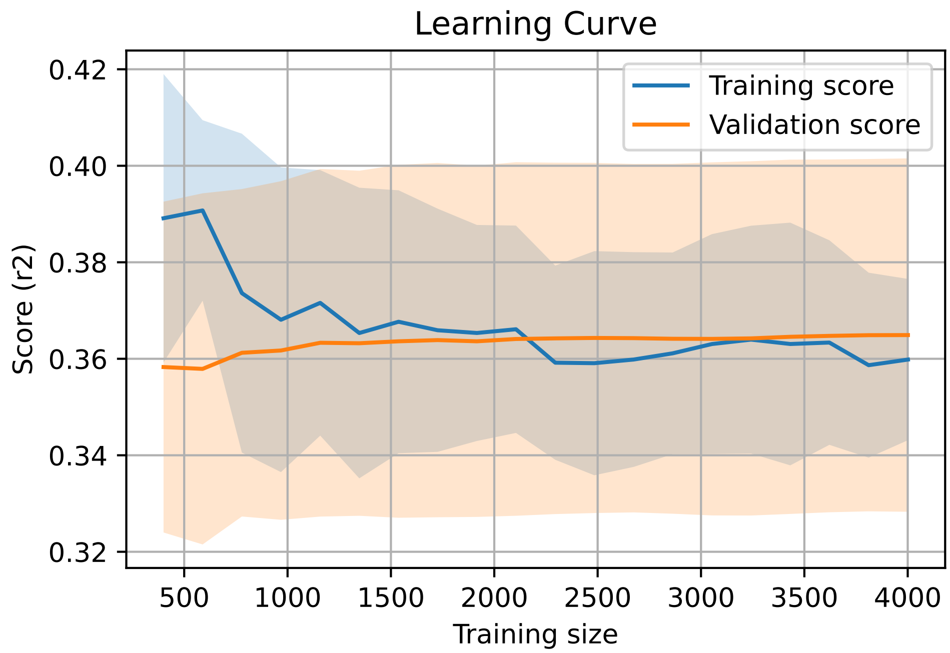
2.3.3 Scatter residuals



2.3.4 Learning curve

$\mathbf{Len(group)}$: 8168, $\mathbf{Len(training)}$: 5718,

$\mathbf{Len(training\_real)}$: 4003, $\mathbf{Len(test\_of\_training)}$: 1715, $\mathbf{Len(test)}$: 2450

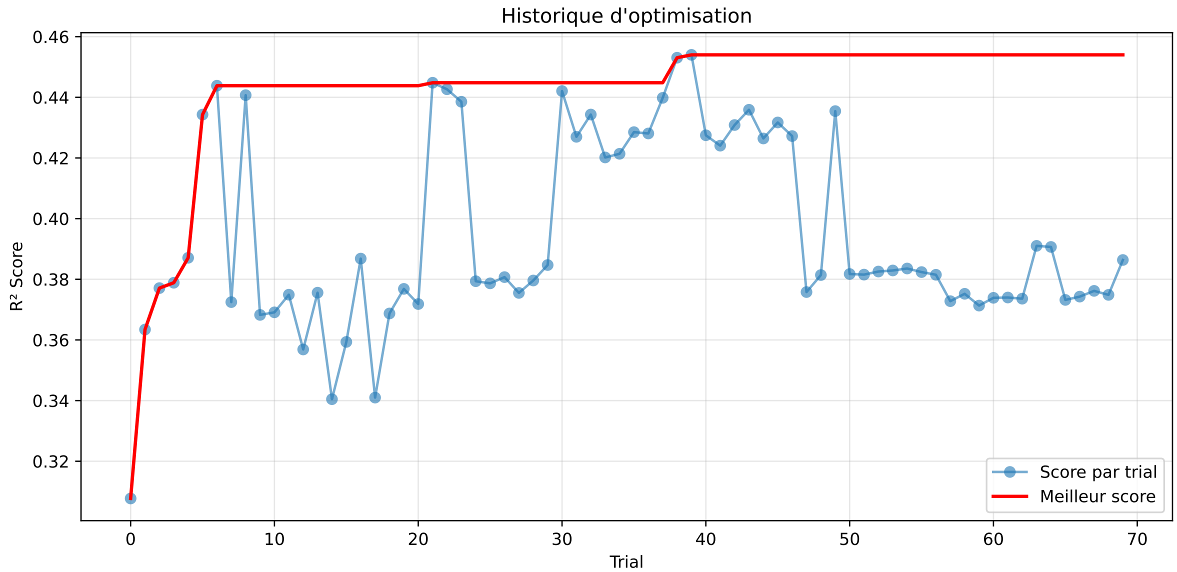


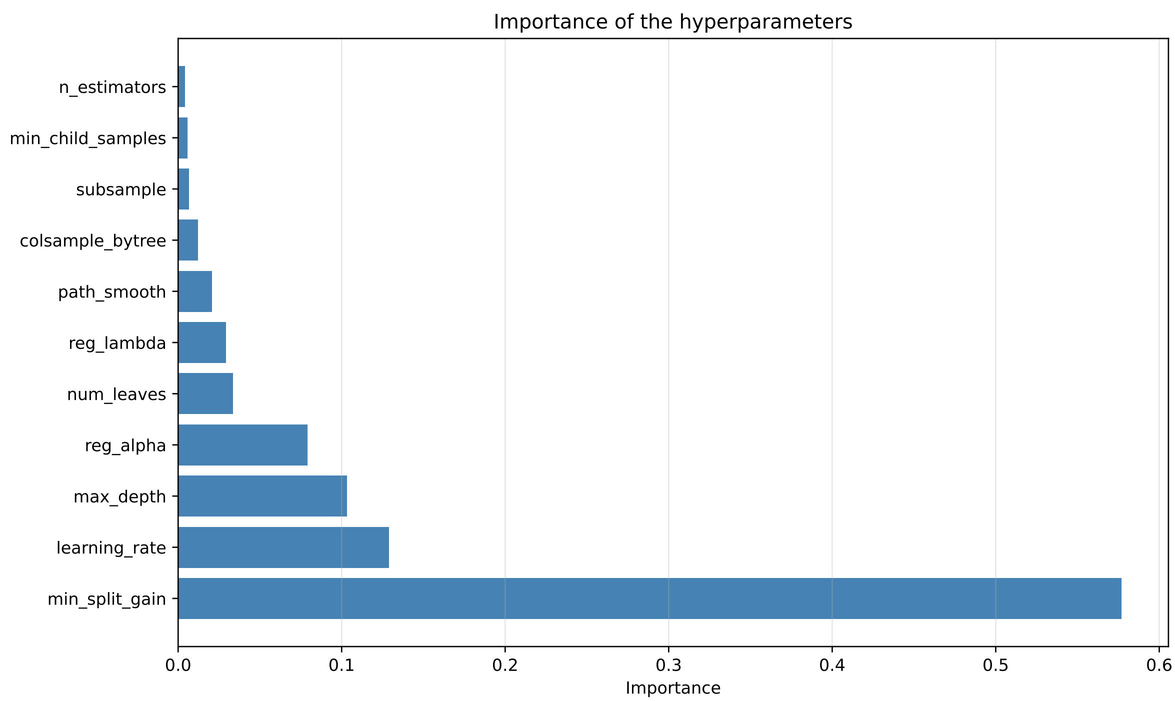
2.4 Model : LGBM

2.4.1 Gridsearch

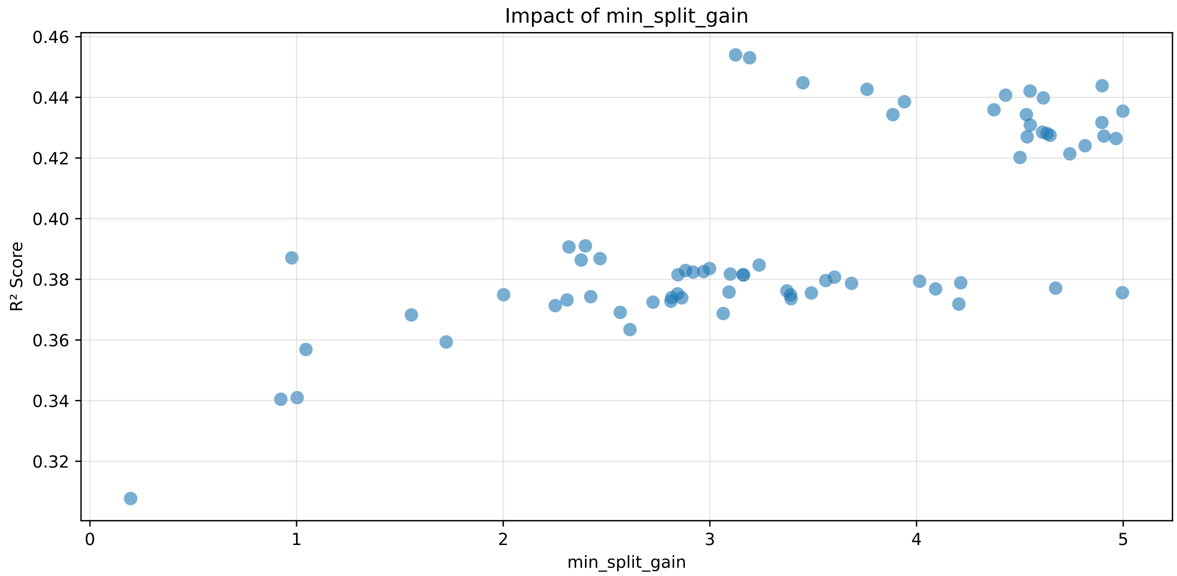
Distribution y_train vs y_test: y_train: mean=2.339, std=0.798, min=0.959, max=13.845 y_test: mean=2.374, std=0.786, min=1.071, max=7.277

==================== TOP Importance FEATURES ==================== feature importance r_CO2 594 r_umidity 549 r_NH3 446 r_THI 302 r_temperature 265

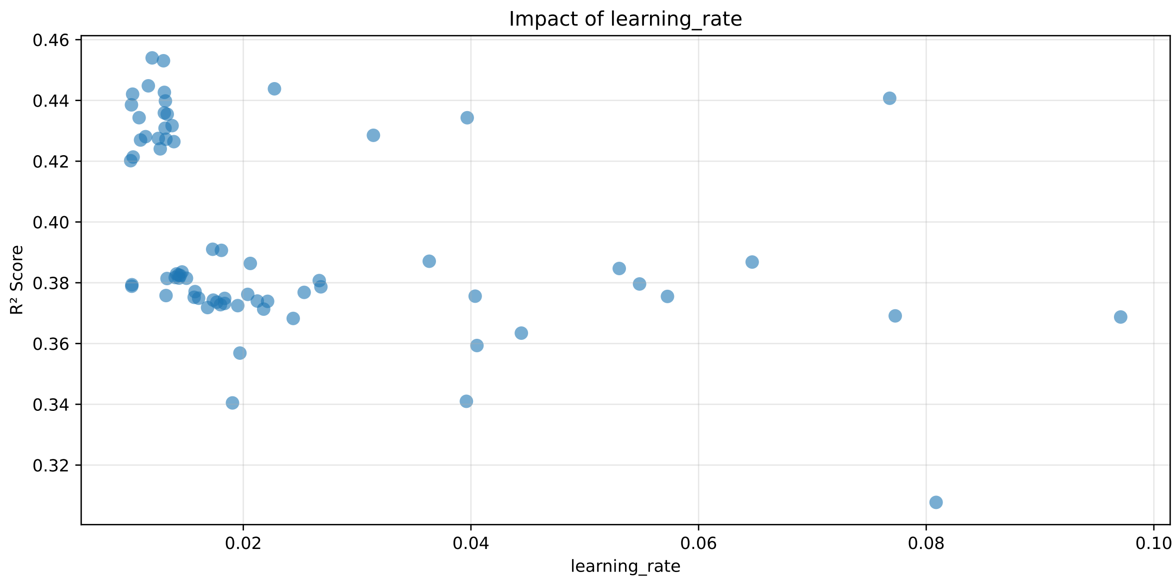




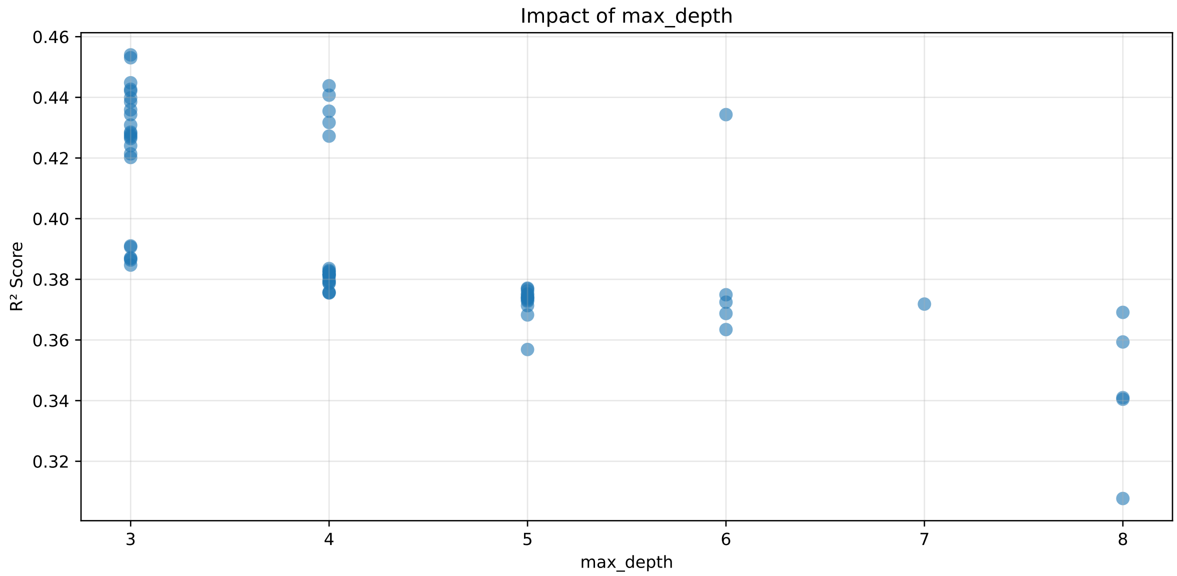
Paramètres avec >5% d'importance : ['min_split_gain', 'learning_rate', 'max_depth', 'reg_alpha']

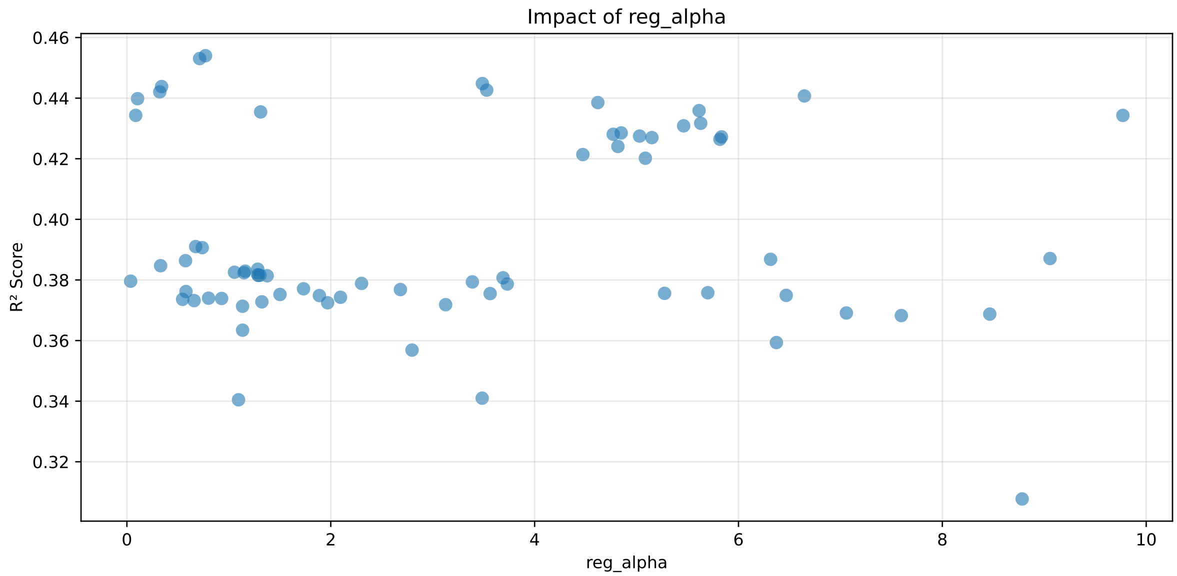


Impact of min_split_gain
R² Score
0.46
0.44
0.42
0.40
0.38
0.36
0.34
0.32
0
1
2
3
4
5
min_split_gain



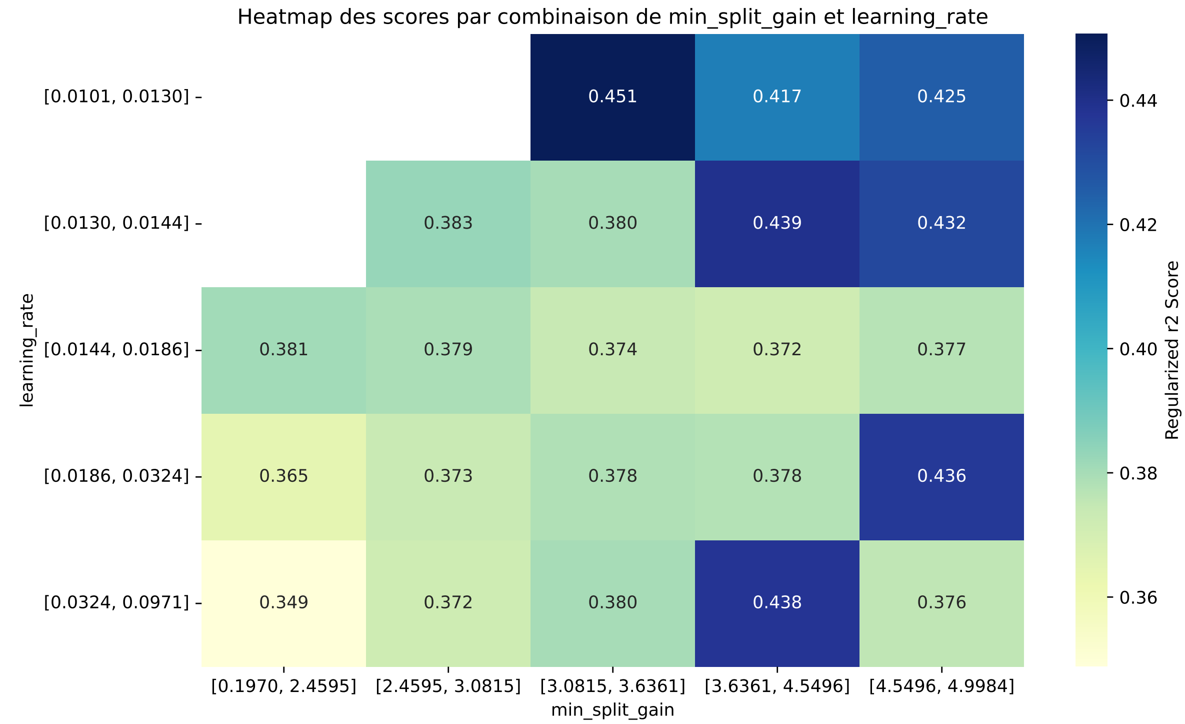
Impact of learning_rate
R² Score
0.46
0.44
0.42
0.40
0.38
0.36
0.34
0.32
0.02
0.04
0.06
0.08
0.10
learning_rate






```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main
```

```
The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4757	0.454	0.4674	0.0359	0.454	0.4909	0.0369	1.0812	1.0478
2	0.4745	0.4531	0.466	0.0364	0.4531	0.4906	0.0375	1.0828	1.0473
3	0.4647	0.4448	0.4565	0.0354	0.4448	0.4823	0.0375	1.0843	1.0448
4	0.4654	0.4438	0.4578	0.0379	0.4438	0.4787	0.0349	1.0787	1.0486
5	0.4618	0.4427	0.4522	0.0383	0.4427	0.4766	0.0339	1.0766	1.0433
6	0.4613	0.4421	0.4547	0.0372	0.4421	0.4799	0.0378	1.0856	1.0434
7	0.4623	0.4407	0.4526	0.0376	0.4407	0.4842	0.0435	1.0986	1.049
8	0.4587	0.4398	0.4494	0.0367	0.4398	0.4705	0.0307	1.0698	1.0429
9	0.4575	0.4385	0.4488	0.0362	0.4385	0.4712	0.0326	1.0744	1.0433
10	0.4543	0.4359	0.4478	0.0363	0.4359	0.4758	0.0399	1.0915	1.0423
---	---	---	---	---	---	---	---	---	---
70	0.5923	0.5211	0.5507	0.0364	0.3077	0.5735	0.0523	1.1004	1.1365

Hyperparameters:

Rank 1: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.01199168437990067, 'reg__num_leaves': 22, 'reg__subsample': 0.5615776645479831, 'reg__colsample_bytree': 0.6819659152036085, 'reg__min_child_samples': 45, 'reg__reg_alpha': 0.7713203912032762, 'reg__reg_lambda': 39.63690534464575, 'reg__min_split_gain': 3.1243897613964666, 'reg__path_smooth': 1.233501469474358}

Rank 2: {'reg___n_estimators': 800, 'reg___max_depth': 3, 'reg___learning_rate': 0.012988181554606818, 'reg___num_leaves': 24, 'reg___subsample': 0.5513892115387549, 'reg___colsample_bytree': 0.6803672790895157, 'reg___min_child_samples': 44, 'reg___reg_alpha': 0.7133344134211523, 'reg___reg_lambda': 39.484381675468235, 'reg___min_split_gain': 3.1922591726154215, 'reg___path_smooth': 1.0380228088989287}

Rank 3: {'reg___n_estimators': 800, 'reg___max_depth': 3, 'reg___learning_rate': 0.011656355459181965, 'reg___num_leaves': 27, 'reg___subsample': 0.5202826350775632, 'reg___colsample_bytree': 0.6693325408784353, 'reg___min_child_samples': 49, 'reg___reg_alpha': 3.4874563128486566, 'reg___reg_lambda': 34.04463106403613, 'reg___min_split_gain': 3.450138350041967, 'reg___path_smooth': 0.18118818725727737}

Rank 4: {'reg___n_estimators': 700, 'reg___max_depth': 4, 'reg___learning_rate': 0.022737156319525612, 'reg___num_leaves': 17, 'reg___subsample': 0.6989599888806316, 'reg___colsample_bytree': 0.7099027962649053, 'reg___min_child_samples': 23, 'reg___reg_alpha': 0.3408243444685466, 'reg___reg_lambda': 37.49228214197993, 'reg___min_split_gain': 4.8981840144302815, 'reg___path_smooth': 3.5760252051295516}

Rank 5: {'reg___n_estimators': 400, 'reg___max_depth': 3, 'reg___learning_rate': 0.013066123673130377, 'reg___num_leaves': 28, 'reg___subsample': 0.5366805913404906, 'reg___colsample_bytree': 0.5214343699048389, 'reg___min_child_samples': 10, 'reg___reg_alpha': 3.5301010402877586, 'reg___reg_lambda': 33.97346991298898, 'reg___min_split_gain': 3.760399507873379, 'reg___path_smooth': 0.24282977669440298}

Rank 6: {'reg___n_estimators': 800, 'reg___max_depth': 3, 'reg___learning_rate': 0.010273940833793l4, 'reg___num_leaves': 40, 'reg___subsample': 0.5587574634331193, 'reg___colsample_bytree': 0.8414755621877715, 'reg___min_child_samples': 47, 'reg___reg_alpha': 0.3223901854967406, 'reg___reg_lambda': 42.06156091810501, 'reg___min_split_gain': 4.549415042357401, 'reg___path_smooth': 3.4455315146122913}

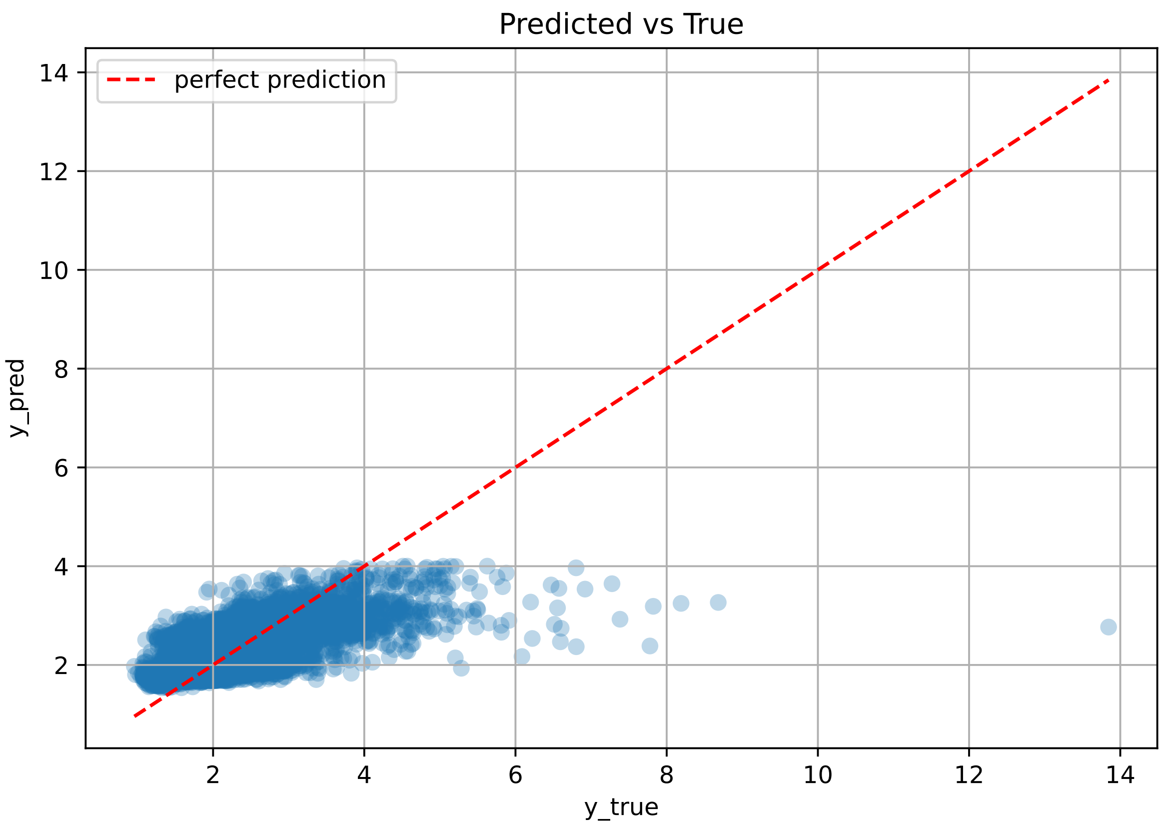
Rank 7: {'reg___n_estimators': 400, 'reg___max_depth': 4, 'reg___learning_rate': 0.07678997050727404, 'reg___num_leaves': 46, 'reg___subsample': 0.5947631275375199, 'reg___colsample_bytree': 0.8166476385594135, 'reg___min_child_samples': 10, 'reg___reg_alpha': 6.646737548985743, 'reg___reg_lambda': 23.360510666866258, 'reg___min_split_gain': 4.430929970920343, 'reg___path_smooth': 2.096576134560337}

Rank 8: {'reg___n_estimators': 300, 'reg___max_depth': 3, 'reg___learning_rate': 0.013157114873697211, 'reg___num_leaves': 23, 'reg___subsample': 0.5504717252485534, 'reg___colsample_bytree': 0.6865036006041869, 'reg___min_child_samples': 10, 'reg___reg_alpha': 0.10501352244014095, 'reg___reg_lambda': 38.78474491422058, 'reg___min_split_gain': 4.613562313705793, 'reg___path_smooth': 3.4470033862432667}

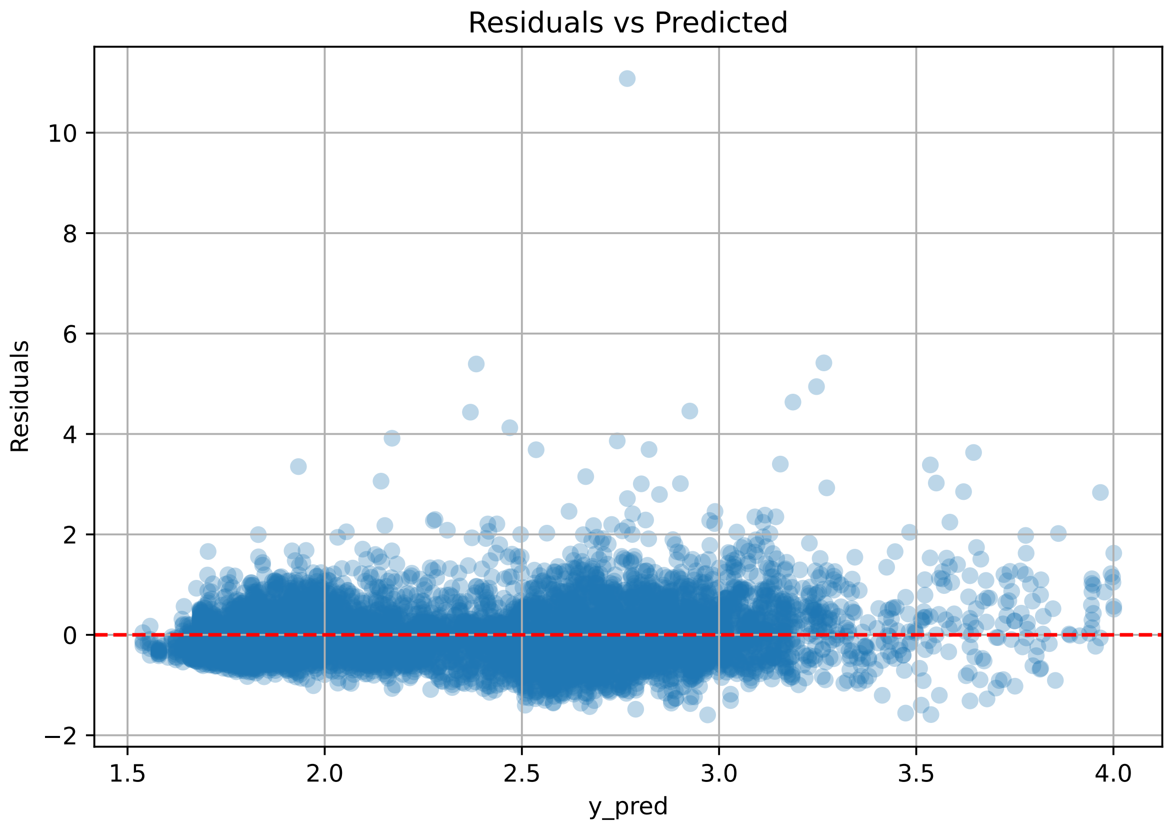
Rank 9: {'reg___n_estimators': 400, 'reg___max_depth': 3, 'reg___learning_rate': 0.010173955292635924, 'reg___num_leaves': 29, 'reg___subsample': 0.5914100946495716, 'reg___colsample_bytree': 0.5130890414128788, 'reg___min_child_samples': 24, 'reg___reg_alpha': 4.620112276614183, 'reg___reg_lambda': 29.768834625531902, 'reg___min_split_gain': 3.9415442176803066, 'reg___path_smooth': 0.3987830795781573}

Rank 10: {'reg___n_estimators': 800, 'reg___max_depth': 3, 'reg___learning_rate': 0.013062609272879309, 'reg___num_leaves': 36, 'reg___subsample': 0.5477846956053634, 'reg___colsample_bytree': 0.7445680453058902, 'reg___min_child_samples': 50, 'reg___reg_alpha': 5.613430613358474, 'reg___reg_lambda': 29.753928817991593, 'reg___min_split_gain': 4.3748995794096235, 'reg___path_smooth': 1.1698548902278627}

2.4.2 Scatter



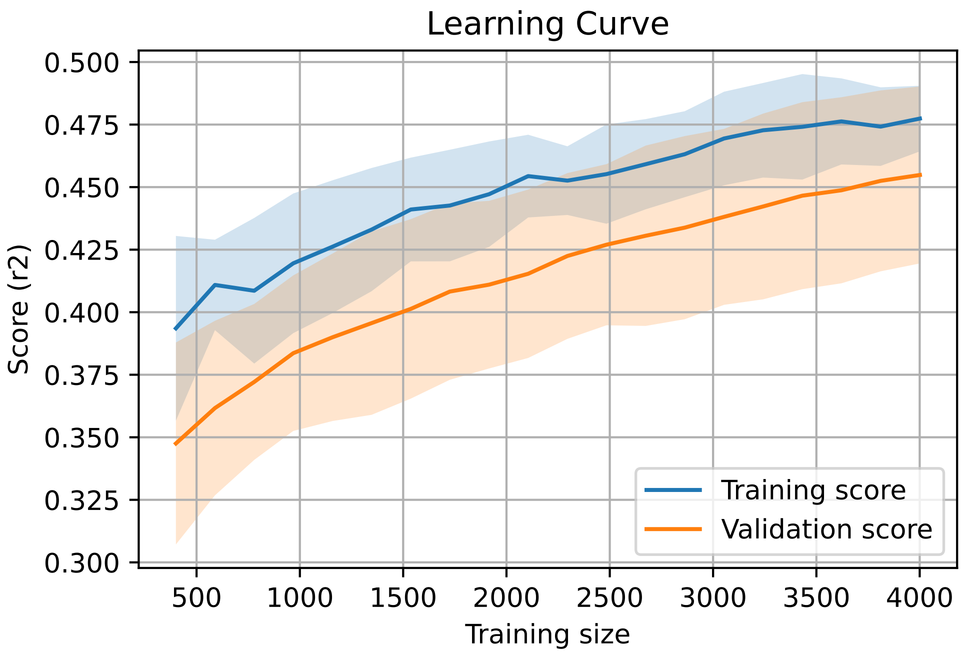
2.4.3 Scatter residuals



2.4.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450

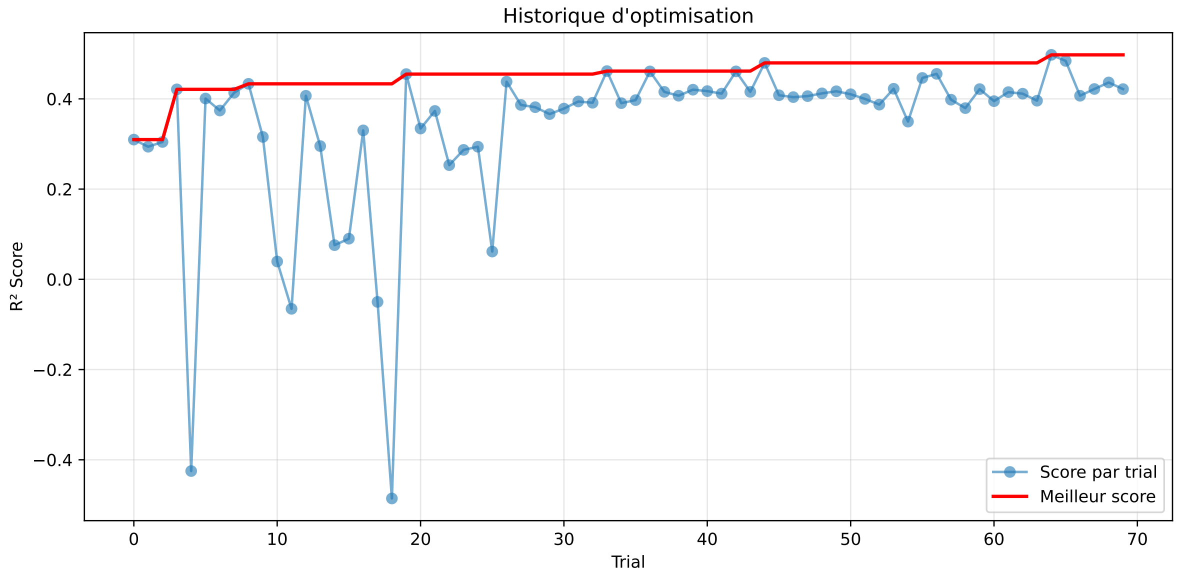


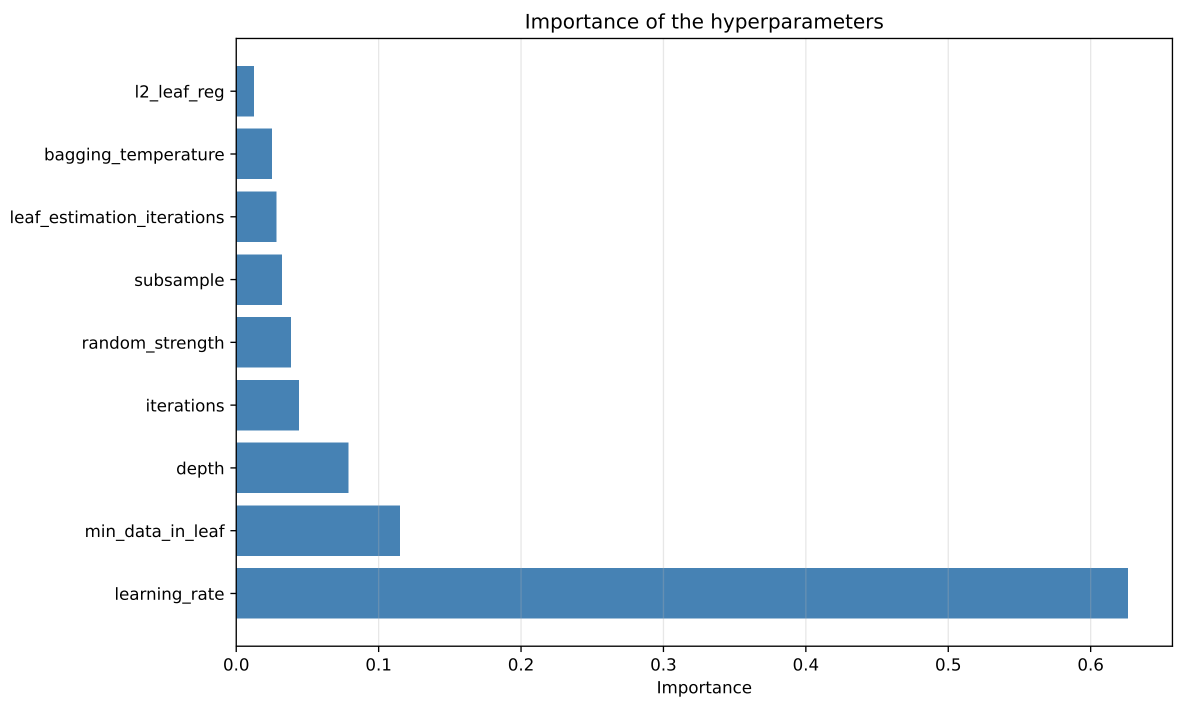
2.5 Model : CatBoost

2.5.1 Gridsearch

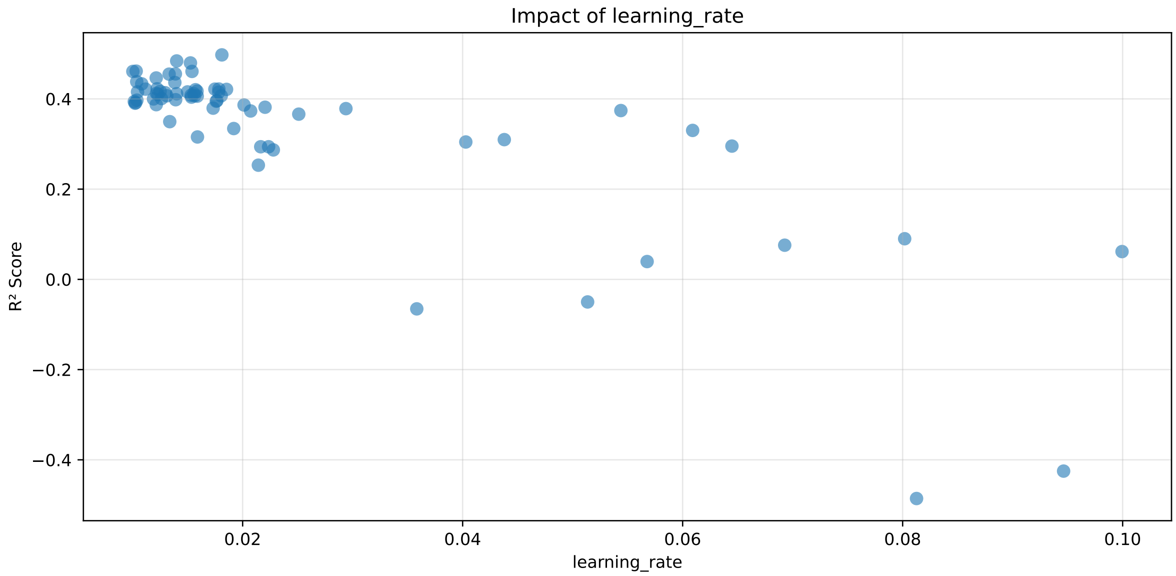
Distribution y_train vs y_test: y_train: mean=2.339, std=0.798, min=0.959, max=13.845 y_test: mean=2.374, std=0.786, min=1.071, max=7.277

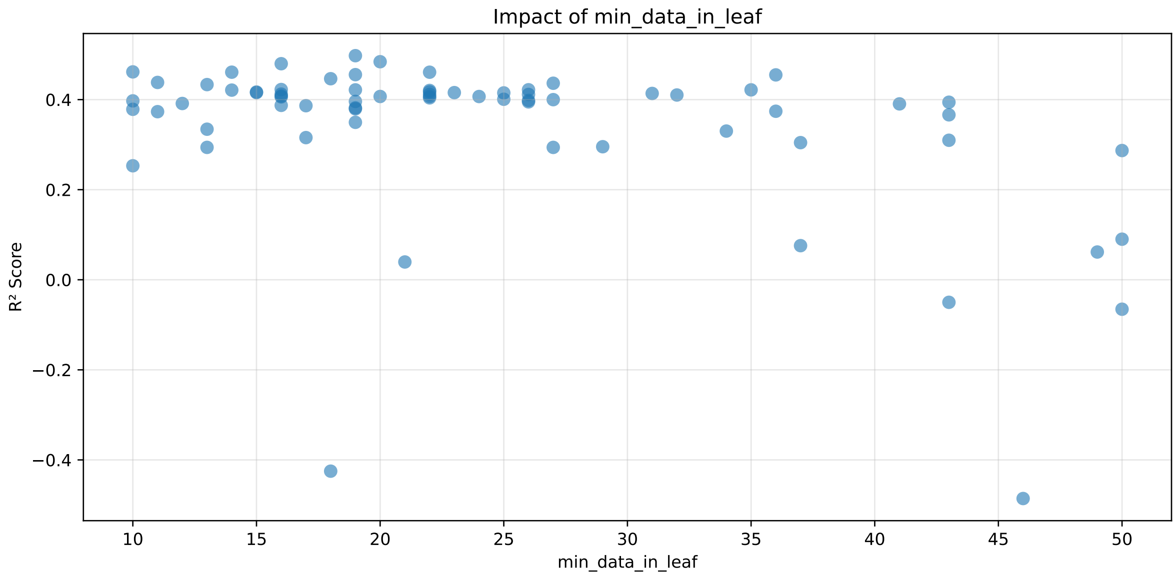
==================== TOP Importance FEATURES ==================== feature importance r_umidity 47.226513 r_NH3 18.878599 r_CO2 16.019557 r_temperature 9.726713 r_THI 8.148617

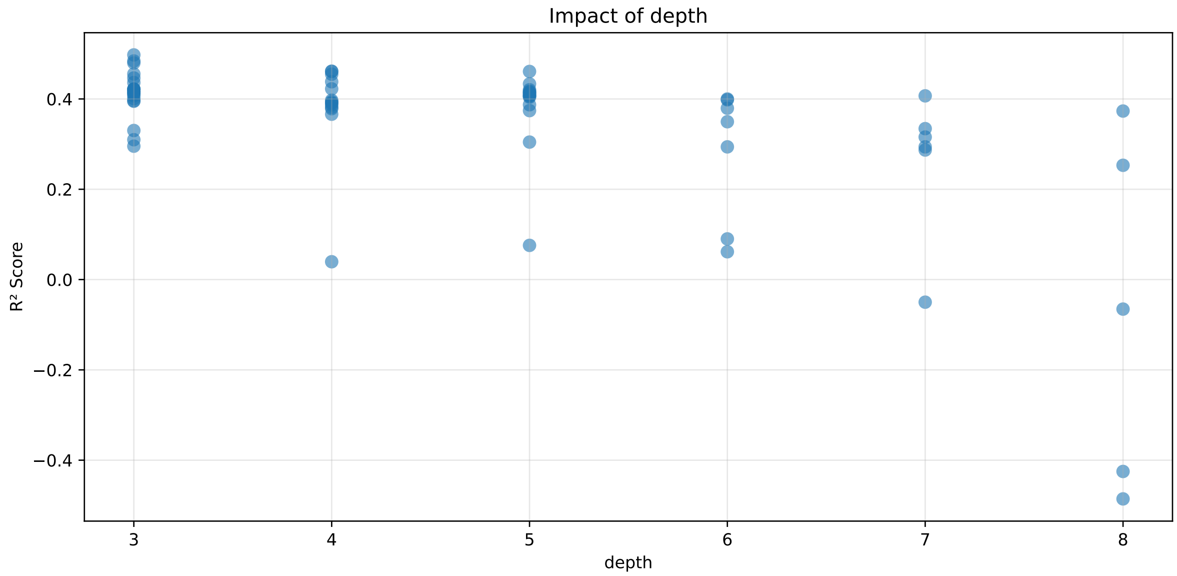




Paramètres avec >5% d'importance : ['learning_rate', 'min_data_in_leaf', 'depth']

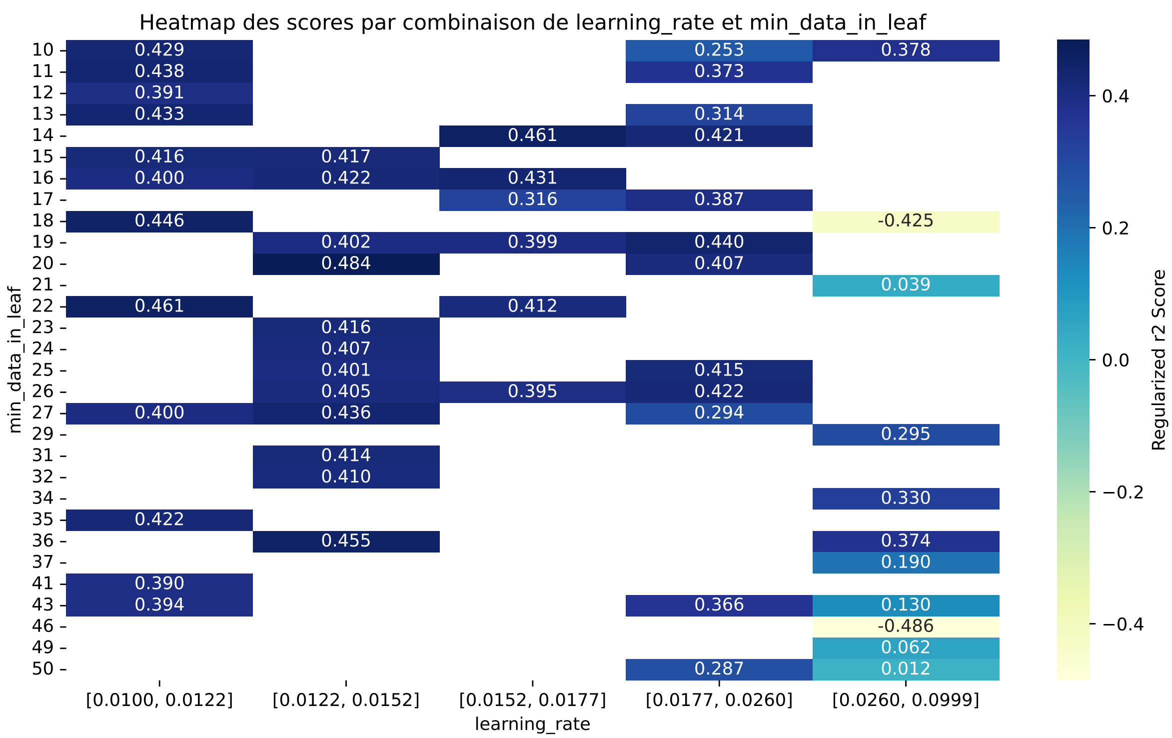

Impact of learning_rate
R² Score
learning_rate
0.4
0.2
0.0
−0.2
−0.4
0.02
0.04
0.06
0.08
0.10


Impact of min_data_in_leaf
R² Score
min_data_in_leaf
0.4
0.2
0.0
−0.2
−0.4
10
15
20
25
30
35
40
45
50




```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5217	0.4976	0.5153	0.0368	0.4976	0.5187	0.0211	1.0424	1.0485
2	0.5034	0.4841	0.4969	0.036	0.4841	0.5008	0.0167	1.0345	1.0398
3	0.4993	0.4797	0.4939	0.0362	0.4797	0.4968	0.017	1.0355	1.0408
4	0.4838	0.4616	0.4735	0.0352	0.4616	0.4779	0.0163	1.0354	1.0481
5	0.4822	0.4609	0.475	0.0364	0.4609	0.4818	0.0209	1.0454	1.0462
6	0.4754	0.4609	0.4742	0.0372	0.4609	0.4786	0.0177	1.0385	1.0315
7	0.4678	0.4554	0.4642	0.0355	0.4554	0.469	0.0136	1.0298	1.0271
8	0.4765	0.4549	0.4671	0.036	0.4549	0.4712	0.0164	1.036	1.0476
9	0.4574	0.4464	0.4547	0.0356	0.4464	0.4601	0.0137	1.0307	1.0246
10	0.4575	0.4381	0.4485	0.0348	0.4381	0.45	0.0119	1.0271	1.0442
---	---	---	---	---	---	---	---	---	---
70	0.9285	0.575	0.6073	0.0366	-0.4858	0.6265	0.0515	1.0896	1.6149

Hyperparameters:

Rank 1: {’reg__iterations’: 600, ’reg__depth’: 3, ’reg__learning_rate’: 0.018112776270640337, ’reg__l2_leaf_reg’: 21.339050218387253, ’reg__bagging_temperature’: 3.4240831055459298, ’reg__random_strength’: 3.37779201204604, ’reg__subsample’: 0.581509345320927, ’reg__min_data_in_leaf’: 19, ’reg__leaf_estimation_iterations’: 3}

Rank 2: {’reg__iterations’: 600, ’reg__depth’: 3, ’reg__learning_rate’: 0.014005604465527814, ’reg__l2_leaf_reg’: 4.358699896033917, ’reg__bagging_temperature’: 3.4853900827907065, ’reg__random_strength’: 3.2064642264037446, ’reg__subsample’: 0.7936203869266801, ’reg__min_data_in_leaf’: 20, ’reg__leaf_estimation_iterations’: 3}

Rank 3: {’reg__iterations’: 500, ’reg__depth’: 3, ’reg__learning_rate’: 0.015256039226130976, ’reg__l2_leaf_reg’: 8.02295055567171, ’reg__bagging_temperature’: 3.73983777006383, ’reg__random_strength’: 2.806559633567974, ’reg__subsample’: 0.6277141834070601, ’reg__min_data_in_leaf’: 16, ’reg__leaf_estimation_iterations’: 6}

Rank 4: {’reg__iterations’: 300, ’reg__depth’: 4, ’reg__learning_rate’: 0.01032377848275197, ’reg__l2_leaf_reg’: 7.805581796902707, ’reg__bagging_temperature’: 4.236724994060335, ’reg__random_strength’: 0.8118916678436594, ’reg__subsample’: 0.5554168901166894, ’reg__min_data_in_leaf’: 10, ’reg__leaf_estimation_iterations’: 8}

Rank 5: {’reg__iterations’: 200, ’reg__depth’: 5, ’reg__learning_rate’: 0.01539656371055835, ’reg__l2_leaf_reg’: 6.39570071949829, ’reg__bagging_temperature’: 3.792578754036, ’reg__random_strength’: 2.693003235372527, ’reg__subsample’: 0.6325578721226011, ’reg__min_data_in_leaf’: 14, ’reg__leaf_estimation_iterations’: 6}

Rank 6: {’reg__iterations’: 400, ’reg__depth’: 4, ’reg__learning_rate’: 0.01001871543394568, ’reg__l2_leaf_reg’: 17.81453242120887, ’reg__bagging_temperature’: 4.3569263394274, ’reg__random_strength’: 2.7196657336453445, ’reg__subsample’: 0.6486632871576355, ’reg__min_data_in_leaf’: 22, ’reg__leaf_estimation_iterations’: 6}

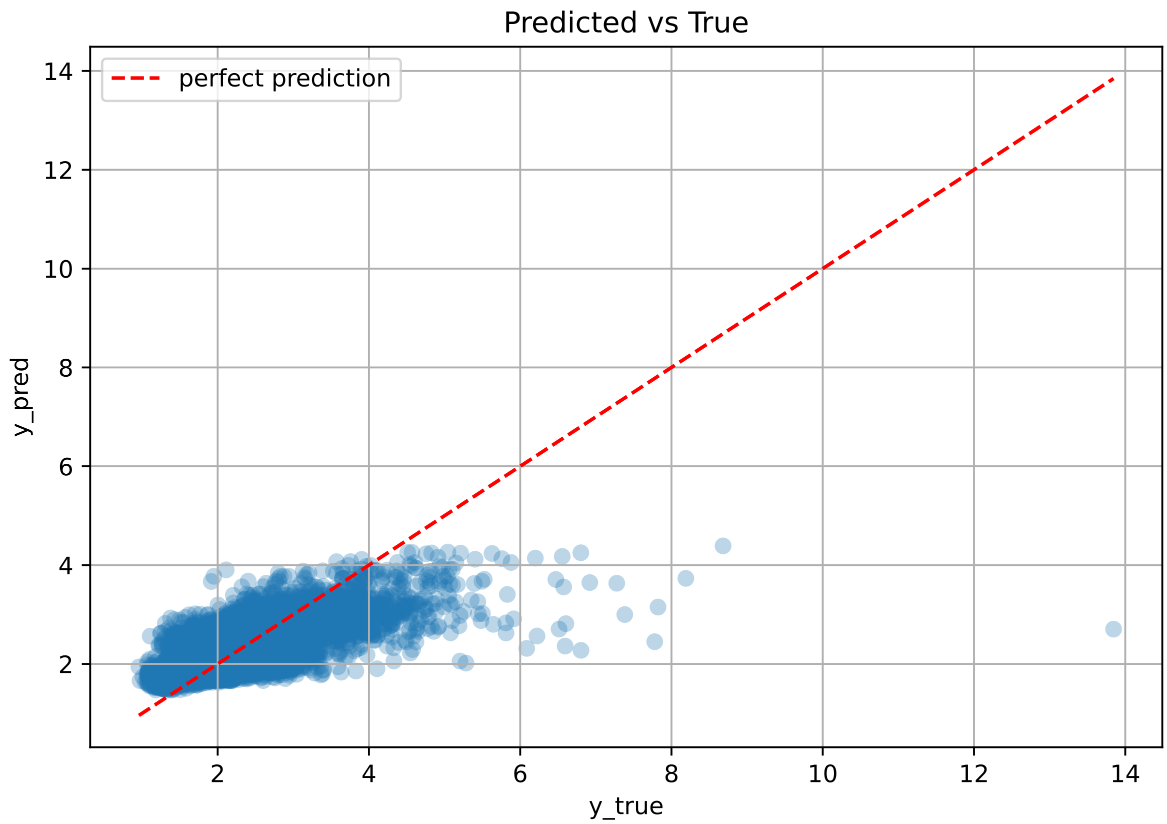
Rank 7: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.013886297416602987, 'reg__l2_leaf_reg': 5.068655881194046, 'reg__bagging_temperature': 3.403974572591717, 'reg__random_strength': 3.1441555238539896, 'reg__subsample': 0.6966201272758933, 'reg__min_data_in_leaf': 19, 'reg__leaf_estimation_iterations': 5}

Rank 8: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.013316533637910964, 'reg__l2_leaf_reg': 13.576935508506308, 'reg__bagging_temperature': 3.164320179096501, 'reg__random_strength': 0.22410012253300526, 'reg__subsample': 0.6132145874841467, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 2}

Rank 9: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.012143828688403005, 'reg__l2_leaf_reg': 8.252307691610573, 'reg__bagging_temperature': 4.564242076054199, 'reg__random_strength': 2.925804775833842, 'reg__subsample': 0.579929573833968, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 5}

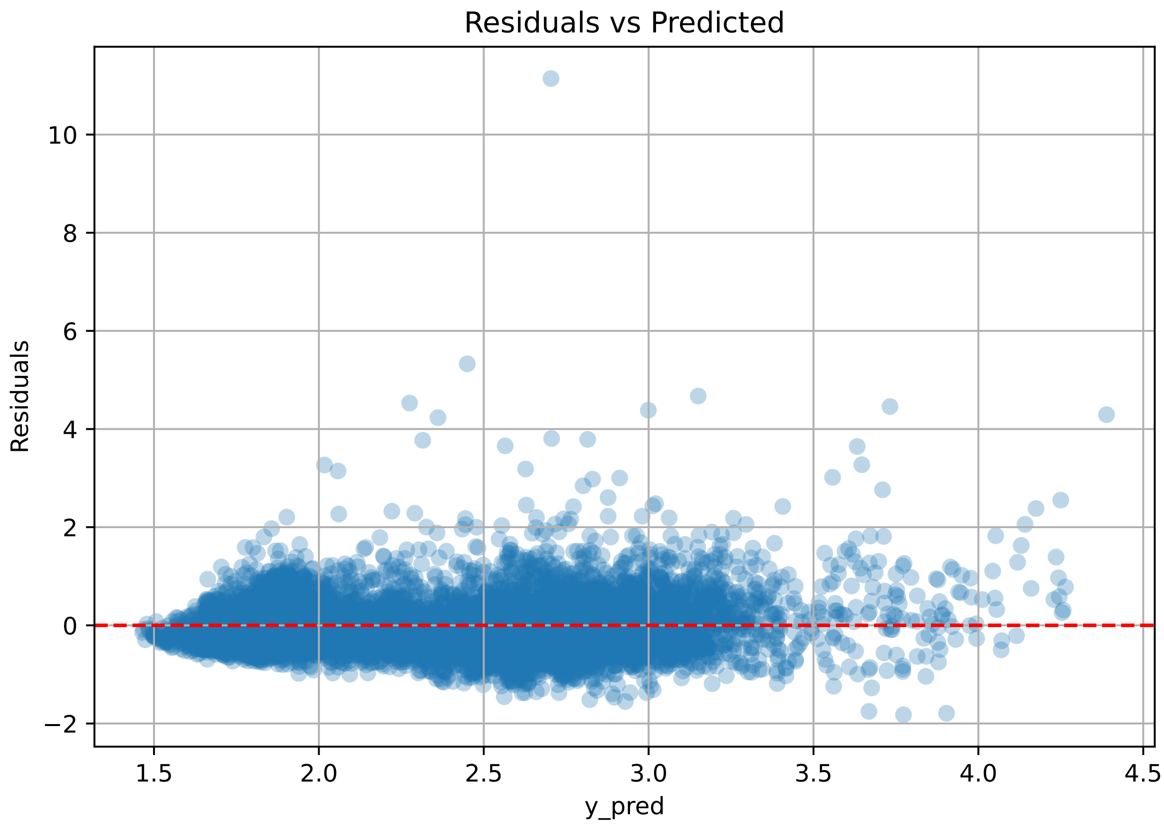
Rank 10: {'reg__iterations': 200, 'reg__depth': 4, 'reg__learning_rate': 0.010372445559157423, 'reg__l2_leaf_reg': 12.18426345555258, 'reg__bagging_temperature': 4.116241432652986, 'reg__random_strength': 0.08820128679547673, 'reg__subsample': 0.5274570202968552, 'reg__min_data_in_leaf': 11, 'reg__leaf_estimation_iterations': 7}

2.5.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

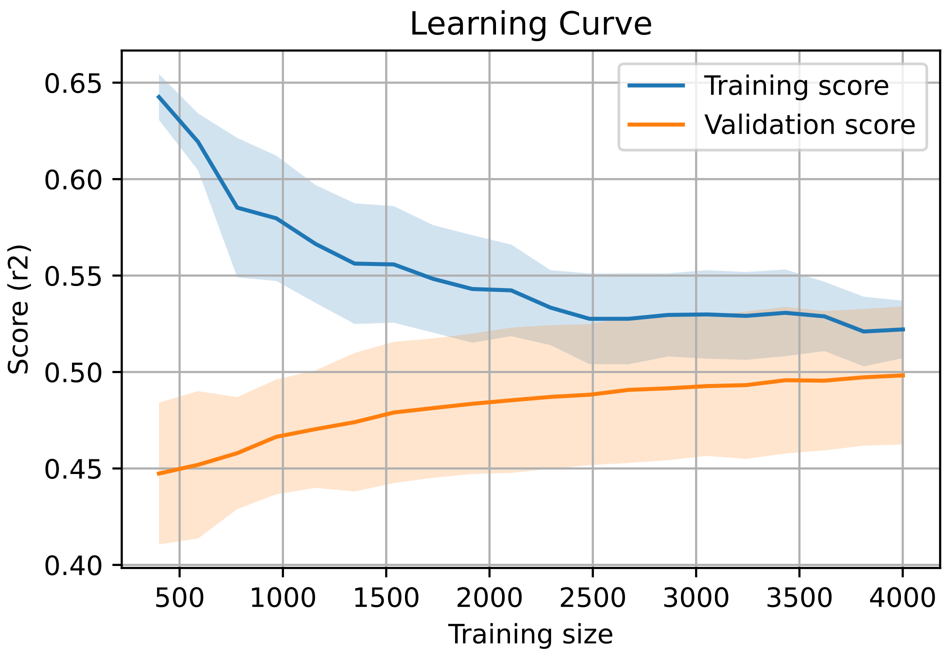
2.5.3 Scatter residuals



2.5.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450



2.6 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.36	0.365	0.037	0.345	0.345	-0.02	0.946	0.986	nan
LGBM	0.476	0.454	0.036	0.467	0.491	0.037	1.081	1.048	0.454
CatBoost	0.522	0.498	0.037	0.515	0.519	0.021	1.042	1.048	0.4976

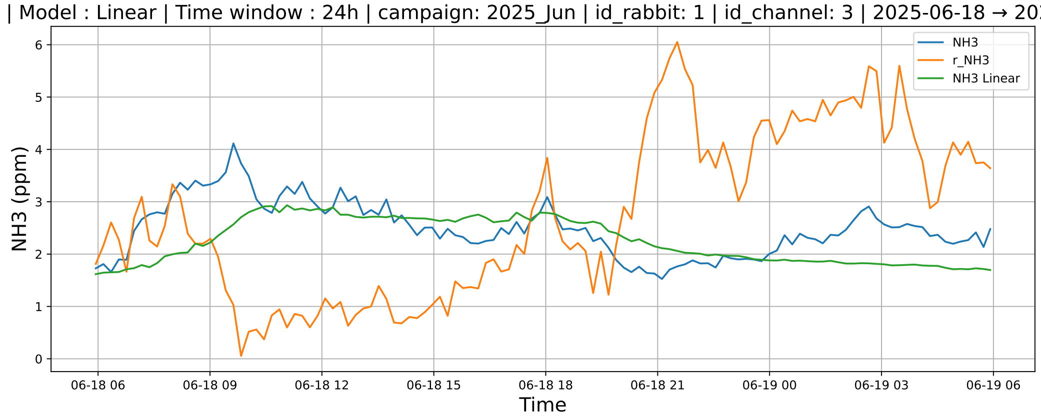
3 Plot

3.1 Standalone plots

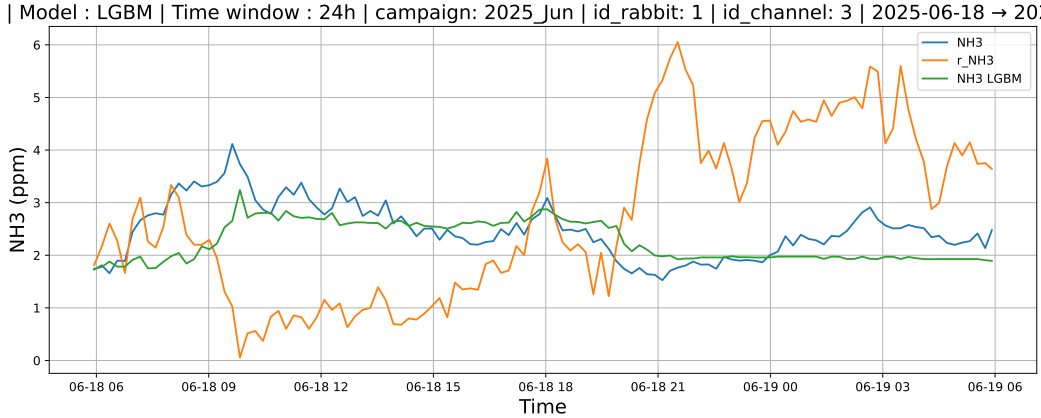
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

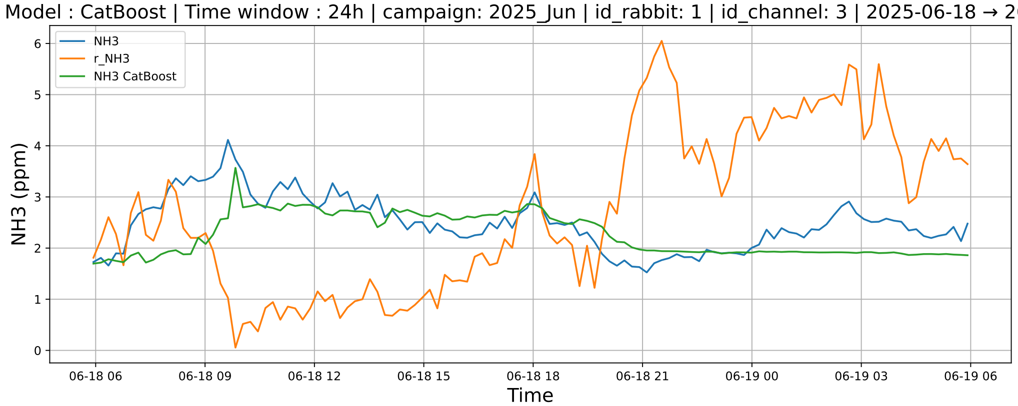
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



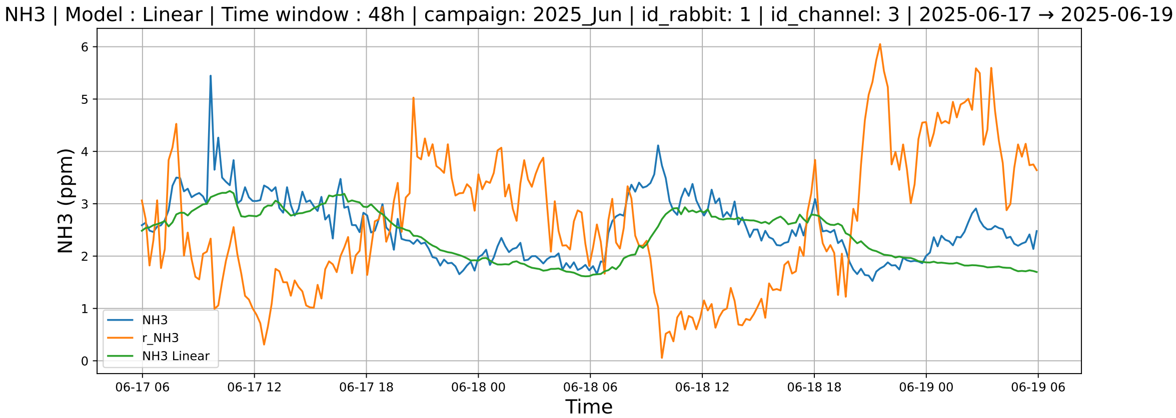
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

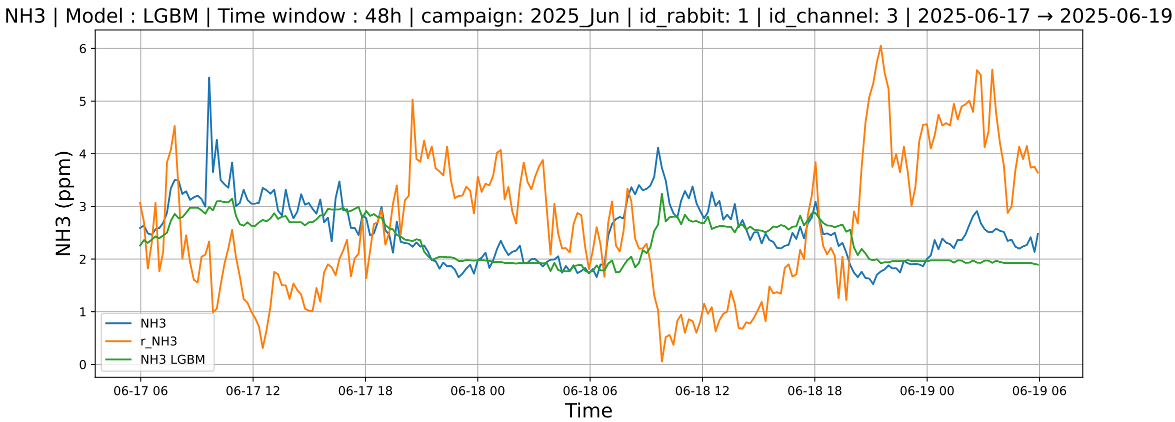


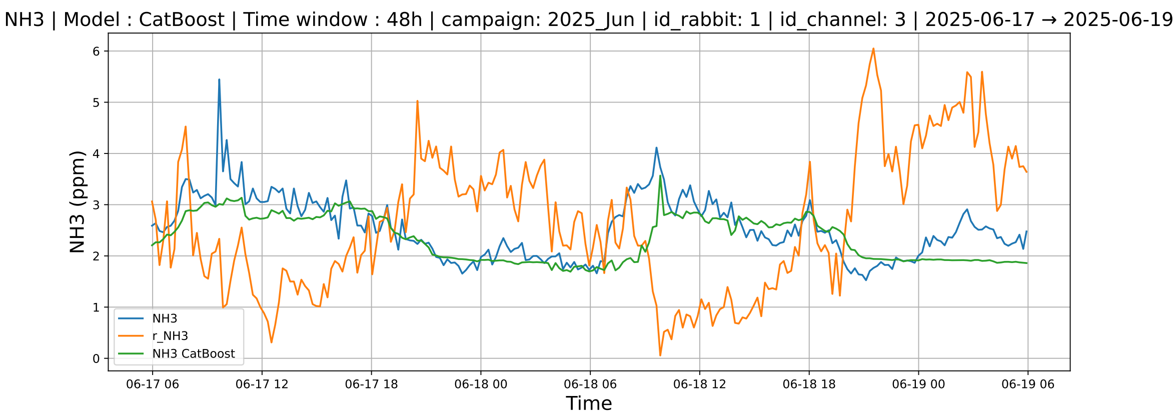
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



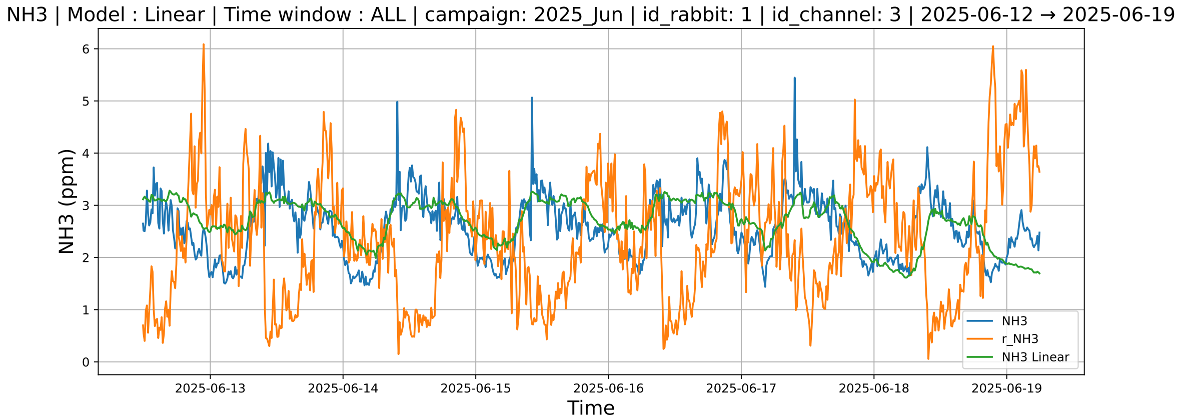
3.1.1.2 Time window : 48

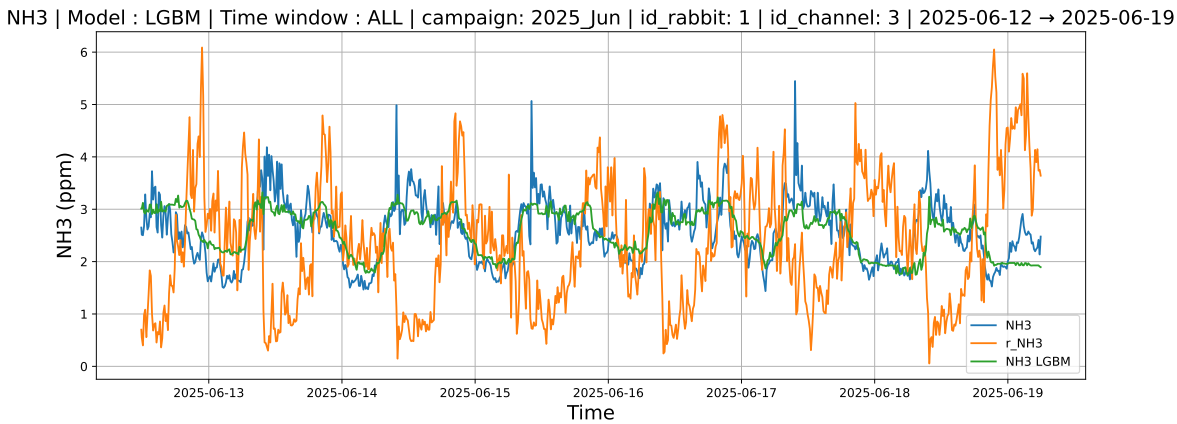


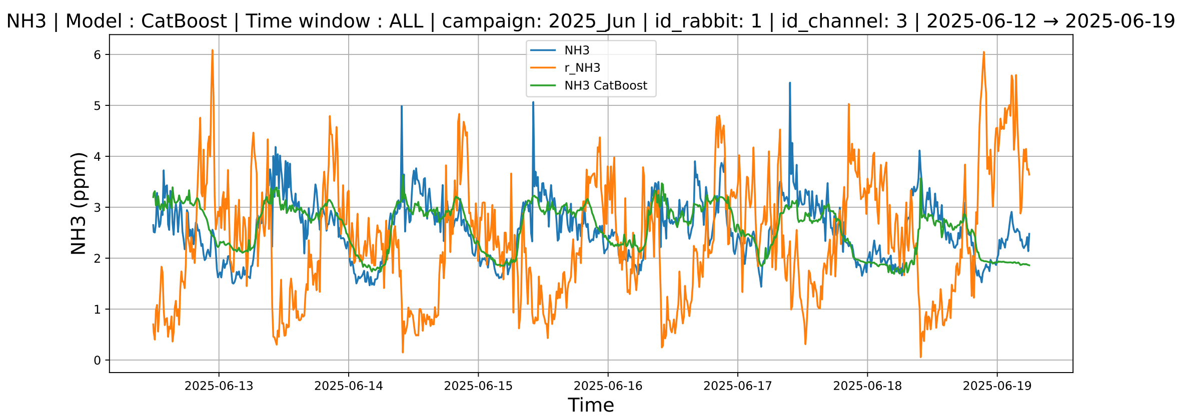




3.1.1.3 Time window : ALL



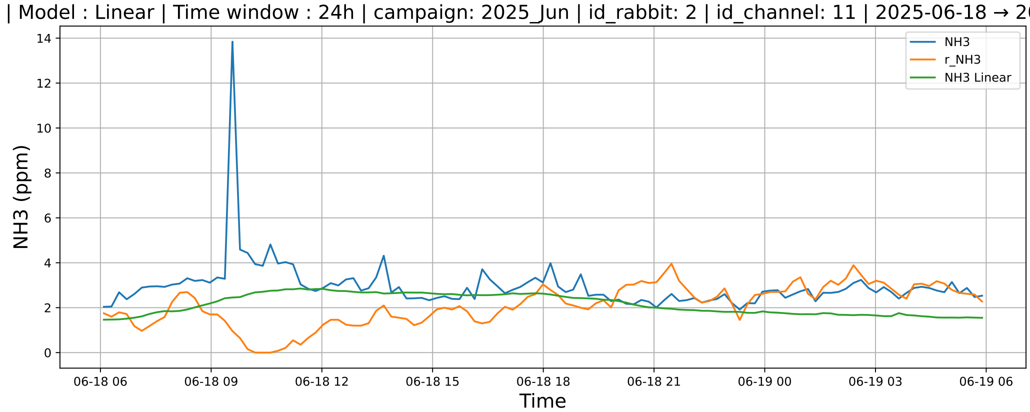




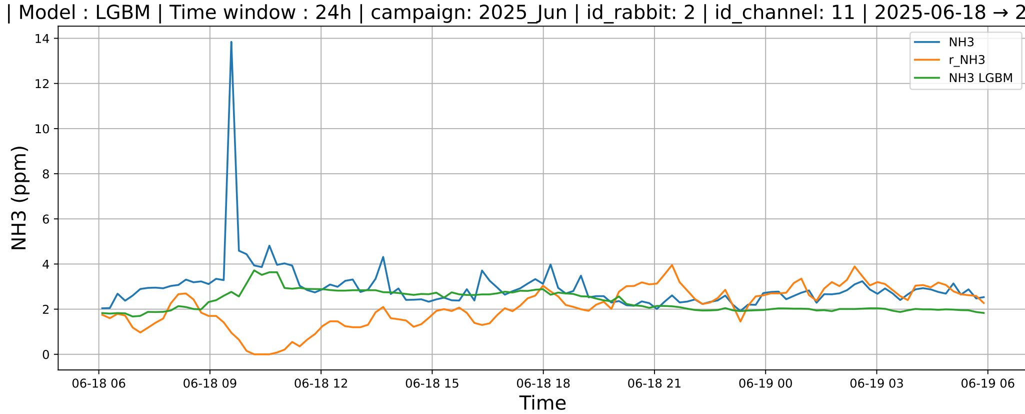
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

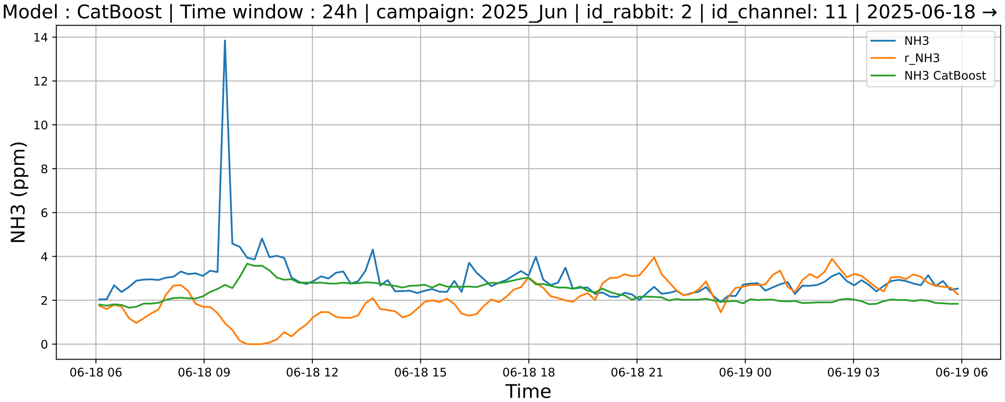
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

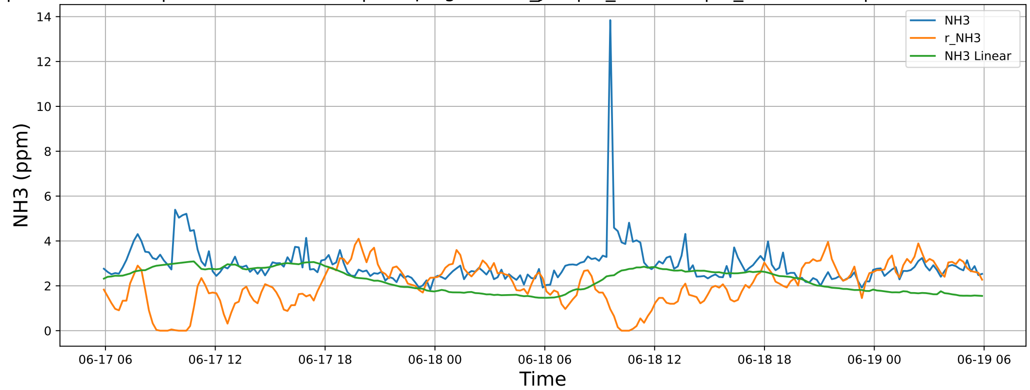


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

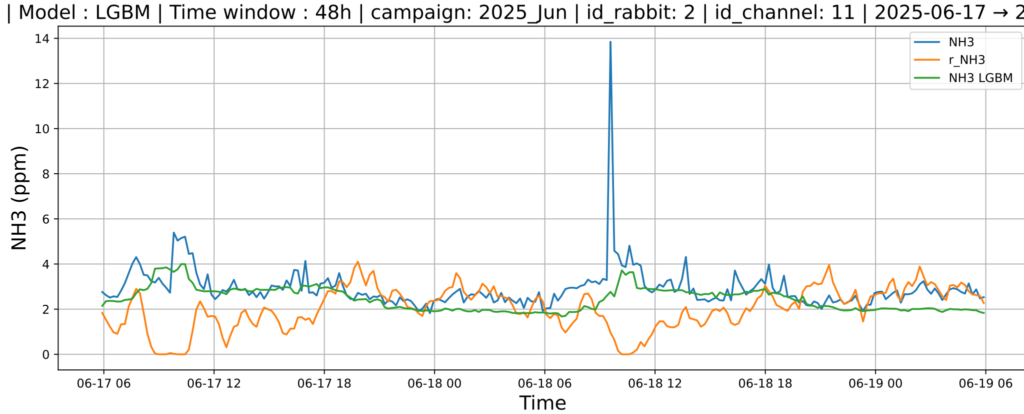


3.1.2.2 Time window : 48

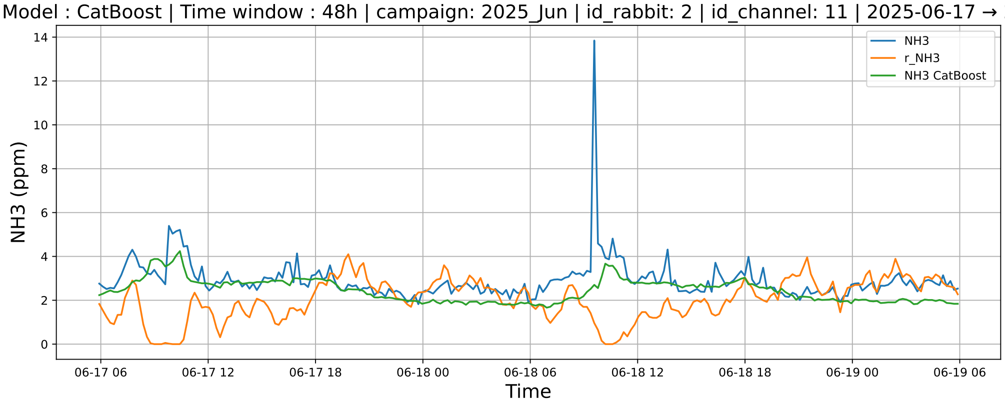
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

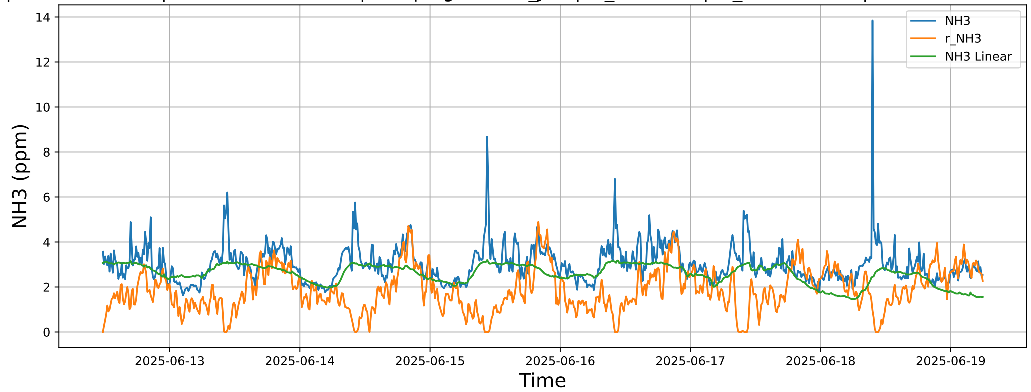


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

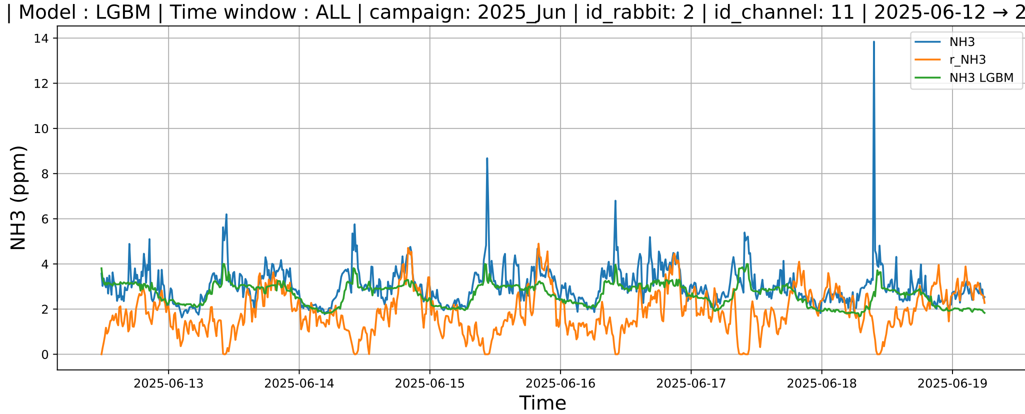


3.1.2.3 Time window : ALL

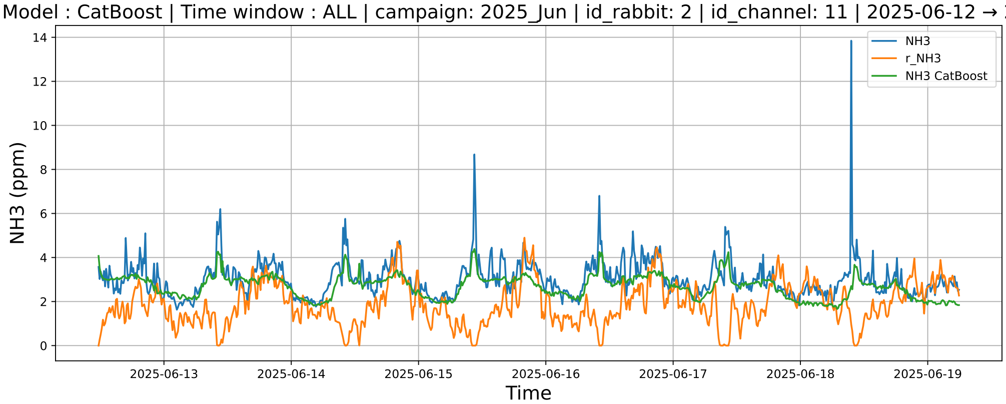
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19

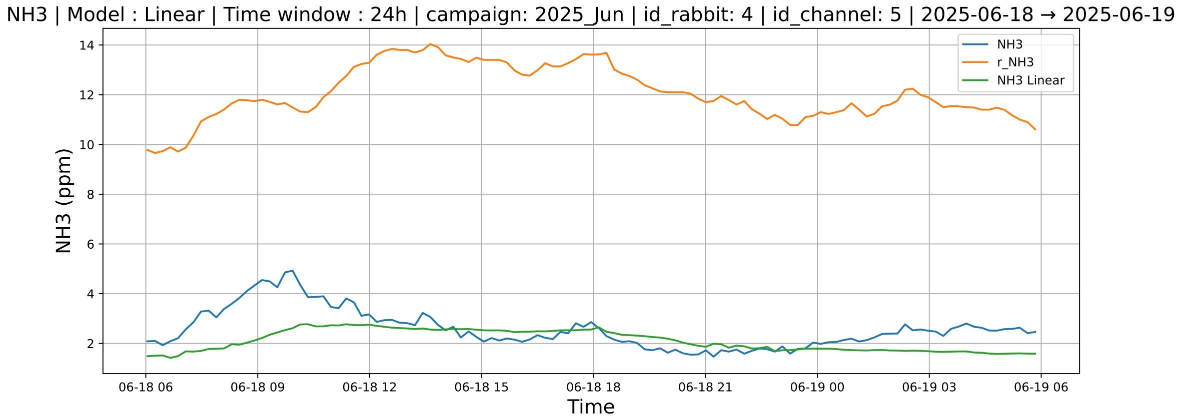


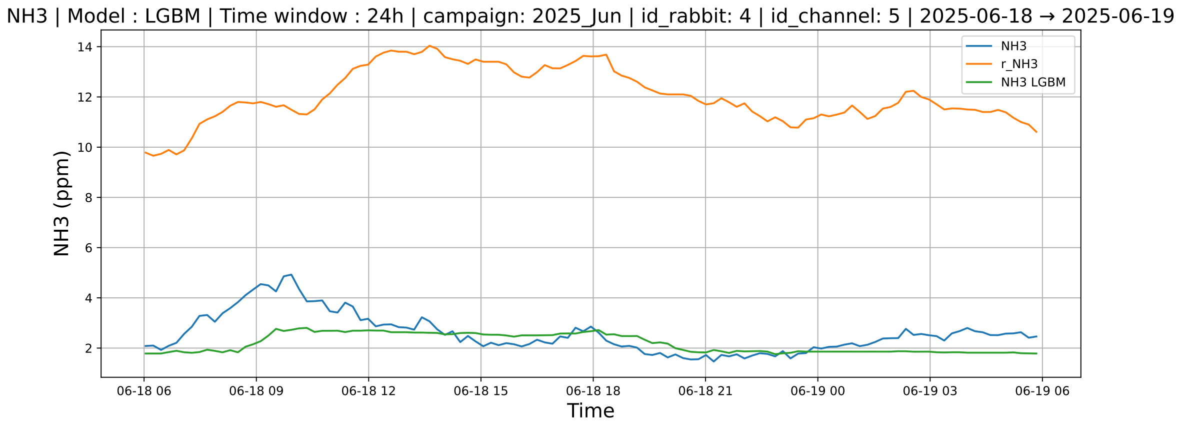
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19

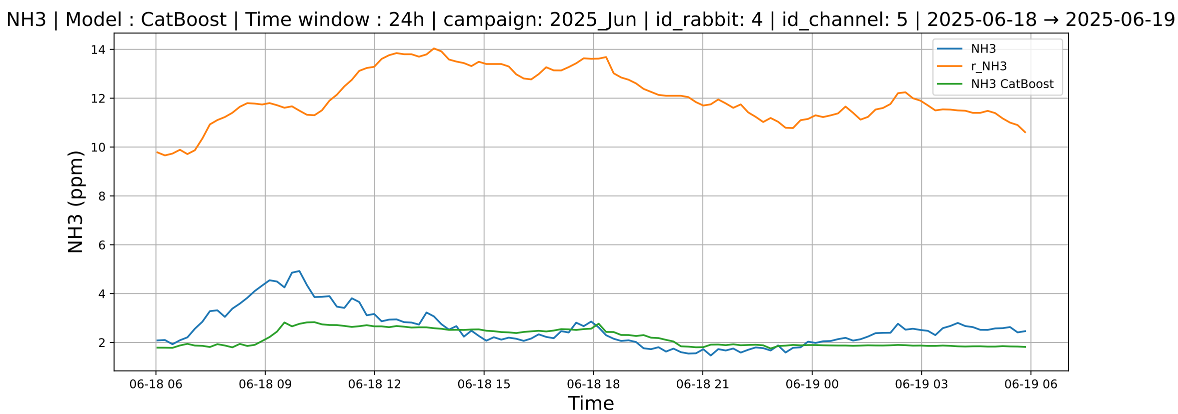


3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

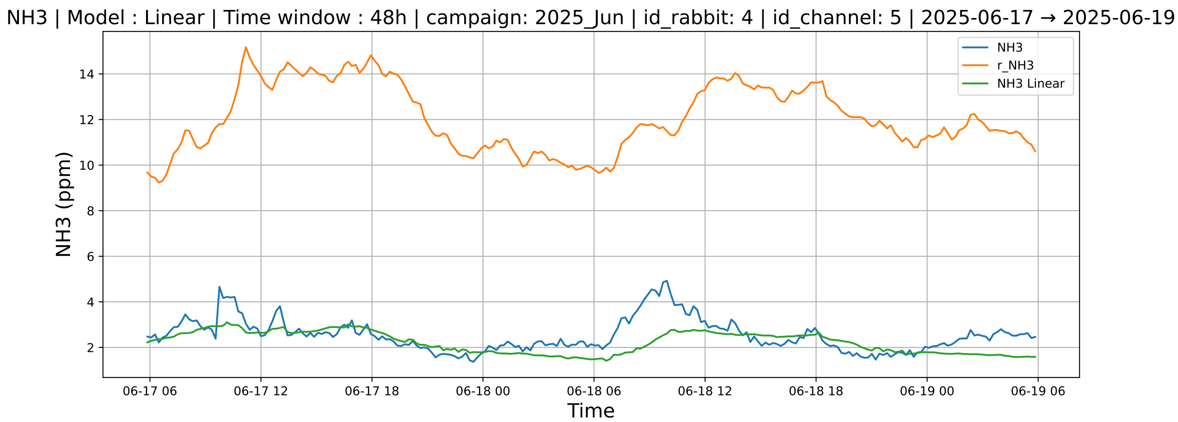
3.1.3.1 Time window : 24

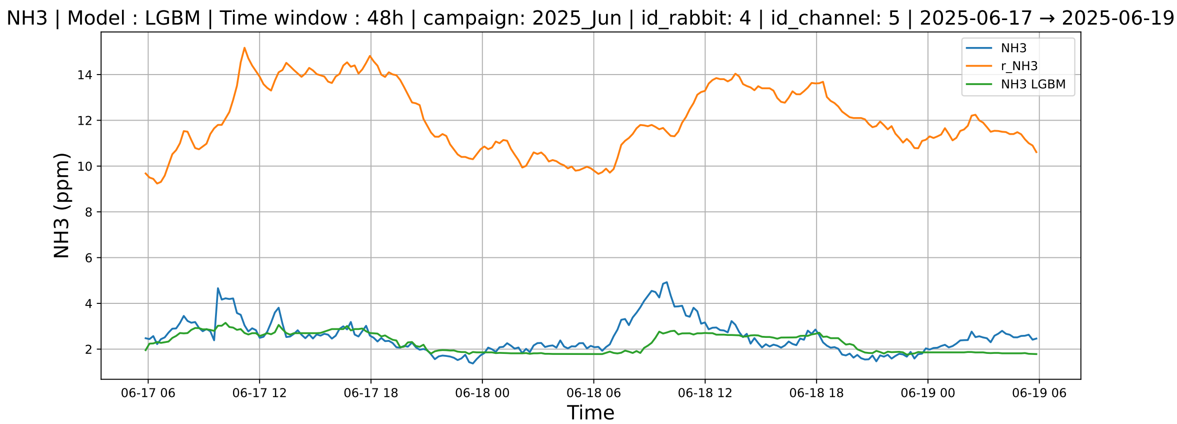


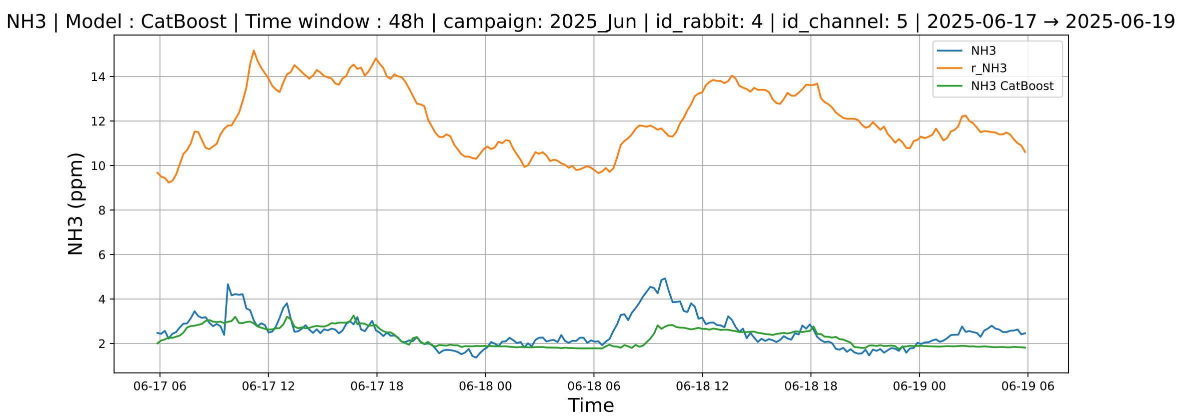




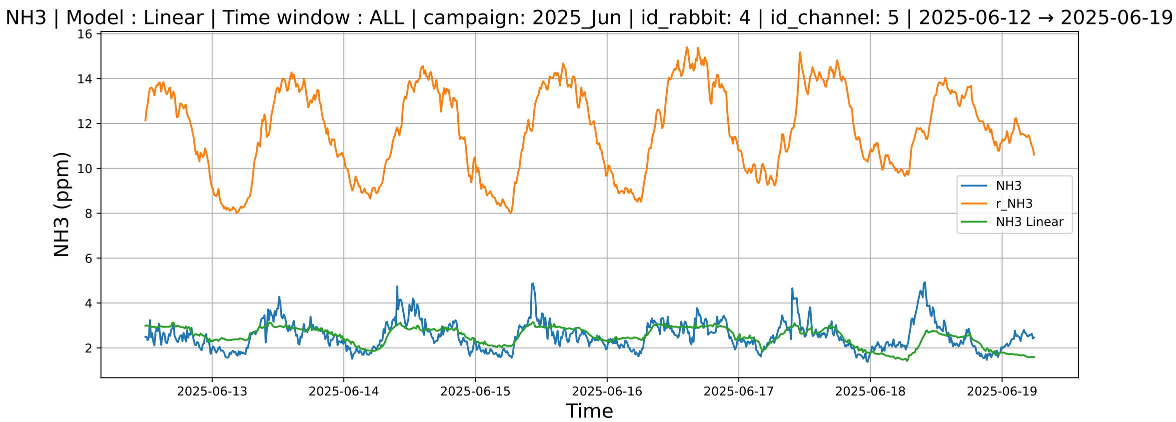
3.1.3.2 Time window : 48

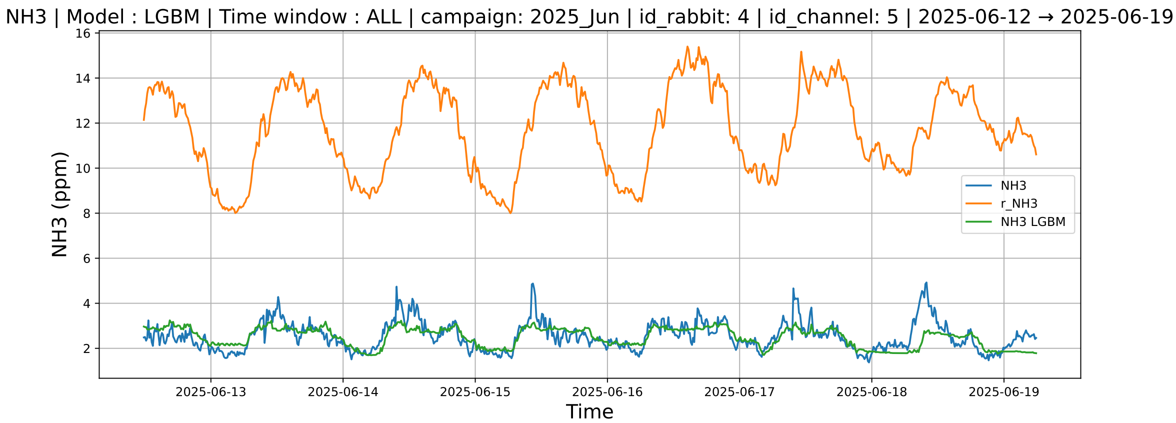


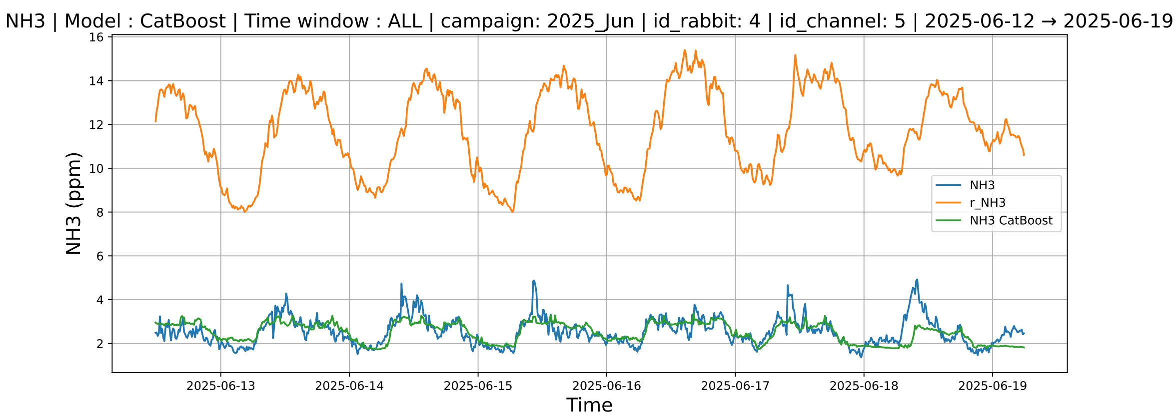




3.1.3.3 Time window : ALL

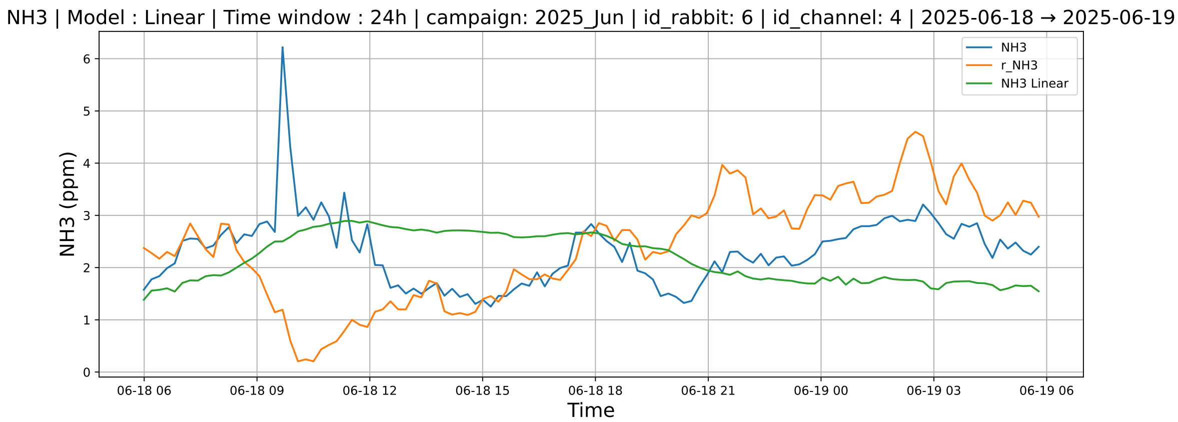


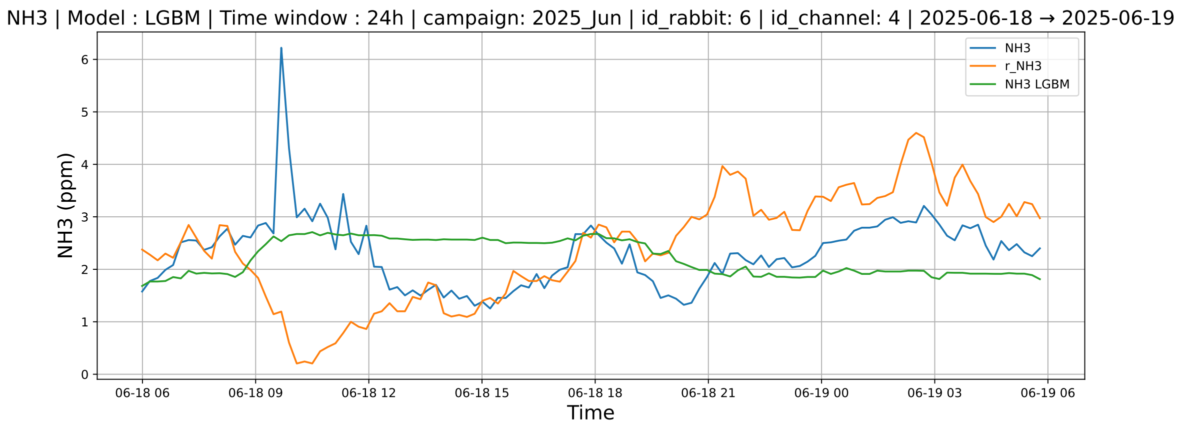


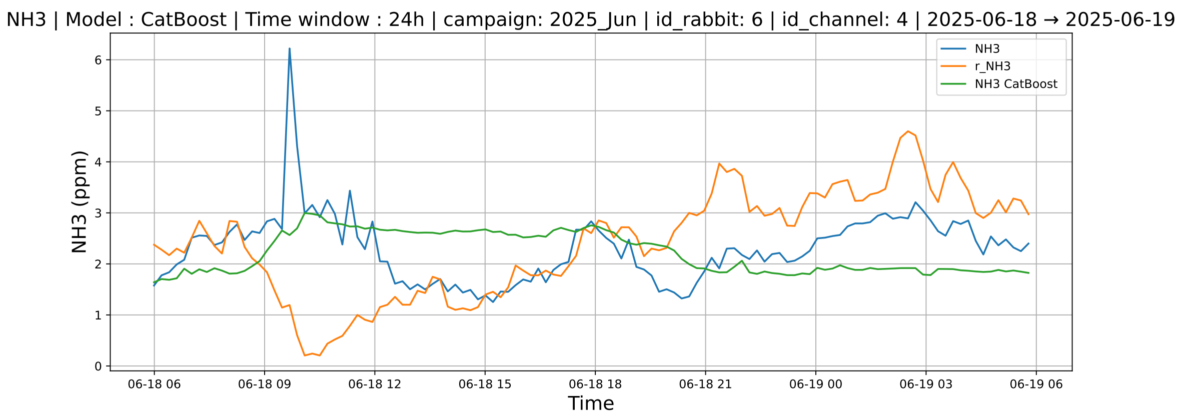


3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

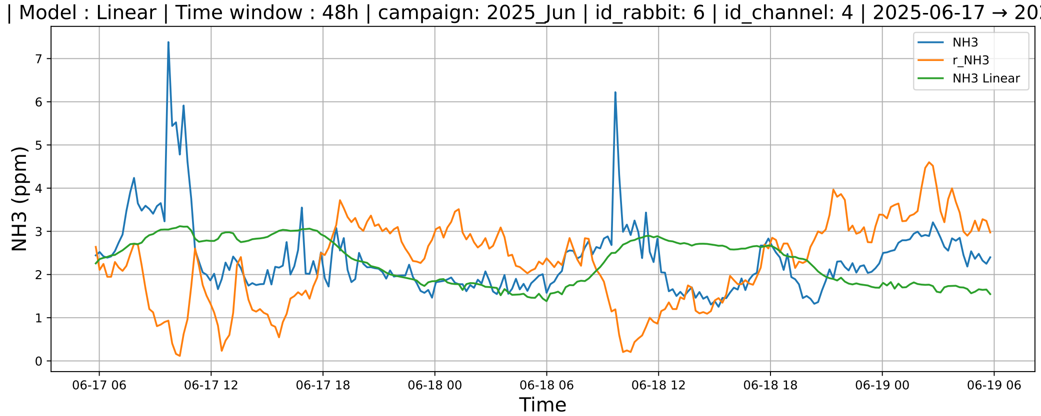




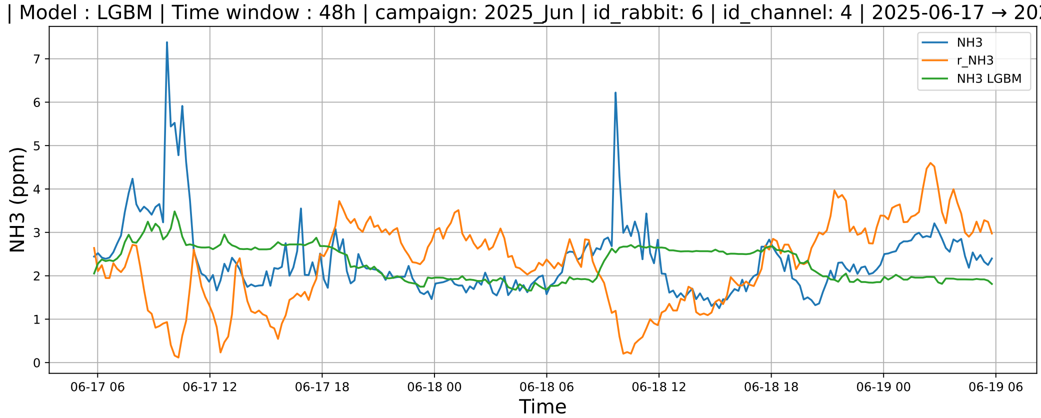


3.1.4.2 Time window : 48

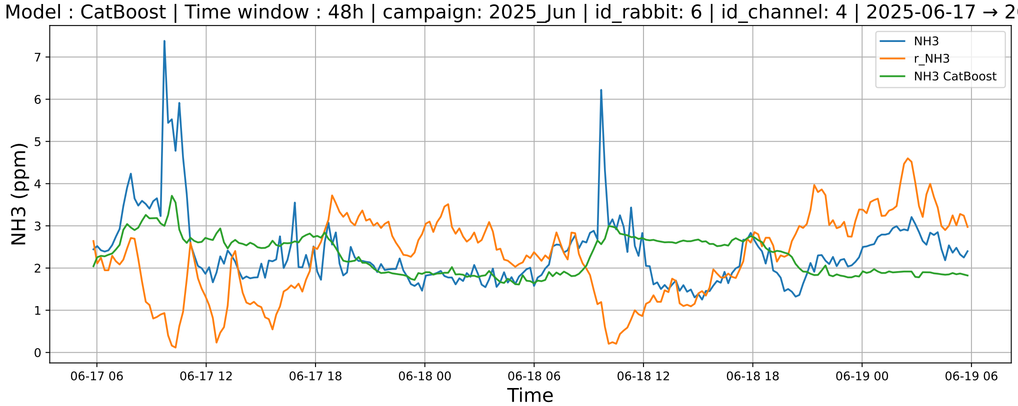
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



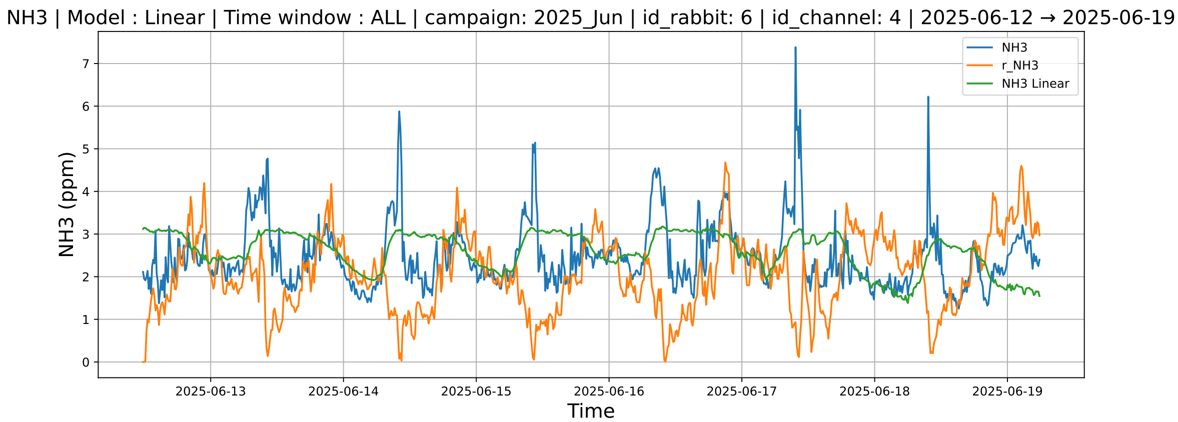
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

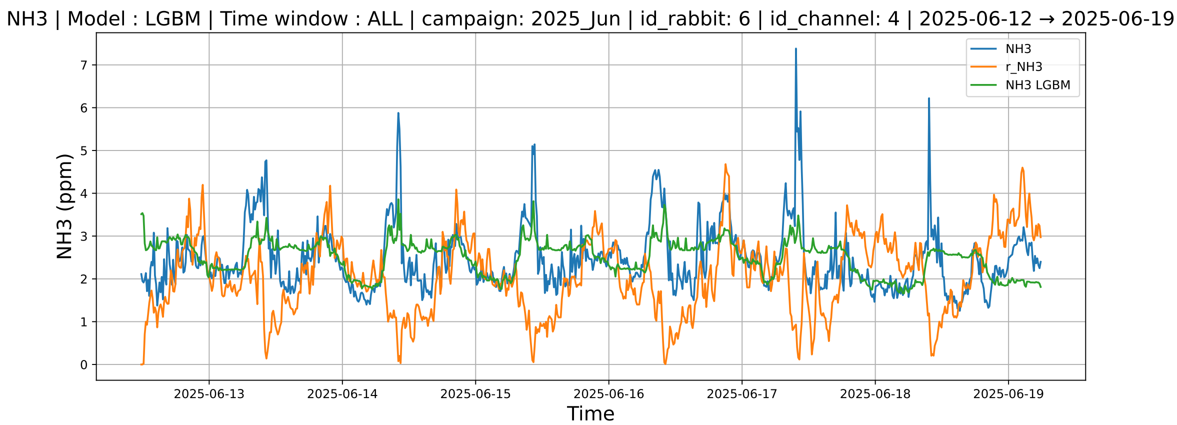


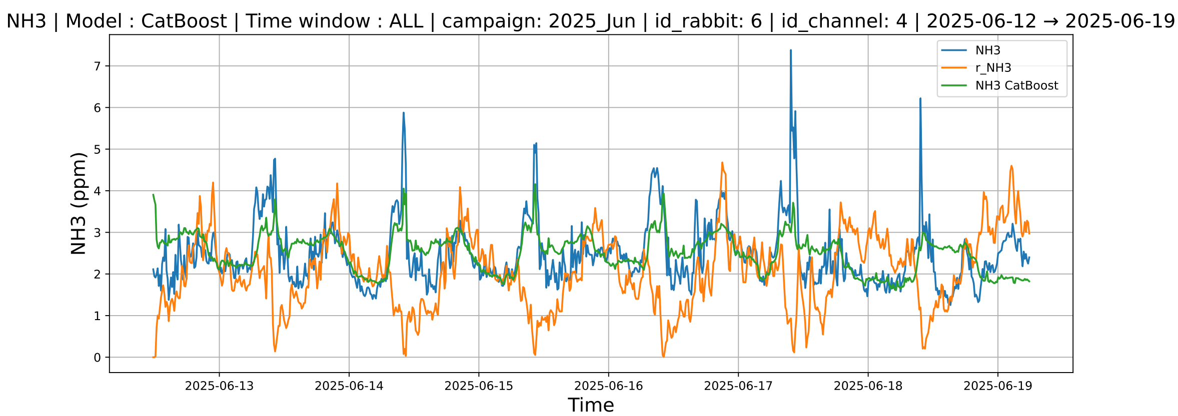
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.1.4.3 Time window : ALL



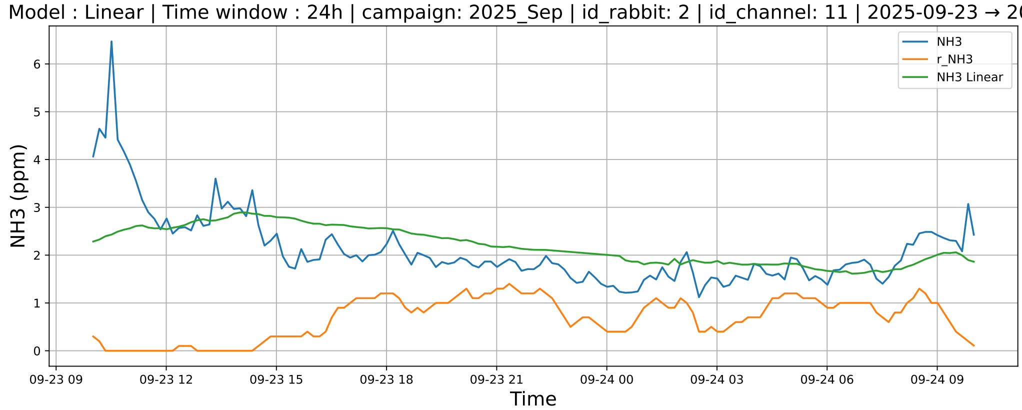




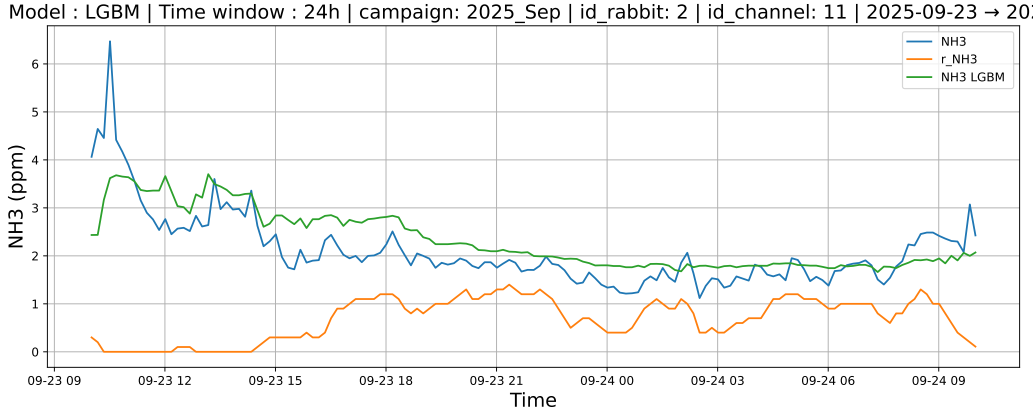
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

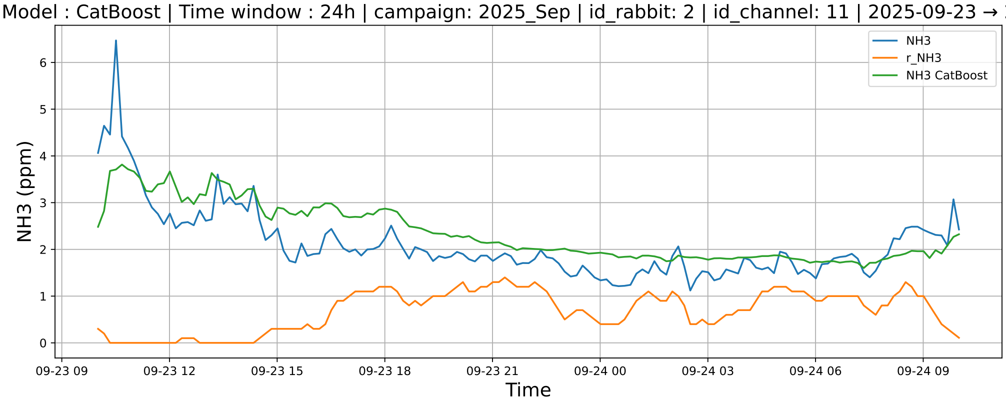
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

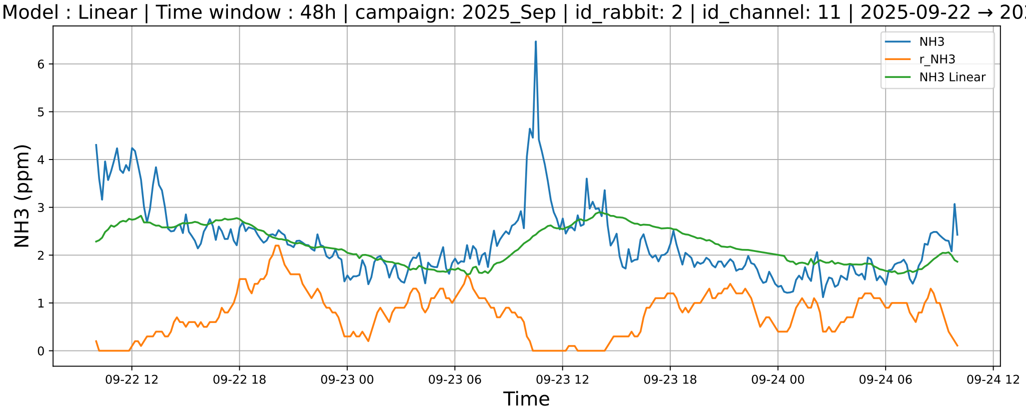


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

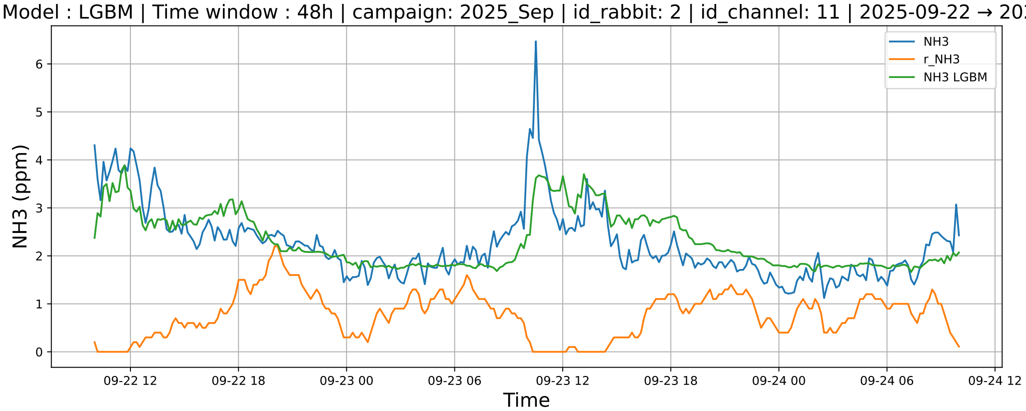


3.1.5.2 Time window : 48

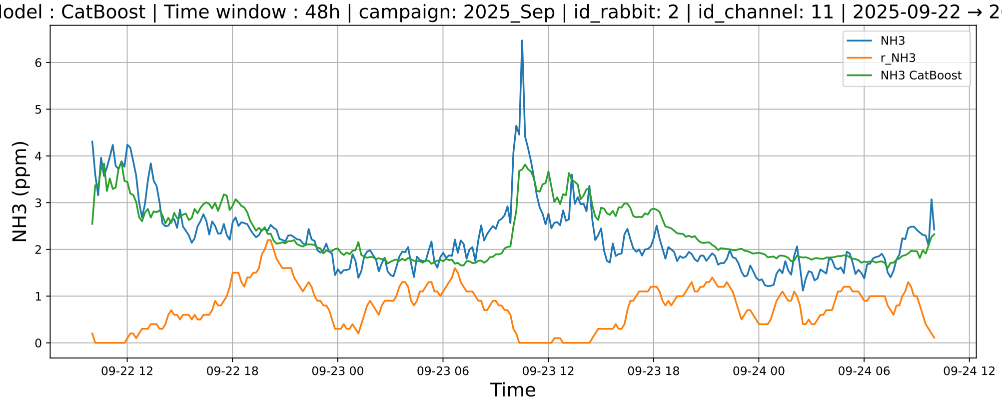
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



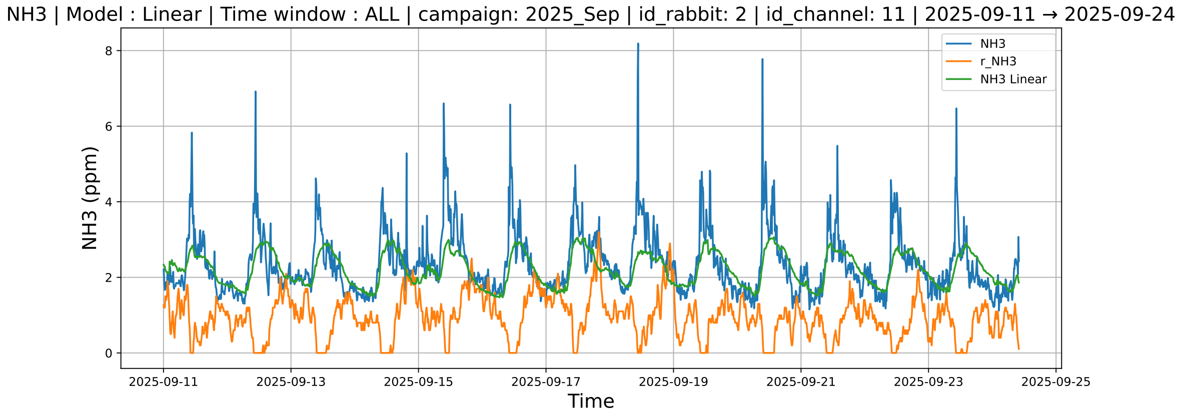
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

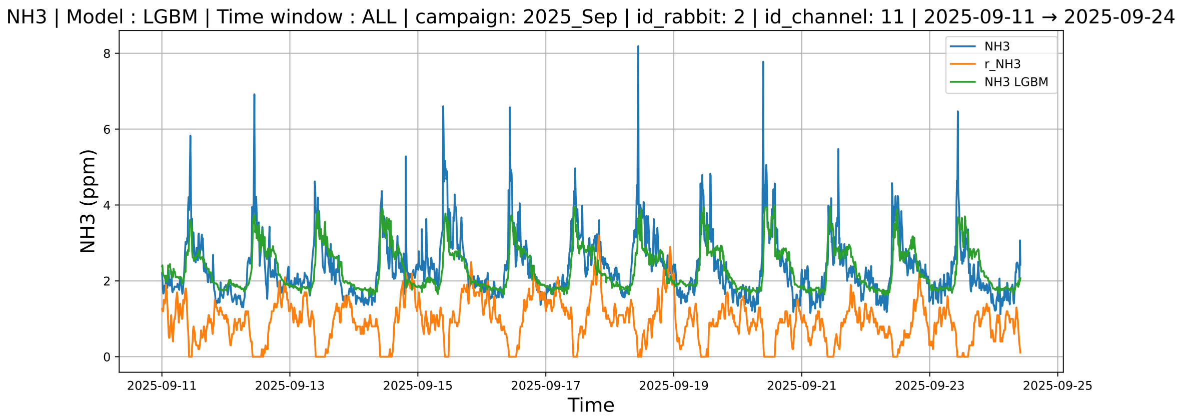


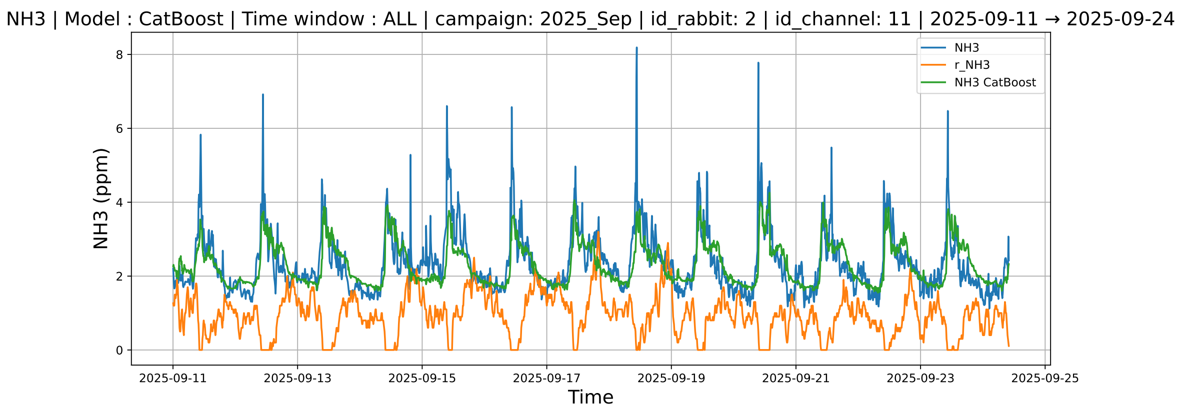
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.1.5.3 Time window : ALL

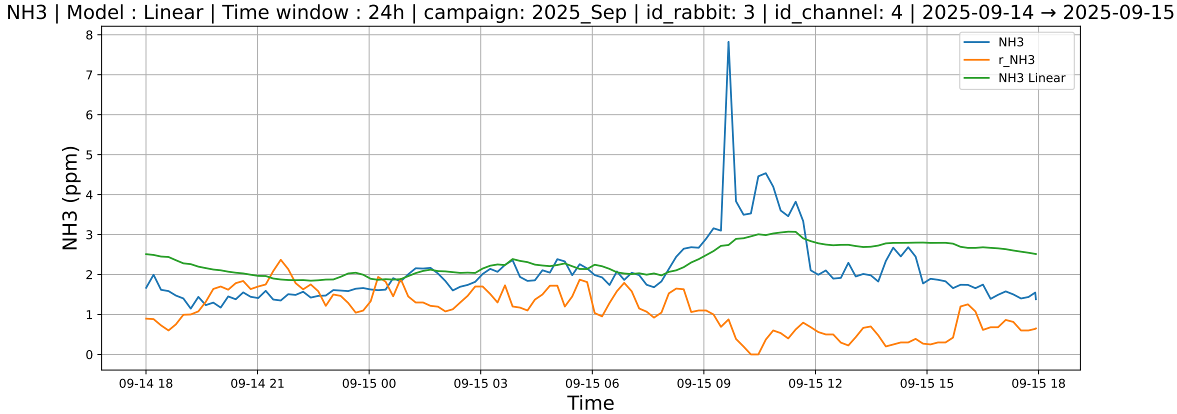


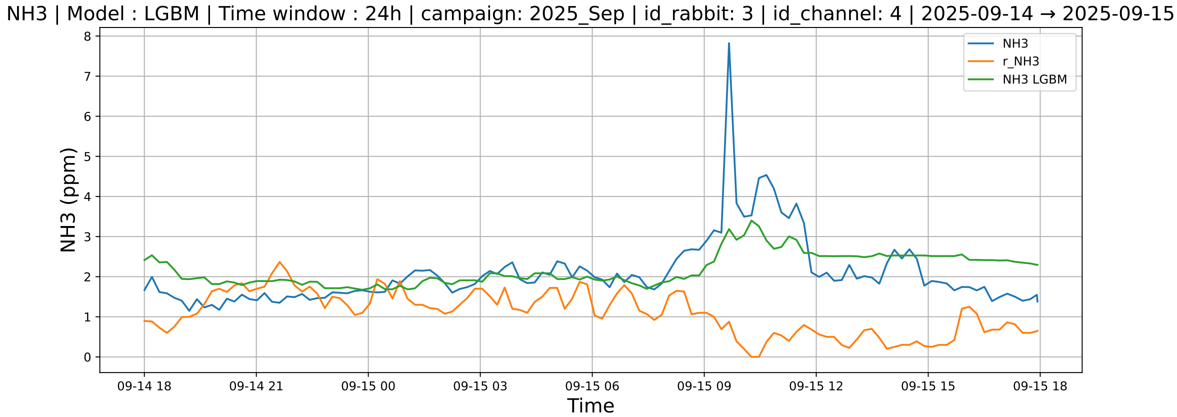


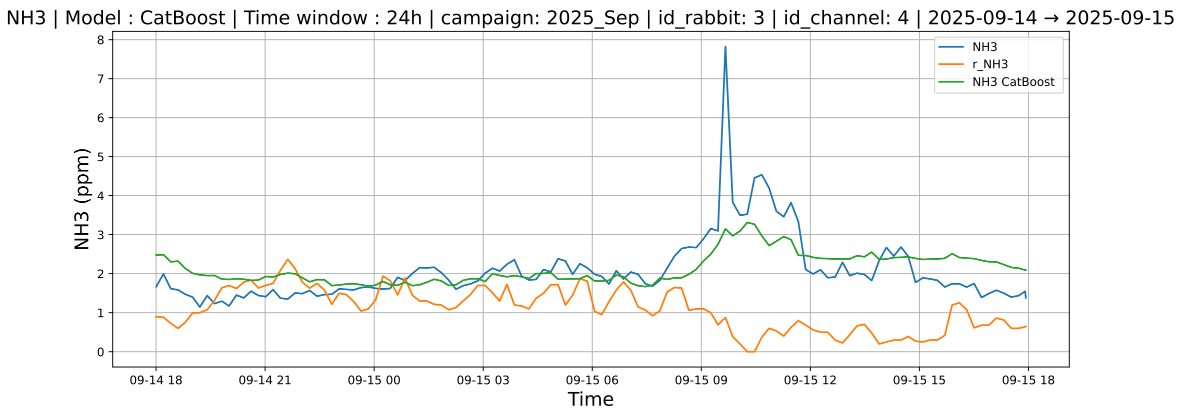


3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

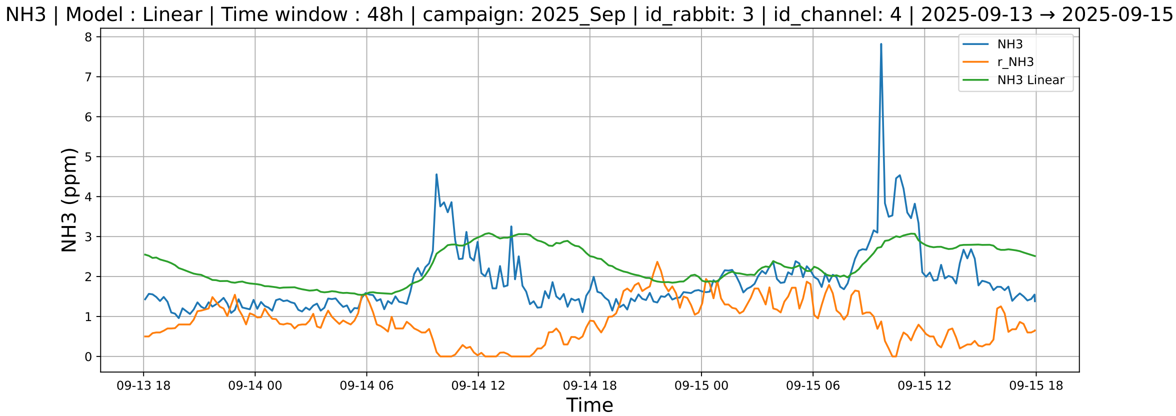
3.1.6.1 Time window : 24

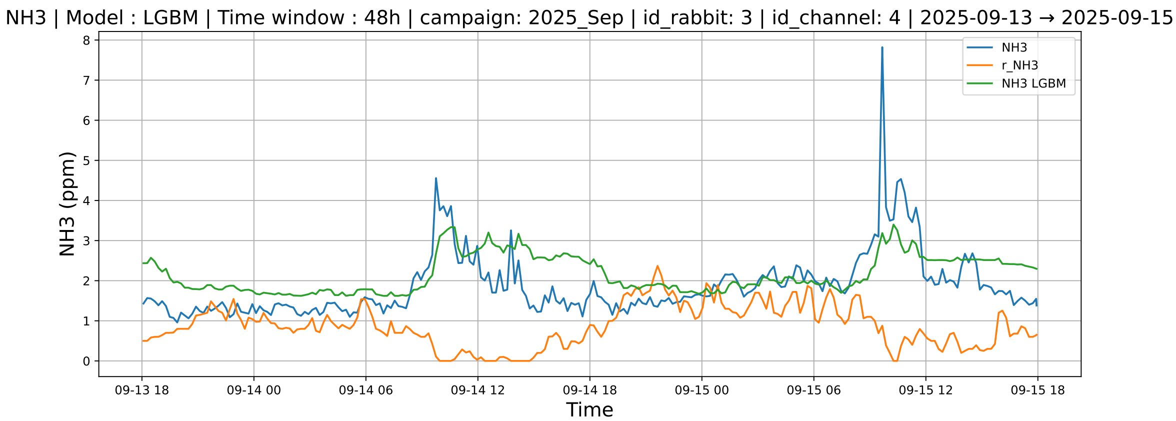


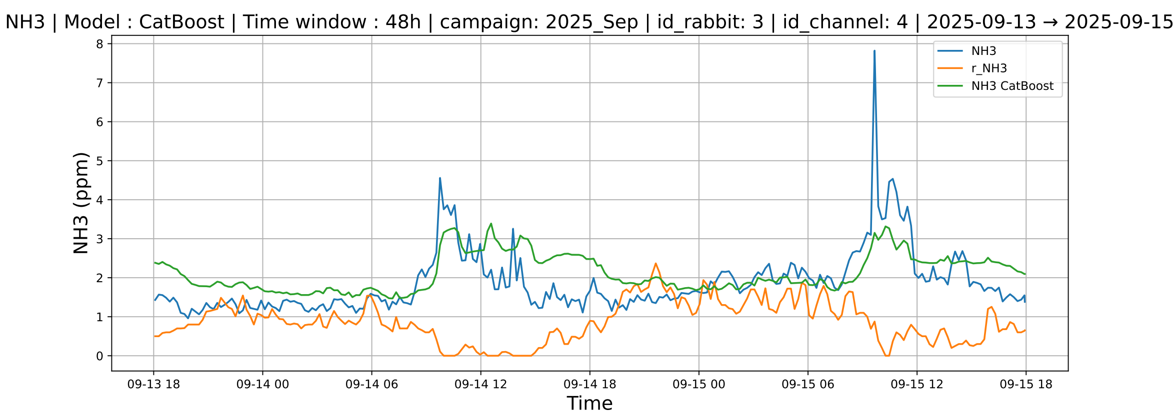




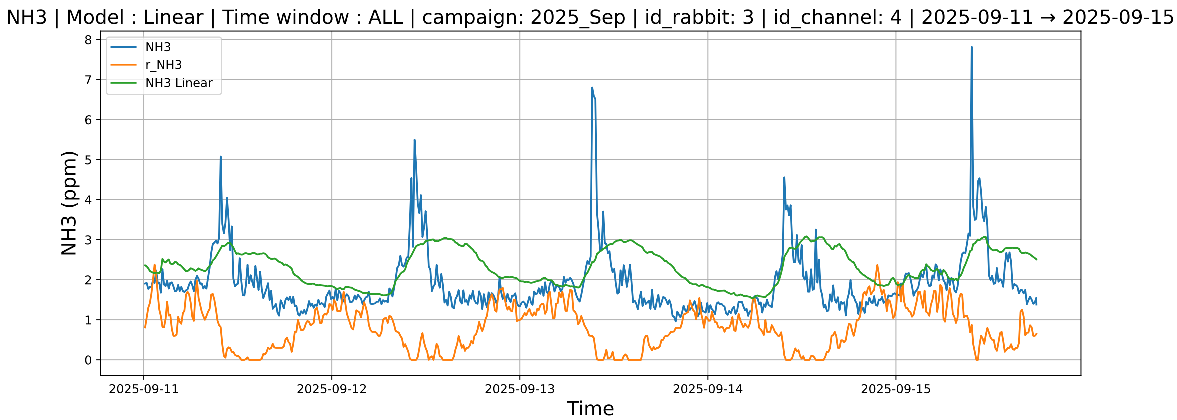
3.1.6.2 Time window : 48

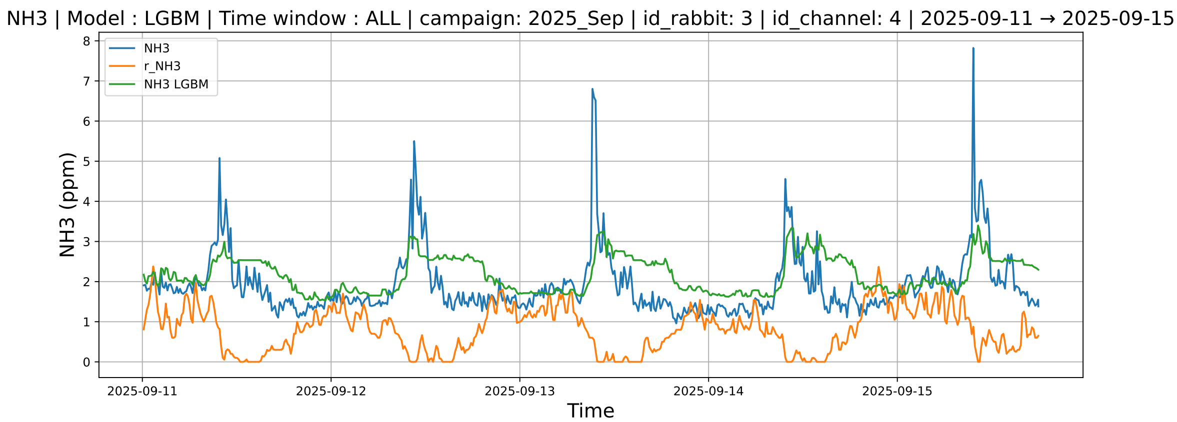


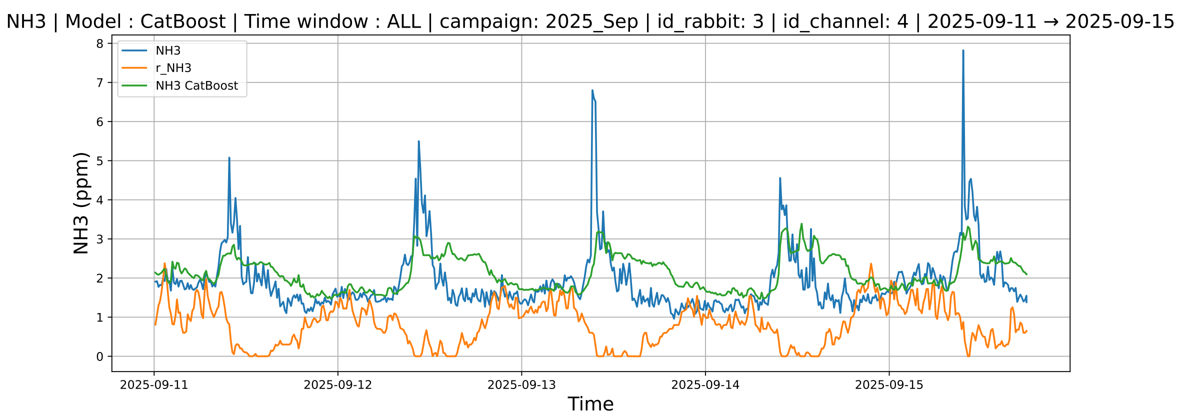




3.1.6.3 Time window : ALL



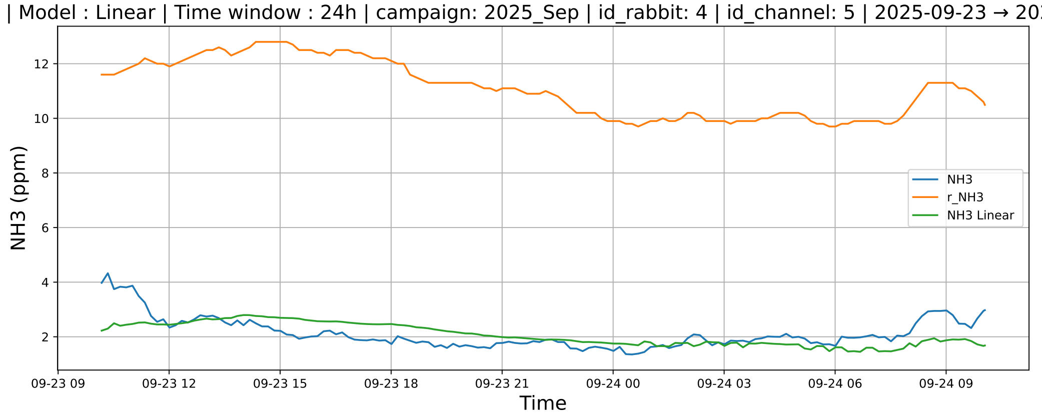




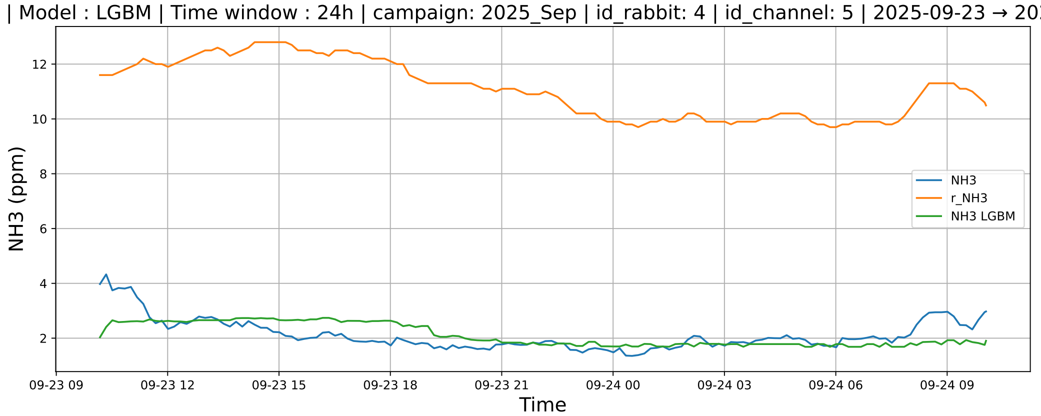
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

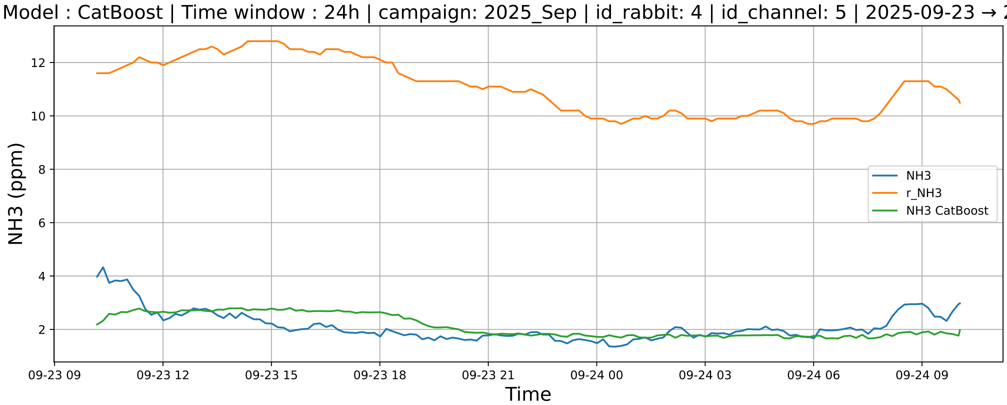
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



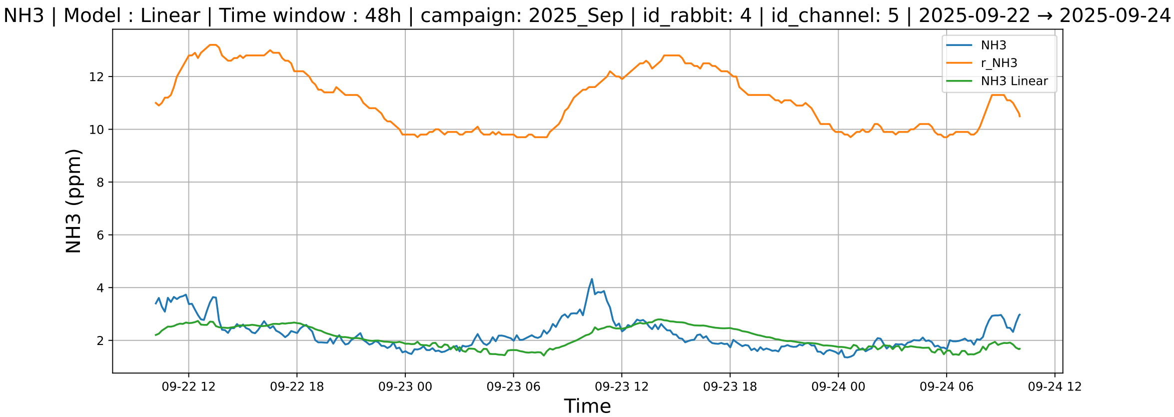
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

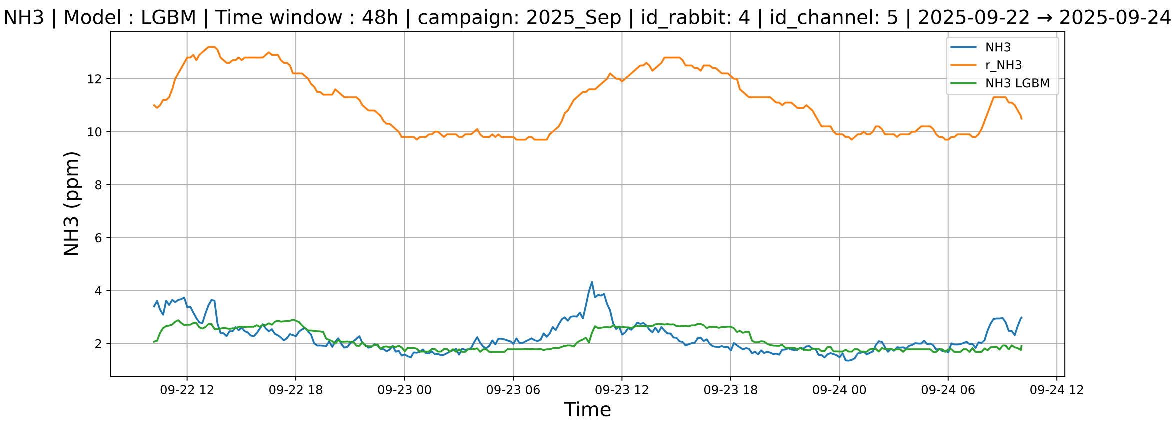


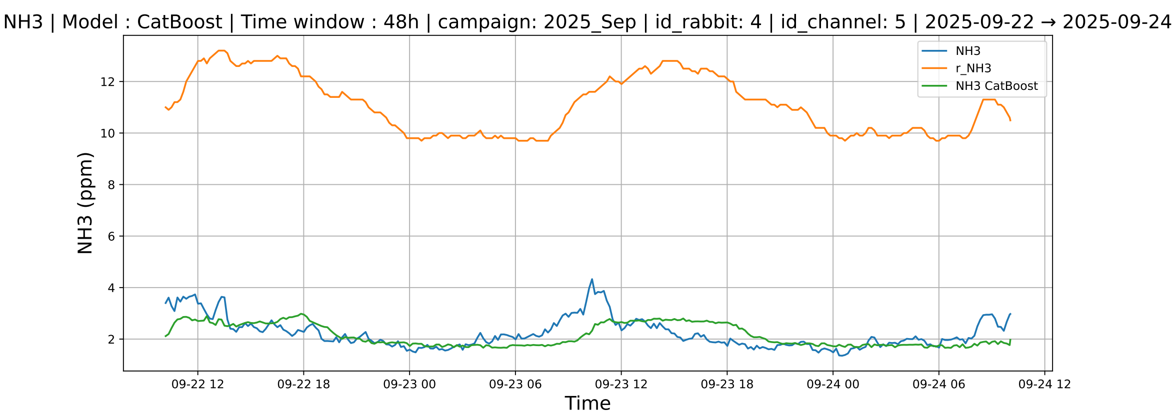
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



3.1.7.2 Time window : 48

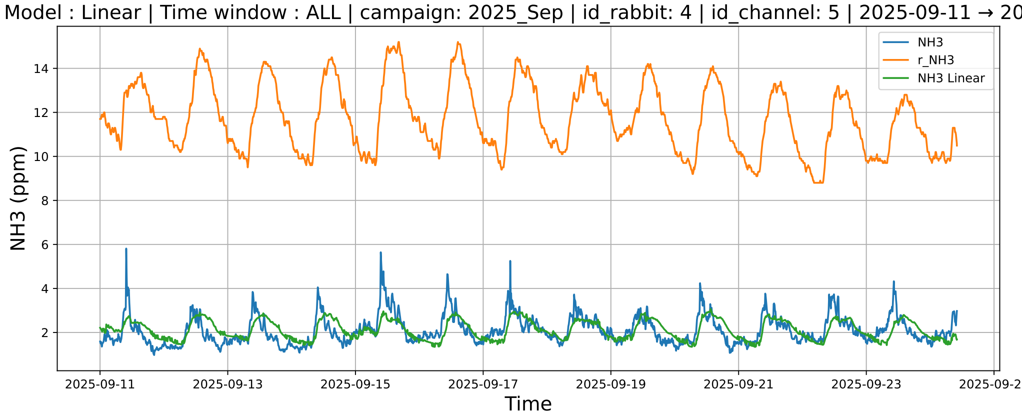




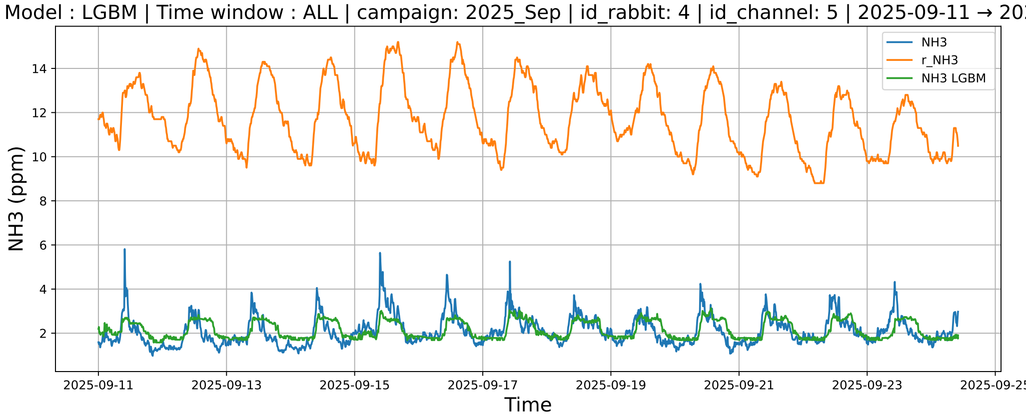


3.1.7.3 Time window : ALL

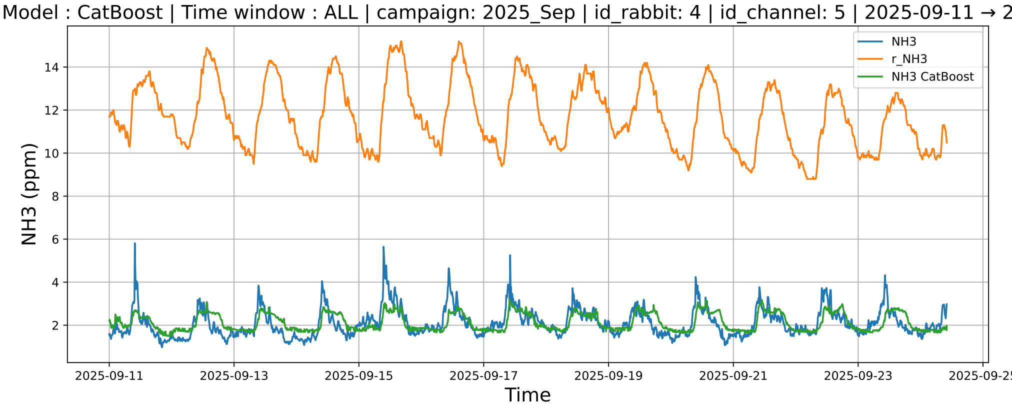
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24

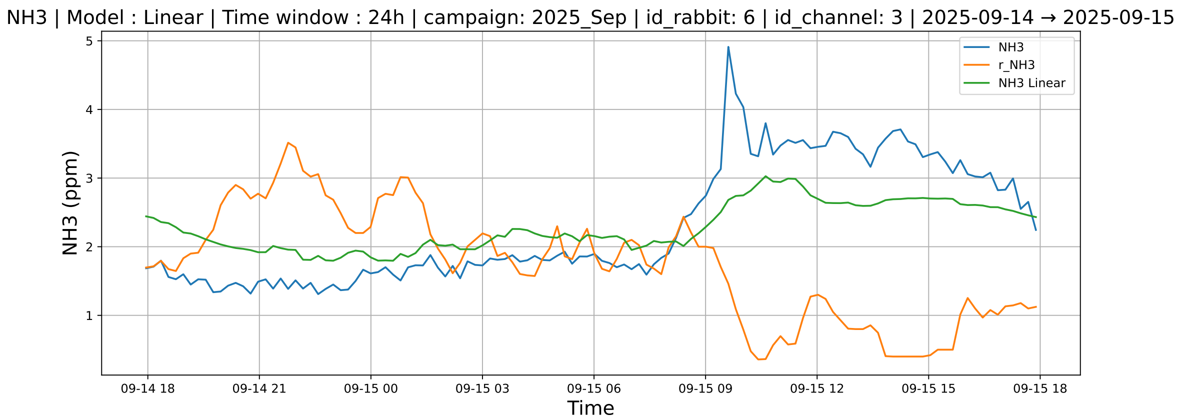


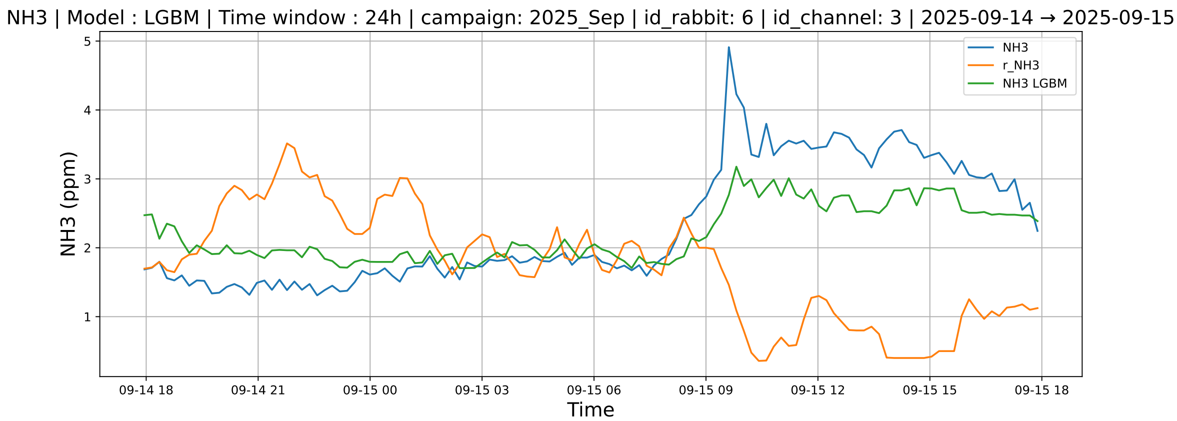
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24

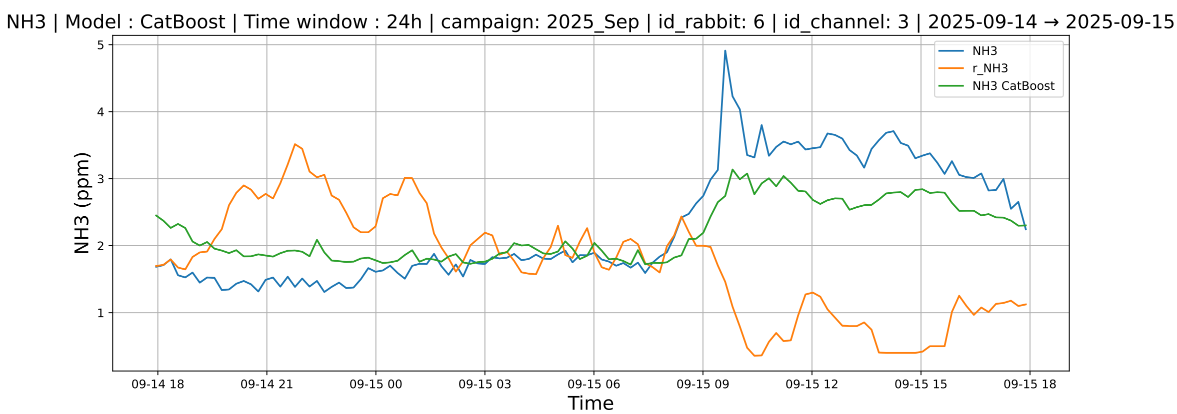


3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

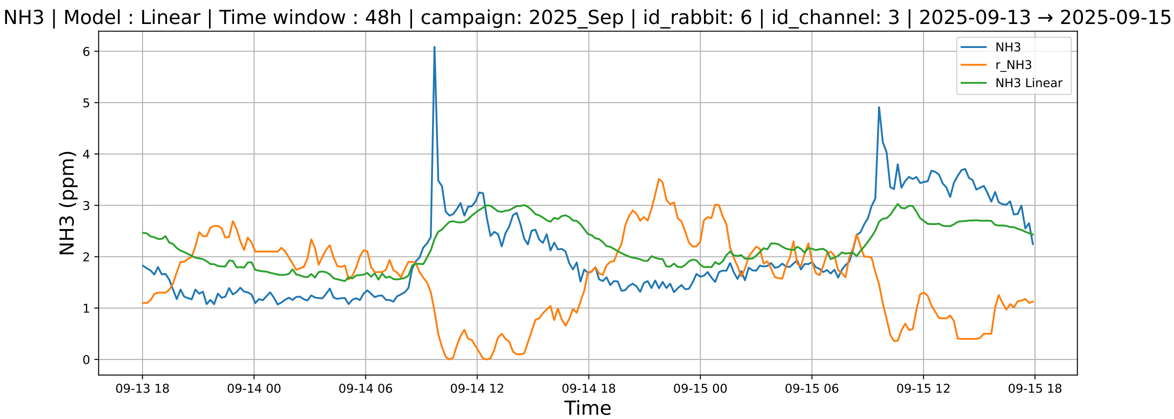
3.1.8.1 Time window : 24

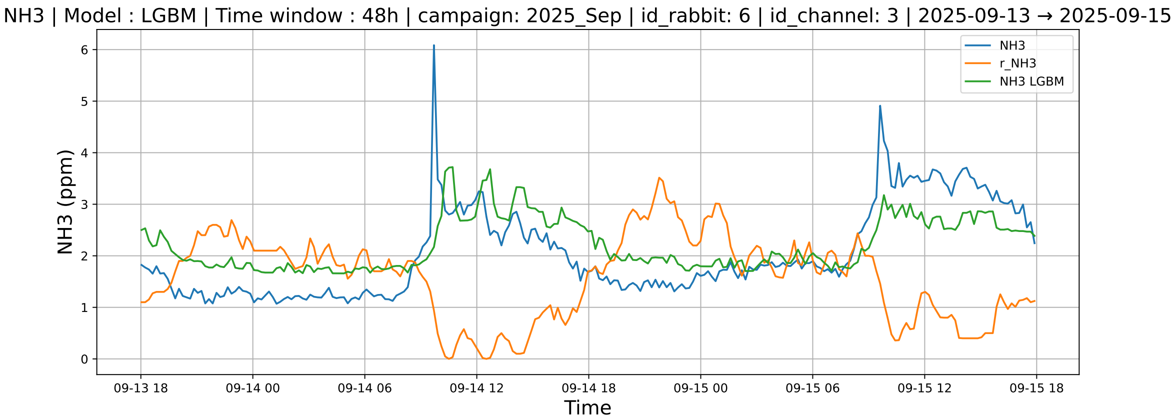


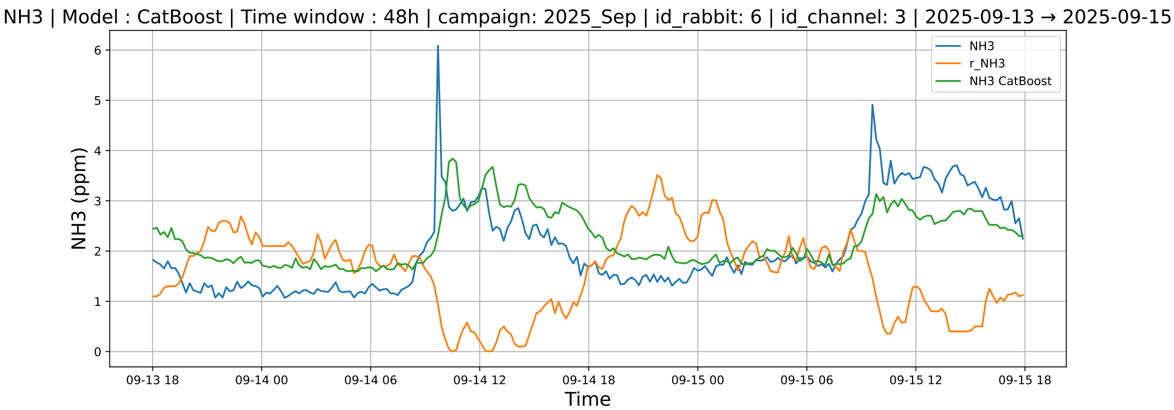




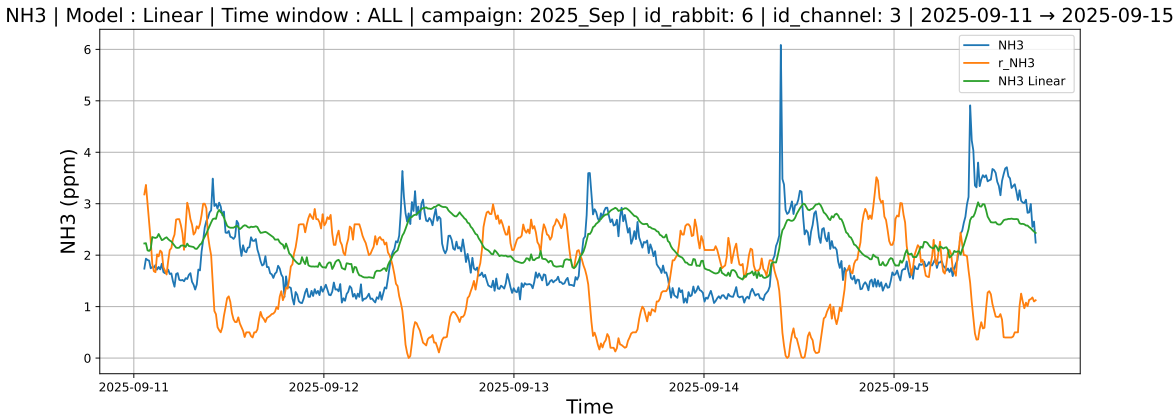
3.1.8.2 Time window : 48

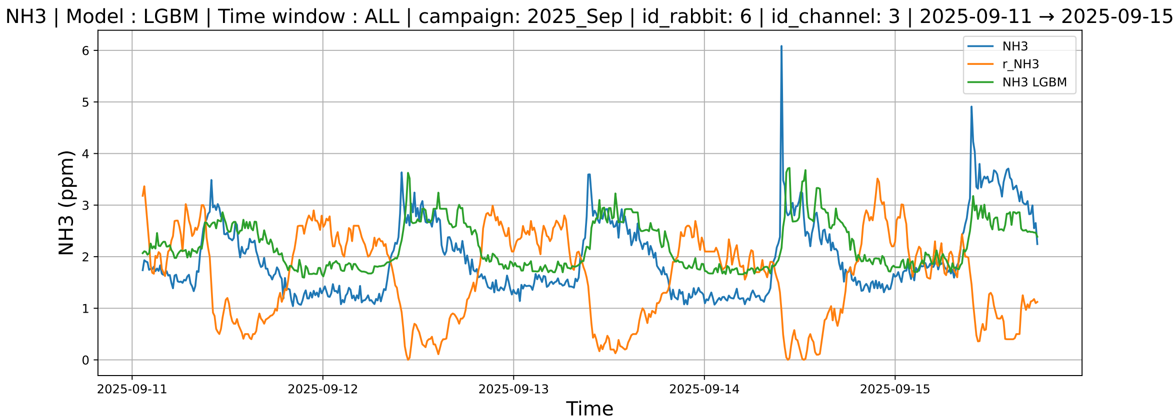


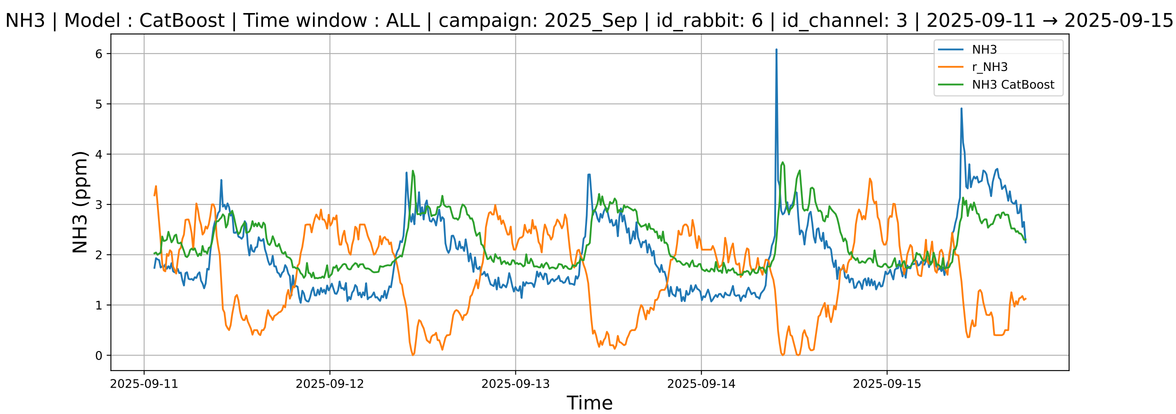




3.1.8.3 Time window : ALL





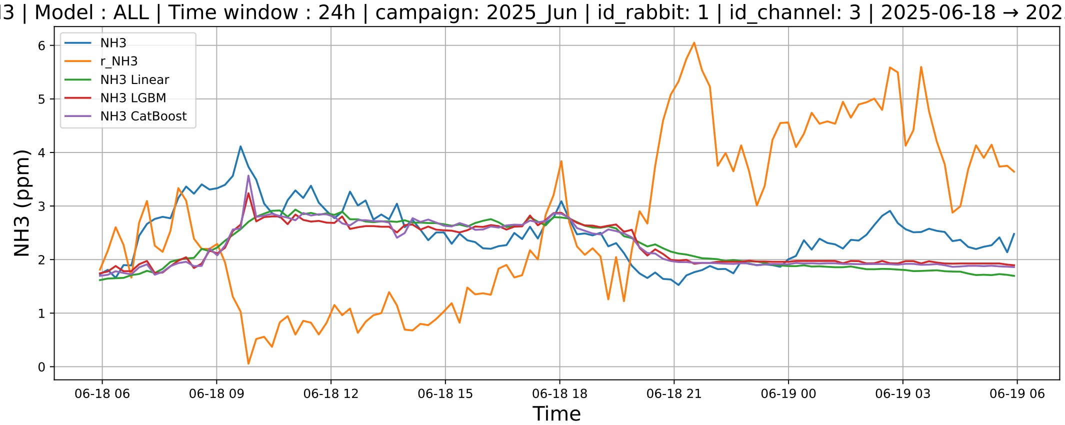


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

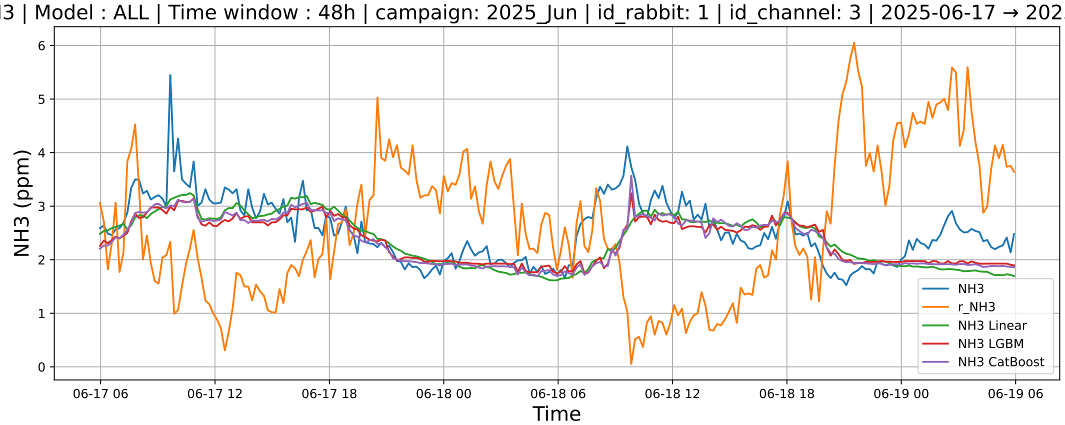
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



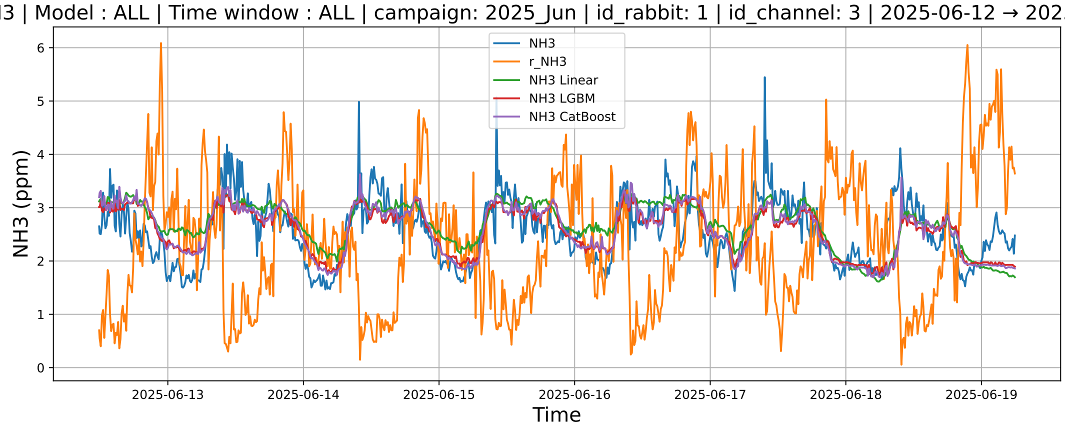
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



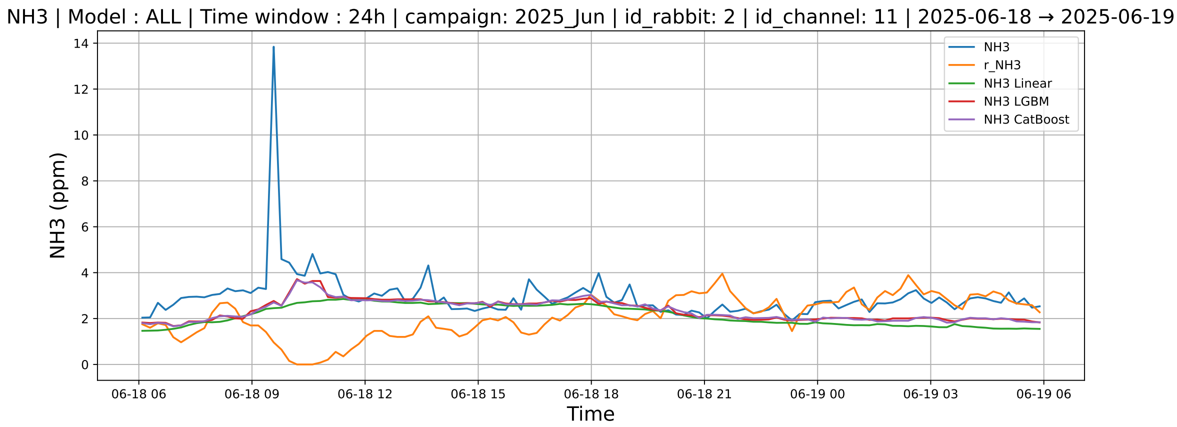
3.2.1.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19

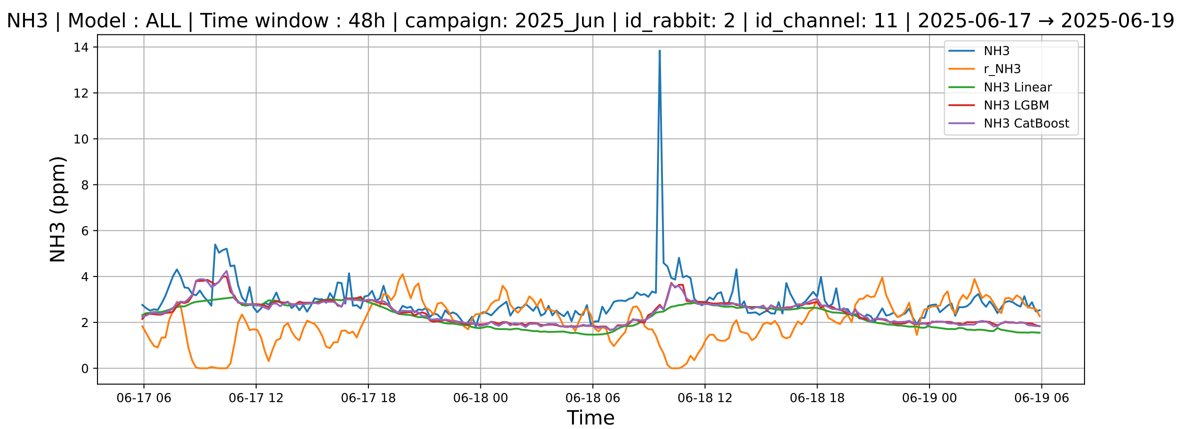


3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.2.2.1 Time window : 24

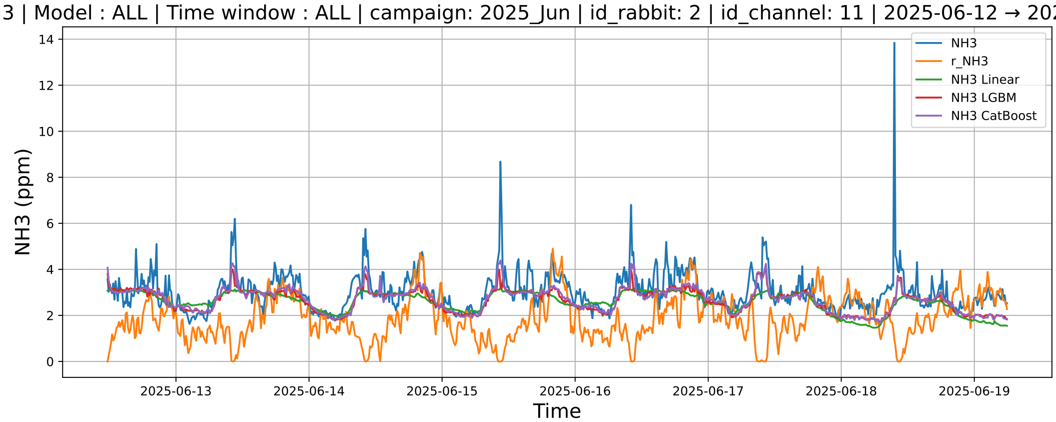


3.2.2.2 Time window : 48



3.2.2.3 Time window : ALL

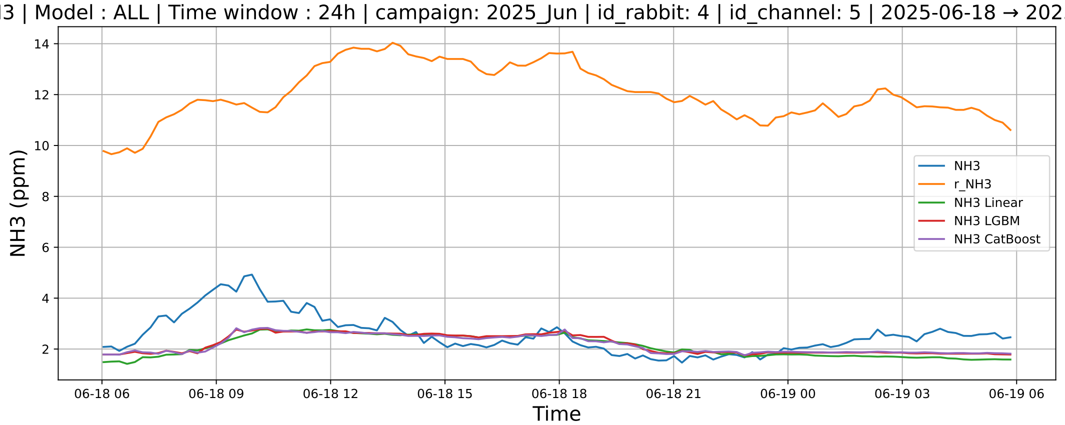
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



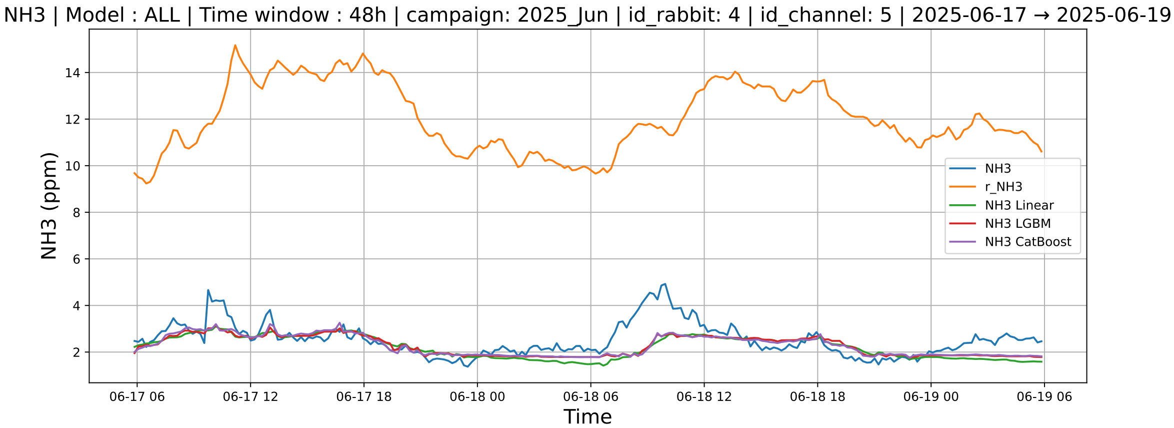
3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.2.3.1 Time window : 24

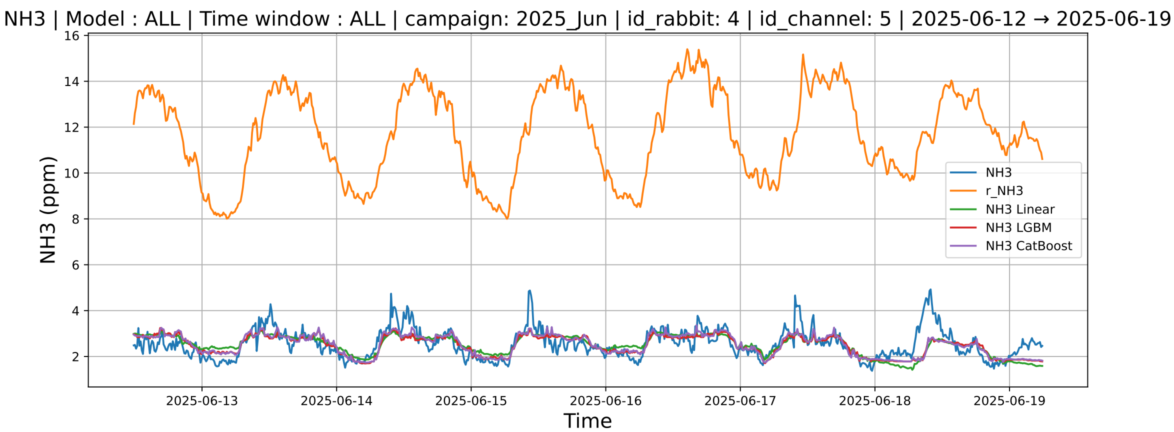
NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



3.2.3.2 Time window : 48

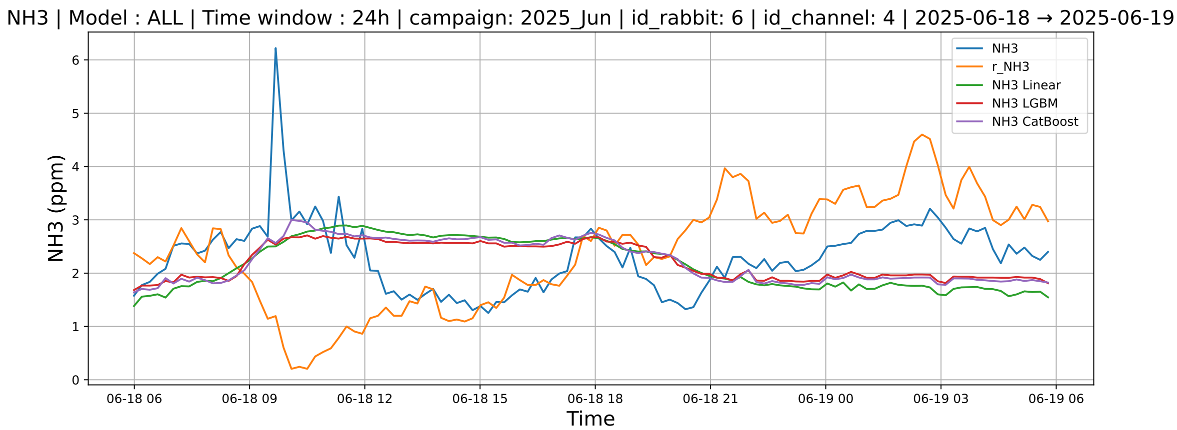


3.2.3.3 Time window : ALL

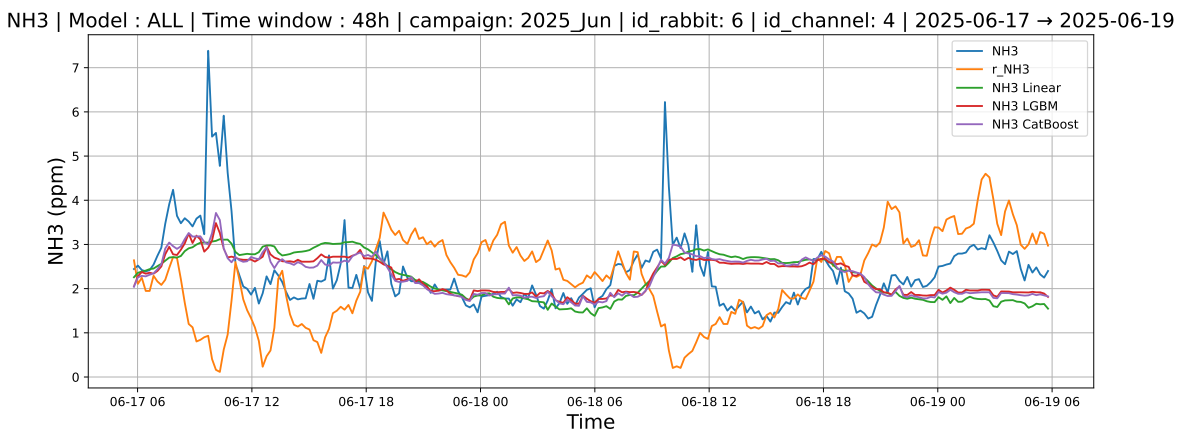


3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

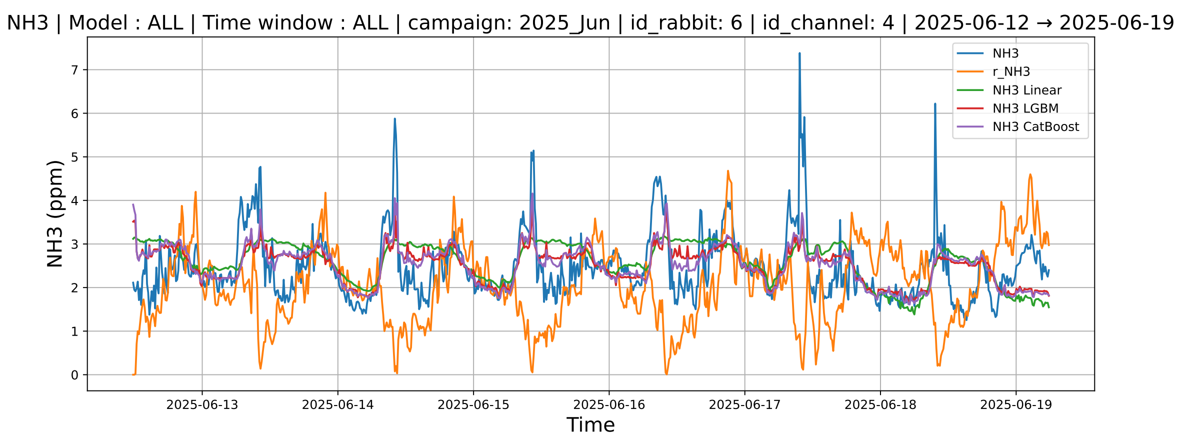
3.2.4.1 Time window : 24



3.2.4.2 Time window : 48

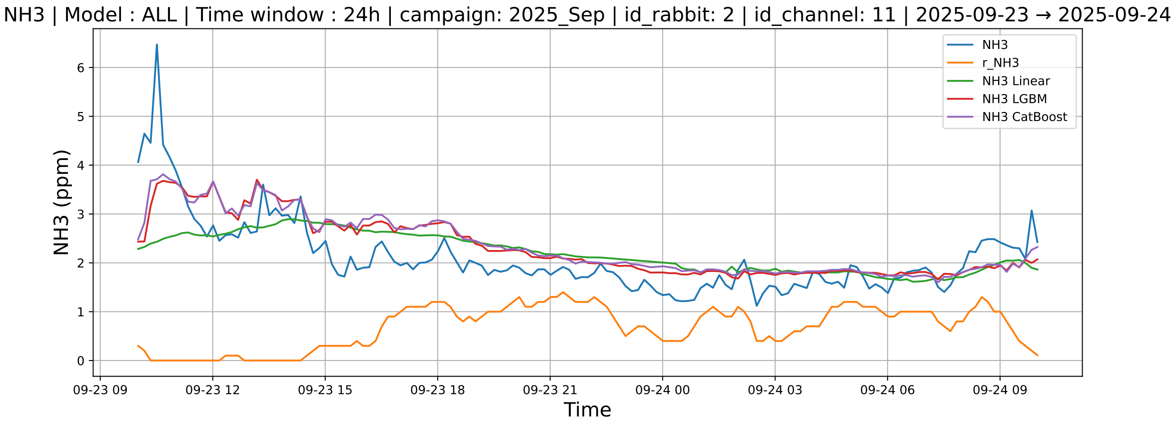


3.2.4.3 Time window : ALL

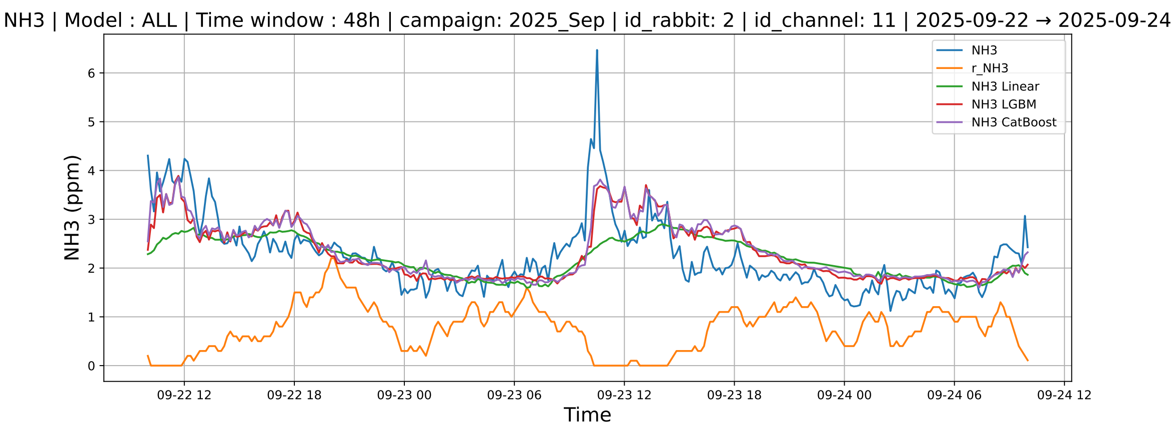


3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

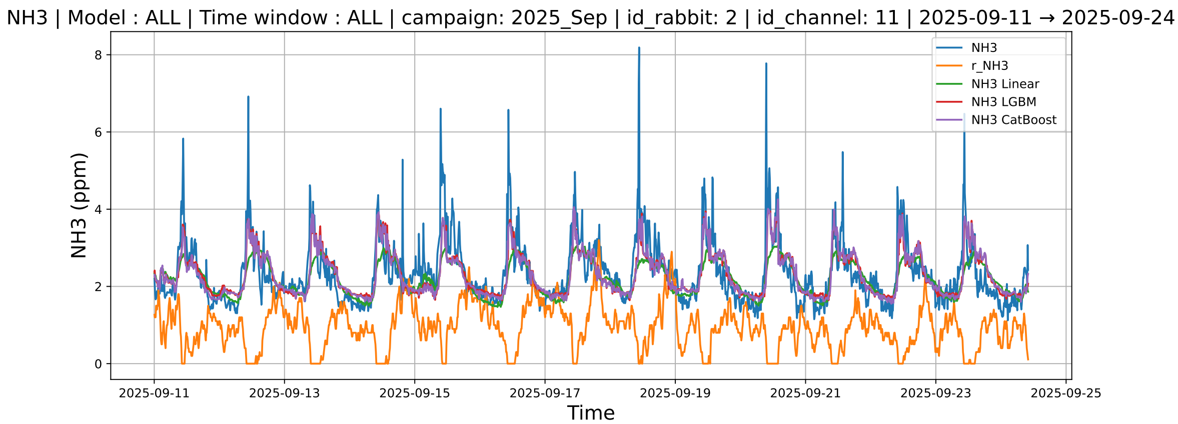
3.2.5.1 Time window : 24



3.2.5.2 Time window : 48

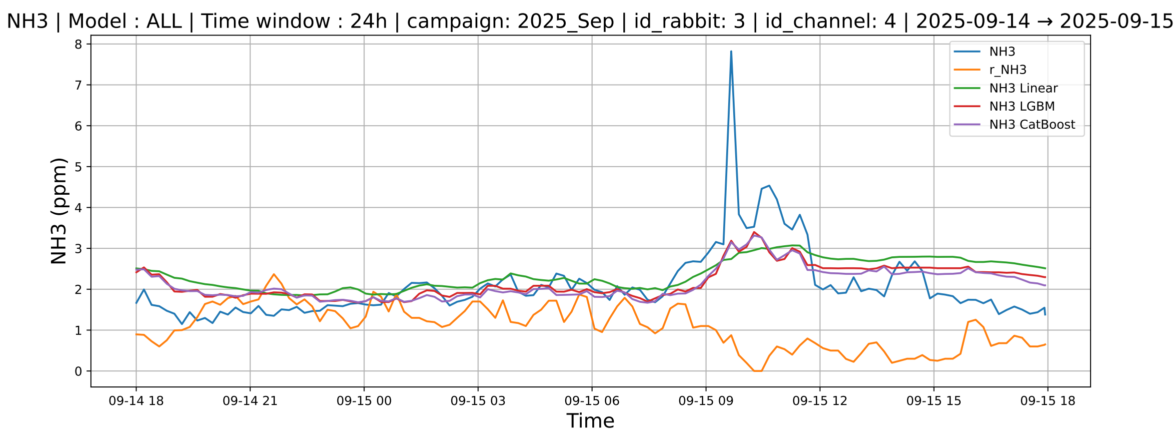


3.2.5.3 Time window : ALL

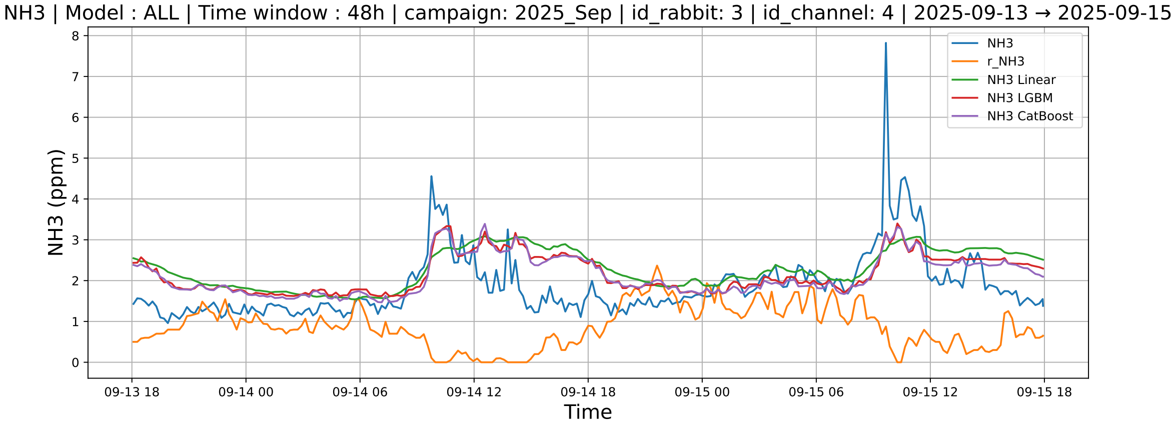


3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

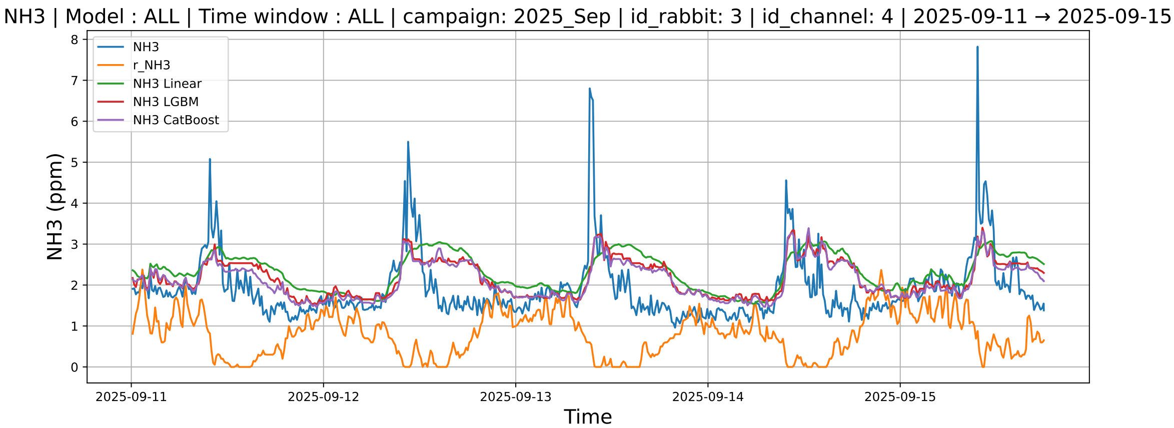
3.2.6.1 Time window : 24



3.2.6.2 Time window : 48



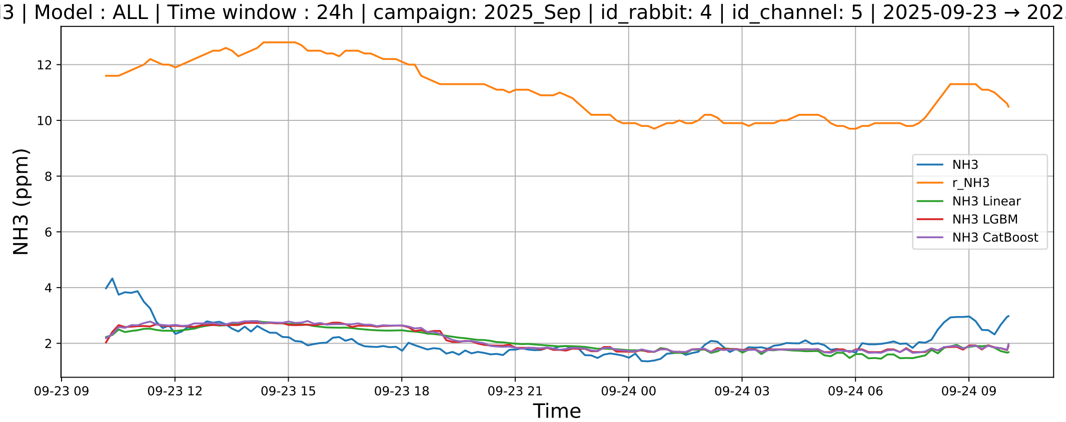
3.2.6.3 Time window : ALL



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

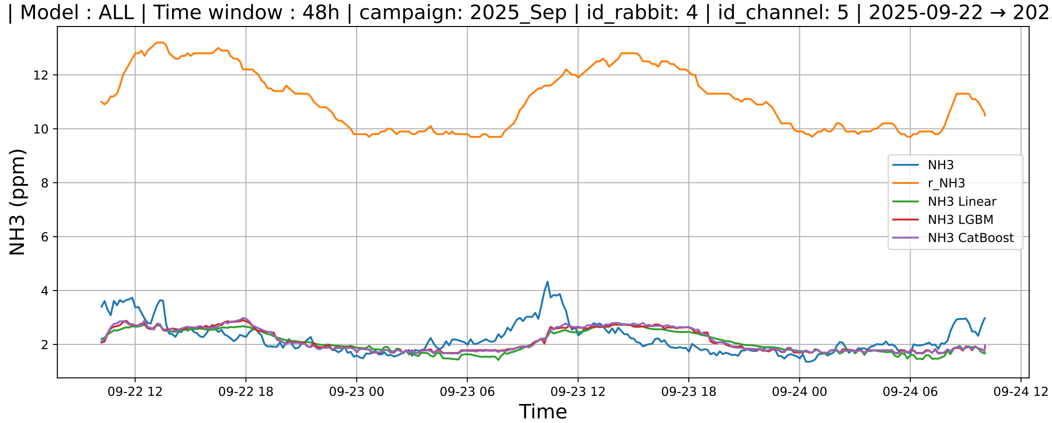
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



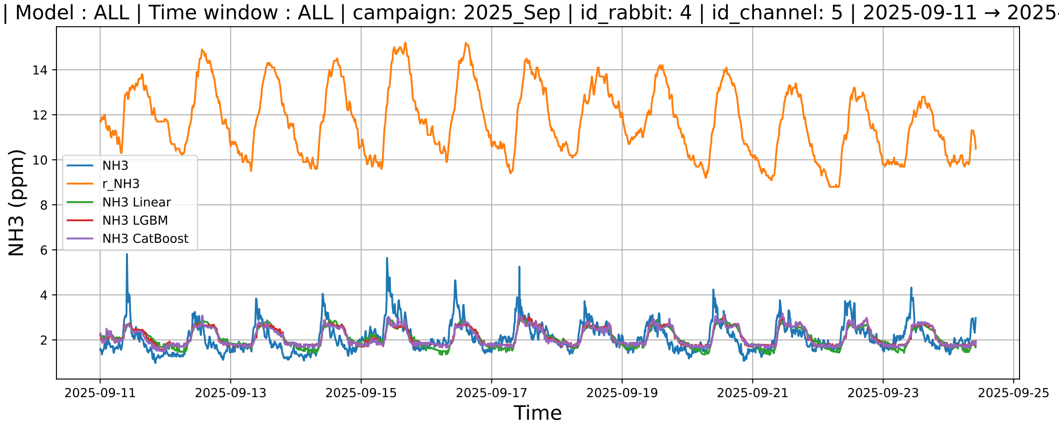
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

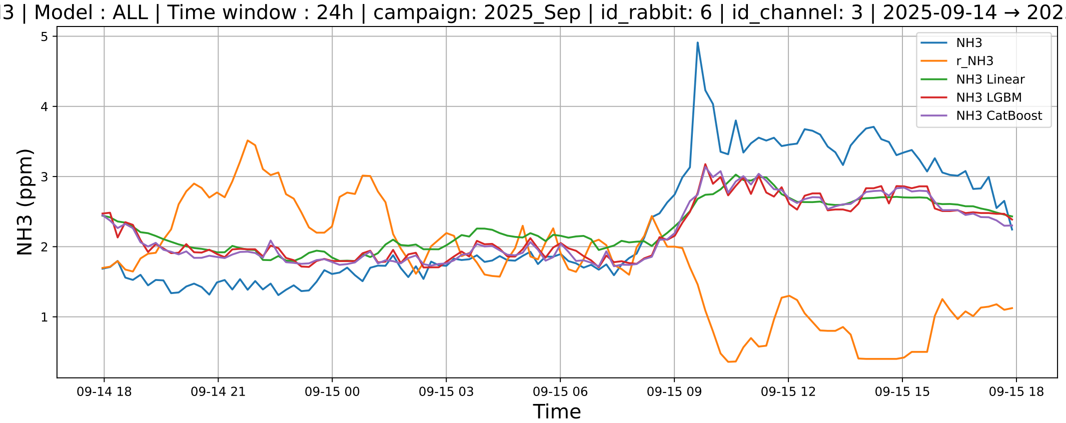
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

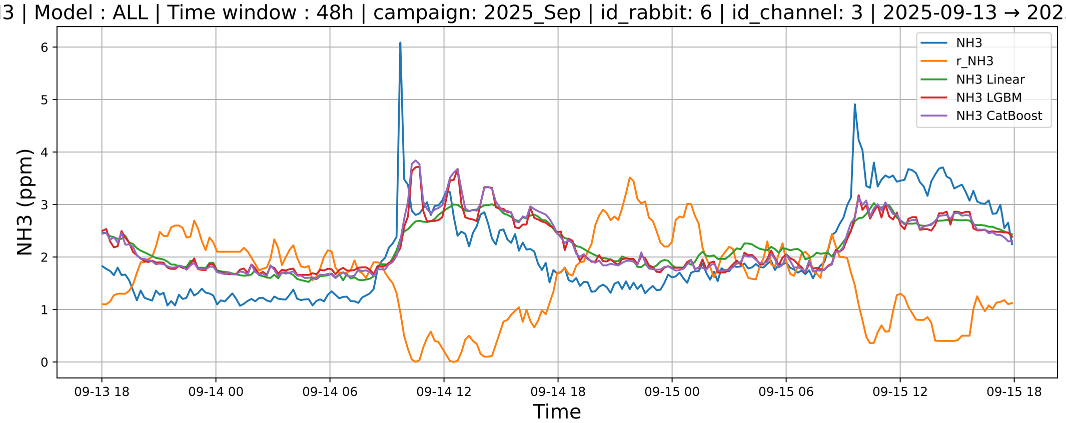
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

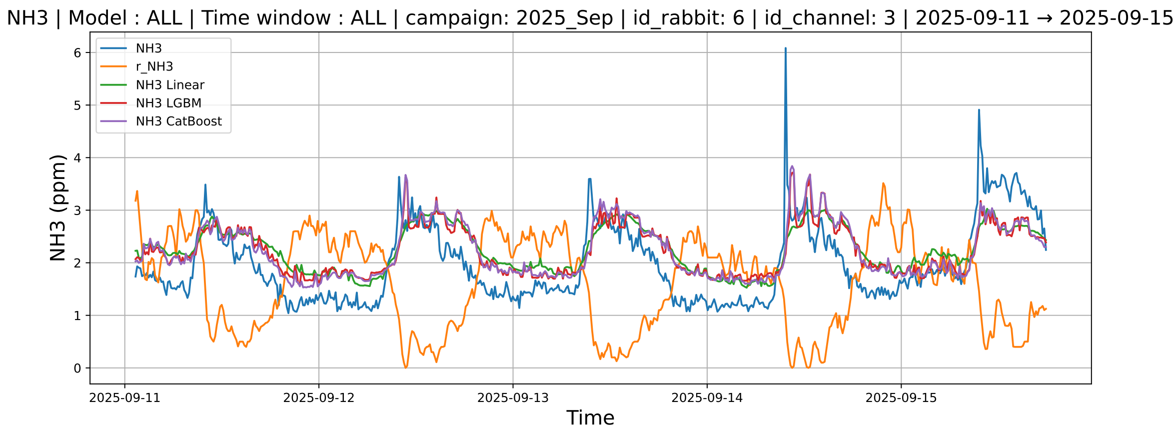


3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL



NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15
NH3
r_NH3
NH3 Linear
NH3 LGBM
NH3 CatBoost
NH3 (ppm)
Time
2025-09-11
2025-09-12
2025-09-13
2025-09-14
2025-09-15